

MANAGED FUTURES

— *Follow the crowd from a distance*



HENRIK HALLENBORG

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Managed Futures – Follow the crowd from a distance

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*Thank you to Sonia, who has shared her home with both me and
the markets throughout all these years.*

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About This Preview Edition

This version is not yet final. An updated and corrected edition will be published as soon as it is ready. If you notice any errors or typographical mistakes, please contact me at **henrik@hallenborgcta.com**

Prologue

I

The Problem

Why Traditional Portfolios Fall Short

Equities

For most investors, saving begins and ends with equities. In long-term investing, whether the goal is retirement or future financial security, the stock market almost always takes centre stage. This is true for both private investors and professional asset managers. Even though pension funds, foundations and other institutional investors normally build broad portfolios containing several asset classes, equities often remain the most important source of long-term returns and therefore a central component of asset management.

There are good reasons for this. Over the past hundred years, equities have generated higher returns than most other traditional asset classes. Although their performance has been far from smooth, the long-term trend has been clear. Companies create value by producing goods and services, investing in new technology, hiring employees and developing their businesses. As the economy grows, corporate profits tend to increase,

something that is eventually reflected in rising share prices.

Unlike many other assets, a share represents ownership of something productive. The buyer becomes a part-owner of the business. If the company succeeds in increasing its profits, expanding into new markets or improving its efficiency, the value of shareholders' capital also rises. This link between economic activity and ownership is the fundamental explanation for the stock market's long-term success.

Another important explanation is the effect of compounding. When profits are reinvested, a cumulative process begins in which earlier returns start generating new returns. The effect may seem insignificant during the first few years, but it becomes increasingly powerful over time. History shows that a large share of the wealth created through equities has not come from spectacular trades or extraordinary forecasts, but from allowing capital to grow over long periods. In many cases, time has been more important than the investment decision itself.

Few people have come to symbolise this philosophy better than Warren Buffett. Through his investment company Berkshire Hathaway, Buffett built one of the world's largest private fortunes over several decades by investing in profitable businesses and holding them through changing market cycles. His success helped reinforce the perception that equities are not merely a good investment, but the most natural investment for anyone seeking to build long-term wealth.

In recent decades, index funds have reinforced this development. An index fund gives investors broad exposure to the entire market at very low cost. The strategy has proved both simple and effective. For millions of savers around the world, index funds have become the standard solution for long-term investing, and the message has been clear: buy the market, reinvest the

returns and give the investments time to grow. Viewed from this perspective, it is easy to understand why equities have acquired such a central role in the investment world.

The problem is that investors tend to focus on long-term average returns and forget the journey required to achieve them. The history of the stock market is not a straight line upwards. It consists of periods of optimism, expansion and prosperity, but also deep crises, long periods of stagnation and recurring disappointments. Although equities have been a successful investment over the very long term, there have been several periods in history when investors have had to wait many years, sometimes decades, to recover their losses.

The stock market crash of 1929 is perhaps the best-known example. After a long period of speculation and optimism, the US stock market collapsed. The decline was so severe that it shaped an entire generation of investors. For many people, it changed their view of equities for the rest of their lives. The event demonstrated that even markets that have risen sharply for a long time can fall much further than most people consider possible. The US stock market did not return to its 1929 peak until roughly 25 years later, showing how long the recovery from a major market decline can take. Adjusted for inflation, the recovery took even longer.

The 1970s revealed a different problem. During this period, the global economy was marked by high inflation, rising energy prices and recurring economic disruptions. Although the stock market did not collapse in the same way as it had in the 1930s, returns remained weak for long periods. At the same time, inflation reduced the purchasing power of money. Investors found that their capital stood still or even declined in real terms despite their equity holdings.

By the beginning of the 1980s, interest in equities had reached record lows. Many members of the older generation still carried memories of the Great Depression and regarded the stock market as risky and unreliable. The younger generation, meanwhile, had grown up during an entire decade of weak market performance and saw little reason to invest in equities. Few could have imagined that one of the strongest and longest bull markets in history was about to begin. Over the following two decades, share prices rose sharply and created enormous wealth. Yet, as before, investors' optimism would eventually be driven to the opposite extreme.

From the peak of the dot-com bubble in 2000 until the beginning of the 2010s, investors once again experienced a stock market that moved sideways for a very long time. First came the sharp decline following the collapse of the technology bubble, which erased much of the gains accumulated during the late 1990s. This was followed by several years of recovery before the world was plunged into a deep financial crisis in 2008. Investors following the Dow Jones Industrial Average had to wait until 2013 before the index established itself firmly above its 2000 peak. Despite recoveries between these events, overall performance was considerably weaker than many had expected. The period was another reminder that an equity portfolio may sometimes need more than a decade to generate genuine growth in wealth again.

Perhaps the clearest example of the drawbacks of equity investing can be found in Japan. During the 1980s, many regarded the country as the future of the global economy. Japanese companies dominated several industries, and optimism about the country's prospects seemed almost limitless. When Japan's asset bubble burst at the end of 1989, the situation changed

dramatically. The stock market fell sharply, and almost 35 years would pass before the Nikkei index again traded above its 1989 peak. Adjusted for inflation, the market has still not fully recovered its losses. Although inflation in Japan has been low compared with the West, 35 years of low inflation still erodes purchasing power. For Japanese investors, this became a painful reminder that even highly successful economies can endure extremely long periods of stagnation.

For someone with many years left before the money will be needed, a major market decline need not be a decisive problem. If the market eventually recovers, the portfolio has time to do the same. The situation is different for someone who has less time remaining until retirement or is already living on accumulated savings. If a severe market decline occurs at that stage of life, there may be no opportunity to recover the losses. History shows that it can sometimes take ten, twenty or even thirty years for a stock market to return to previous highs. The decisive question is therefore not whether the recovery will eventually come, but whether the investor will live long enough to benefit from it.

This is precisely why diversification is not merely about reducing fluctuations in a portfolio. Another purpose is to reduce the risk of a major decline at the wrong stage of life. A high long-term average return offers little comfort if the capital must finance retirement before the market has recovered.

This does not mean that equities are a poor investment. On the contrary, there are strong arguments for equities continuing to form the foundation of most long-term portfolios. History also shows that, despite periods of weak performance, equities have often provided better protection against inflation than many other asset classes. When prices in the economy rise, corporate revenues, profits and asset values tend to rise with them over

time, enabling the stock market in many cases to preserve and increase investors' purchasing power over long periods.

Some of the rise in market indices to which investors have become accustomed in recent years can also be explained by inflation. When the money supply grows faster than the production of goods and services, the purchasing power of money tends to decline over time. As a result, more kronor, dollars or euros are required to purchase the same goods and assets. Over the long term, this tends to contribute to higher nominal prices for both goods and real assets, including equities. Part of the stock market's advance therefore reflects not only increased productivity and higher real profits, but also changes in the value of money over time.

History shows that equities are not a universal solution. Their success is closely linked to economic growth, corporate profits, investors' appetite for risk and the stability of the financial system. When these factors change, returns are also affected, sometimes for very long periods. Although equities have proved an effective hedge against inflation over the long term, this does not mean that they always protect investors from periods of weak or absent real returns. This creates a dilemma for the investor. Equities have historically been one of the best routes to long-term wealth, yet they have also shown that they can endure extended periods of weak performance. The question is not whether equities belong in the portfolio, but whether they are sufficient on their own.

The central problem is that many investors assume the future will resemble the past. When the market environment changes, a portfolio that has worked well for a long time can suddenly prove vulnerable. This realisation has led investors to seek other asset classes and methods of managing risk.

Bonds

If equities cannot always deliver the stable and positive long-term performance that investors hope for, a natural question arises: how can risk be reduced without giving up the return potential of the stock market altogether? The most common answer has been to complement equities with other asset classes. For most of modern financial history, bonds have been the most widely used alternative.

A bond is essentially a loan. When an investor buys a government or corporate bond, the investor lends capital in return for a promise of future interest payments and repayment of the principal at maturity. Unlike equities, whose value depends on future profits and market expectations, bonds are based on contractual cash flows. This generally makes them considerably less volatile than equities.

In modern portfolio theory, equities and bonds have been regarded as natural complements. Equities have provided growth, while bonds have provided stability. When the economy expands and corporate profits rise, equities are expected to perform well. When the economy weakens, uncertainty increases and investors seek safety, capital has tended to flow into the bond market.

This idea gave rise to the classic 60/40 portfolio, in which roughly 60 per cent of the capital is invested in equities and 40 per cent in bonds. For many years, the model became something of a standard solution for long-term investing. It did not offer the same return potential as an all-equity portfolio, but it produced smaller fluctuations and more limited declines. To pension funds, foundations and private investors, it appeared to offer a reasonable compromise between return and risk.

Behind its success lay several assumptions that were rarely discussed because they had long been taken for granted. The first was that inflation would remain relatively low and stable. The second was that interest rates would continue to fall over time, or at least would not rise sharply. The third was that equities and bonds would move in opposite directions when market anxiety increased.

The final assumption is particularly important. If the stock market falls while the bond market rises, a natural shock absorber is created within the portfolio. Losses in one part are partly offset by gains in the other. This negative correlation between equities and bonds became one of the most important building blocks of modern portfolio management. From the beginning of the 1980s until around 2020, the model worked unusually well. Inflation gradually fell from the high levels that had characterised the 1970s. Central banks gained credibility in their fight against inflation, and long-term interest rates declined for several decades.

When interest rates fell, the value of existing bonds rose. The explanation is simple: a bond issued with a coupon of, for example, 6 per cent becomes more attractive when newly issued bonds offer only 3 per cent. Because the higher coupon is fixed for the life of the bond, investors are willing to pay more than its original face value to receive the higher interest payments. In such a scenario, a ten-year bond originally purchased for 100 kronor could rise to around 125–130 kronor when the market interest rate was halved. The bond's price rose until its effective yield was once again in line with the new market rate. Bond investors therefore received not only their regular interest payments but could also realise substantial capital gains when they sold their holdings.

The four-decade decline in interest rates created an unusually favourable environment. Equities benefited from economic growth, globalisation and rising profits, while bonds were supported by falling interest rates. When the stock market faltered, bonds usually provided protection, and when the economy recovered, equities propelled the portfolio forward. Most investors have never experienced a prolonged period in which rising interest rates presented a persistent problem. In practice, they have lived with an almost constant tailwind from the bond market—a historically unusual period that is unlikely to be repeated.

The problem was that these favourable conditions led many investors to regard the relationship between equities and bonds as a law of nature rather than the result of a specific historical environment. Increasing numbers assumed that bonds would always protect against falling stock markets, regardless of economic conditions. History shows that this is not always the case. The negative correlation to which investors had become accustomed is not permanent. It depends on the level of inflation, the actions of central banks and the economic environment.

During some periods, equities and bonds have fallen at the same time. When inflation rises rapidly, the bond market risks significant losses, while the stock market comes under pressure from higher interest rates and a deteriorating economic outlook. This means that bonds, despite their important role in portfolio management, do not always provide the protection investors expect. They perform extremely well in some market environments and considerably worse in others. Like equities, they depend on economic conditions that change over time.

For several decades, it was easy to believe that combining

equities and bonds had solved the diversification problem. The inflation shock and rise in interest rates in 2022 showed, however, that the model could falter when equities and bonds fell simultaneously. This leaves the fundamental problem that investors have sought to solve since modern portfolio management took shape: how can a portfolio be built to cope with different economic environments without depending on the repetition of a particular historical pattern?

The Diversification Problem

If equities do not always perform as investors hope and bonds do not always provide the expected protection, diversification appears to be the natural solution. Almost every investment book, adviser and fund manager stresses the importance of spreading risk. The principle is far older than modern portfolio theory. As early as the second part of Miguel de Cervantes' novel *Don Quixote*, published in 1615, readers were warned not to risk all their eggs in one basket. Four hundred years later, the same image is used to explain how investors can reduce the risk of large losses and achieve more stable performance over time. The problem is that diversification is often more difficult than it first appears.

Most investors consider themselves well diversified. A portfolio may contain Swedish equities, US equities, European equities and emerging markets. It may be spread across several industries and countries, as well as a large number of individual companies. On the surface, this looks like a broadly diversified portfolio in which risk should be evenly distributed. Under normal market conditions, it appears to be exactly that. The relative strength of markets varies from year to year. Sometimes

US technology companies lead the advance; at other times, European industrial companies or Asian emerging markets take the lead. These variations create the impression that the capital is spread across several independent investment opportunities.

But genuine diversification is tested only during a crisis. When markets come under severe stress, investor behaviour changes. Uncertainty rises, risk appetite declines and capital begins to leave risky assets. Investments that previously appeared independent start moving in the same direction. Swedish equities fall, US equities fall, European equities fall and emerging markets fall. The differences between them suddenly matter less than the fact that they all belong to the same broad category of risky assets.

The investor then discovers something that was not apparent during good times. The portfolio may contain many holdings, but they are fundamentally exposed to the same economic forces. This brings us to one of the most important concepts in portfolio management: correlation.

Correlation describes the extent to which different assets tend to move together. If two investments rise and fall at the same time, they are said to have a high positive correlation. If one tends to rise when the other falls, they are said to have a negative correlation. The lower the correlation between different holdings, the greater the genuine diversification benefit. Many investors make a common mistake here: they focus on the number of holdings rather than the risks they are actually taking. Owning fifty different shares is not necessarily more diversified than owning five if all of them are affected by the same underlying factors. The number of positions matters less than how those positions respond when the market environment changes.

The 2008 financial crisis illustrated this clearly. When the crisis struck, equities from different sectors and regions of the world fell simultaneously. Assets that had displayed clear differences under normal conditions began moving in the same direction. The crisis demonstrated that diversification across many equities does not protect against risks that affect the entire financial system. This type of risk is generally described as systemic risk.

Behind systemic risk lie common forces with broad effects. Economic growth is one such force. When the global economy expands, many companies and markets benefit. When the economy weakens, many of them suffer at the same time. Interest rates, credit conditions, inflation and investors' willingness to take risk affect many markets in similar ways. Although assets may appear different on the surface, they can depend on the same underlying factors.

This means that a portfolio consisting of Swedish equities, US equities, European equities and emerging markets may in practice be a concentrated exposure to global economic growth. As long as growth continues, the strategy works well. When conditions change, all these markets risk being affected.

From this perspective, it becomes clear that genuine diversification is not about owning many different assets, but about owning assets that respond differently in different market environments. The objective is not to maximise the number of holdings, but to reduce dependence on individual economic scenarios. This is also why investors have long sought strategies that do not depend on rising stock markets or falling interest rates. If equities and bonds are ultimately influenced by the same broad economic forces, the question remains whether other types of investment can generate returns even when these

traditional asset classes are struggling.

It was in the search for this type of diversification that many of the strategies later grouped under the term Managed Futures found a place in modern portfolios.

The Fundamentals of Managed Futures

Unknown but Established

Having established that stocks and bonds do not always provide the diversification investors expect, a natural question remains: what alternatives are available? One of the most established is Managed Futures. Although the asset class manages hundreds of billions of dollars globally and is used by many of the world's largest pension funds, it remains relatively unknown outside professional investment circles. To many private investors, the term sounds technical, complicated, and difficult to place within the traditional division between stocks, bonds, and funds.

The explanation lies partly in the way investments have traditionally been viewed. For most of the twentieth century, asset management was dominated by stocks and bonds. Although the two asset classes differ, they are based on the same fundamental principle: the investor buys an asset in the hope that its value will rise over time. A stock is purchased because the company is expected to prosper, a bond to provide future interest payments,

and a property because rental income and property values are expected to increase. What they share is that their returns ultimately depend on a positive future scenario.

Managed Futures represents a somewhat different approach. Rather than starting with which assets should be owned, the strategy starts with how markets actually move. The primary focus is not on whether a stock is cheap or expensive, whether the economy is growing quickly or slowly, or whether central banks will raise or lower interest rates. The focus is on price movements and the ability to adapt to them. Markets develop trends in both directions. The manager's task is to identify and follow those trends as they unfold. This difference in perspective makes Managed Futures fundamentally different from traditional asset management.

In theory, this means that a Managed Futures portfolio can profit from rising stock markets, falling stock markets, rising interest rates, falling interest rates, stronger currencies, weaker currencies, or sharp moves in commodities. What matters is not the direction, but whether a sufficiently clear trend exists. This approach may seem unusual because it differs from the way investments are normally presented. Financial media focus on forecasts, analysts try to estimate future earnings, and economists discuss business cycles and central banks. Investors become accustomed to the idea that success depends on predicting what will happen next. Managed Futures is based on a different philosophy.

No one knows with certainty what will happen tomorrow, next month, or next year. That is true of investors, economists, central bankers, and politicians alike. The Managed Futures manager addresses this uncertainty through a process designed to adapt to different future scenarios. If inflation rises, the

economy enters a recession, commodity prices surge, or the stock market falls, the portfolio should be able to adjust to the new reality. What if the most important thing for an investor is not knowing what will happen, but being prepared for what actually happens? The focus shifts from forecasting to action: from the question “What will happen?” to “How should the capital be managed when events unfold differently than expected?”

This characteristic makes Managed Futures one of the few asset classes that genuinely differs from traditional investments. The strategy does not depend on rising stock markets, falling interest rates, or any particular economic outcome. Managed Futures is not based on optimism about a particular market, but on the price movements that actually occur.

This characteristic has led many pension funds, foundations, and institutional investors to include Managed Futures in their portfolios. The purpose is not necessarily to achieve higher returns than the stock market. Instead, the aim is to build a portfolio that can function across a wider range of market environments than traditional stock-and-bond portfolios.

Terms such as CTA, Commodity Pool, and trend following may sound technical and unfamiliar to someone encountering the asset class for the first time. To understand how the industry works, we need to begin with its most important building blocks.

The CTA Manager

CTA stands for Commodity Trading Advisor, a designation rooted in the US futures market. Today, however, the name is somewhat misleading.

The word *commodity* readily brings to mind raw materials,

which can easily give the impression that a CTA trades primarily in markets such as oil, gold, wheat, or coffee. Historically, there was some truth to this. During the early development of futures markets, trading was heavily concentrated in agricultural products and other physical commodities. It was once jokingly said that a commodity was something that would hurt if you dropped it on your foot.

Today the picture is different. Modern CTA managers still trade commodities, but they also operate across a wide range of financial markets, including stock indexes, government bonds, interest rates, and currencies. For many of the largest CTA firms, financial markets now account for a larger share of their activity than traditional commodities. The name, however, has endured for historical and regulatory reasons.

The word *advisor* can also be somewhat misleading. A CTA rarely acts like a traditional adviser who makes investment recommendations. The CTA manager makes the investment decisions and executes the trades directly under a predefined mandate. In many respects, the role therefore resembles that of a discretionary fund manager more than an adviser in the everyday sense of the word. The investor entrusts the capital or trading authority to the manager, who then assumes responsibility for the ongoing investment decisions.

A CTA can be described most simply as a professional asset manager who uses futures markets to manage capital. The approach may be entirely systematic, entirely discretionary, or a combination of the two. Many of today's largest CTA firms rely primarily on systematic models in which buy and sell decisions are generated according to predetermined rules. Others combine quantitative models with human judgment, while some base their entire operation on discretionary market assessments.

The management is not passive in the same sense as an index fund. On the contrary, it requires extensive work involving analysis, risk management, portfolio construction, and continuous market monitoring. The difference is that decisions are based largely on established principles rather than subjective forecasts about the economy or individual markets.

In a traditional equity fund, the manager spends much of the time analyzing companies, their earnings, balance sheets, and future prospects. A CTA generally works differently and relies more heavily on market price movements. Signal design, position sizing, and the portfolio's overall risk are central elements of the CTA manager's work.

Another important difference is that a CTA manager can normally operate in both directions. If a market develops a positive trend, the manager can buy. If the same market instead develops a negative trend, the manager can take a short position and potentially profit from the decline. The same principle applies to stock indexes, interest rates, currencies, and commodities. This gives the CTA manager considerably more flexibility than many traditional fund managers, who depend primarily on rising markets.

The CTA industry has grown substantially in recent decades and now includes managers overseeing tens of billions of dollars. At the same time, CTA does not denote a specific strategy. Two CTA managers may use entirely different methods and still belong to the same category. Some base their decisions on trend following, others on statistical models, macro strategies, or various forms of relative-value trading. What they share is not precisely how they invest, but the use of futures markets and professional risk-management processes as the foundation of their operations.

The Futures Market

To understand how a CTA operates, one must first understand the instrument that underpins almost the entire Managed Futures industry: the futures contract.

A futures contract is essentially an agreement to buy or sell an asset at a future date at a price determined in advance. Futures should not be confused with options. A futures contract obligates both parties to fulfill the agreement if the position is held to expiration, whereas an option gives the buyer the right, but not the obligation, to buy or sell the underlying asset. They are two different financial instruments.

Historically, futures contracts emerged in agriculture. A farmer could, for example, lock in the selling price of an upcoming harvest, while a mill owner could secure future purchasing costs. Through the futures contract, both parties could protect themselves against price changes.

An early example can be found in Japan. At the Dojima Rice Exchange in Osaka, organized trading in contracts and rice certificates developed in the early eighteenth century, making it possible to determine the price of rice before final delivery took place. Producers and merchants could thereby reduce the risk of a sharp price change between the agreement and delivery. The Dojima Rice Exchange was officially authorized in 1730 and is commonly described as the world's first organized futures market. Futures contracts are therefore not a modern product created for speculation. They grew out of a practical need to manage uncertainty about future prices.

Over time, futures markets expanded far beyond agriculture. Today, futures contracts are traded on a very broad range of assets, including crude oil, gold, coffee, milk, butter, and cheese,

as well as government bonds, interest rates, currencies, stock indexes, freight rates, and emissions allowances. For many of these assets, futures exchanges are the leading venues for price discovery and trading.

If a futures contract is held until expiration, it will normally result in delivery of the underlying asset. Someone who has bought a wheat futures contract, for example, will receive the quantity and quality specified by the contract. Delivery takes place through an exchange-approved warehouse or another approved delivery location, in accordance with the rules governing the particular contract.

Financial futures work somewhat differently. Depending on the design of the contract, settlement takes place either through delivery of the underlying financial asset, such as a government bond, or through cash settlement, in which only the change in value between the contract price and the market price is settled.

In practice, most financial-market participants close their positions before the contract expires. Selling a purchased contract or buying back a sold contract terminates the obligation. Because around 97 percent of all futures contracts are closed before delivery, only a small proportion actually results in physical delivery.

For a speculator, such as a CTA manager, futures contracts serve a different purpose than they do for producers and consumers who use them to hedge prices or arrange a future purchase or sale of the underlying asset. The speculator uses futures as an efficient way to gain exposure to price movements, without intending to hold the contract through physical delivery or final settlement. A CTA that buys a crude-oil futures contract normally has no intention of taking delivery of thousands of barrels of oil. Likewise, a CTA trading interest-rate futures has

no intention of receiving or delivering government bonds. The objective is to benefit from the price movements that occur in the market.

These speculators also perform an important economic function. By accepting the price risk that producers and consumers seek to reduce, they provide liquidity to the market. This makes it easier for companies to hedge future prices and contributes to more efficient price discovery. Without this transfer of risk, the cost of hedging future prices would probably be higher, which could ultimately increase companies' costs and, in turn, prices for end consumers.

The great advantage of futures markets is that they provide access to a very broad range of markets through standardized and liquid infrastructure. From a single trading account, a manager can trade US stock indexes, European interest rates, Japanese government bonds, currencies, energy, metals, and agricultural products. This flexibility laid the foundation for the entire Managed Futures industry. In practice, CTA managers are often described as global macro managers. They are not confined to a particular region or asset class, but can seek opportunities wherever the strongest trends emerge.

CTA Programs and Commodity Pools

We have now looked at who makes the investment decisions and which instrument is used to implement them. The next question is how the investment itself is structured.

There are two main ways for investors to engage a CTA. The first is through an individually managed futures account, in which the investor owns the account but grants the CTA manager trading authority to execute transactions according to a defined

CTA program. This is known as a *managed account*. The second is through a pooled investment structure, known as a Commodity Pool, in which the capital of several investors is managed collectively. Both models are used within the Managed Futures industry and are based on the same fundamental investment philosophy, but they differ in how the capital is organized and administered.

An individually managed CTA program offers a high degree of transparency. Because the investor owns the account, every trade, position, and account movement can be monitored directly through brokerage statements. This type of arrangement is aimed primarily at larger investors and, in many cases, requires a substantially higher minimum investment than a Commodity Pool. For smaller investors, a Commodity Pool is therefore the most practical route into Managed Futures.

A Commodity Pool can most easily be compared to a fund. Several investors contribute capital to a common structure that is then managed according to a defined strategy. Profits and losses are allocated among the investors in proportion to their ownership interests, in much the same way as in many traditional investment funds.

The basic principle is relatively simple. Instead of each investor opening a separate futures account, joining an individual CTA program, and maintaining a separate relationship with the broker, the capital is combined in a single Commodity Pool.

The management can then be organized in different ways. The pool may invest in one or more external CTA programs, or it may allow a CTA manager, who will usually also be the CPO (Commodity Pool Operator), to trade directly on its behalf. Under either model, investors gain access to markets and strategies that would otherwise be difficult or impractical to

manage independently. Administration, reporting, and ongoing management are thereby simplified because everything takes place within a single structure.

Within the pooled structure, trading, risk management, administration, auditing, and reporting are handled centrally. The investor therefore does not need to manage these functions separately. The main difference from a traditional equity fund lies in the markets used and the regulatory frameworks surrounding the activity. Whereas an equity fund invests primarily in stocks, a Commodity Pool generally invests through futures contracts and other derivatives. This makes it possible to gain exposure to a very large number of markets through a single account.

Another important difference is that futures markets require margin rather than full payment for the underlying asset. When a CTA manager takes a position in crude oil, government bonds, or a stock index, for example, only around 5–15 percent of the contract's value is normally required as margin. Full payment is required, or the underlying asset must be delivered, only if the contract is held until delivery or final settlement, depending on the structure of the contract. The remaining capital can be used for other positions or held in liquid assets. This enables CTA managers to build highly diversified portfolios with exposure to many markets at the same time.

Historically, the Commodity Pool structure has been particularly common in the United States, where it developed alongside the CTA industry and the growth of futures exchanges. Many of the world's largest Managed Futures programs are still organized as Commodity Pools, although comparable structures now exist in many different legal forms around the world. For the investor, however, the practical meaning is straightforward.

A Commodity Pool is the legal and economic framework that brings investors' capital together, while the CTA manager is responsible for how that capital is invested. Together, they form the basic structure underlying much of the Managed Futures industry.

In the United States, CTA activity is overseen by the Commodity Futures Trading Commission, or CFTC. Managers are normally registered with the National Futures Association, or NFA, and are subject to requirements governing areas such as reporting, documentation, and marketing. Registration and ongoing oversight help create structure and confidence in the industry.

Investment Strategy

Now that we have examined who manages the capital, which instrument is used, and how the investment is structured, the most important question remains: how are the investment decisions made?

The strategy most strongly associated with Managed Futures is trend following. The basic premise is that markets sometimes develop persistent price movements that can last for months or even years. The manager does not try to predict when these movements will begin or end, but identifies trends once they have already become established. If a market is rising, a long position is established. If the market is falling, a short position is established to profit from the decline. When the trend weakens, the position is closed and the capital is shifted to other markets where stronger movements are developing.

The strategy does not depend on the manager knowing why prices are rising or falling. It is enough to recognize that a trend

exists and then adapt to it. One of the main differences between trend following and many other investment strategies is that the manager does not try to determine what a market ought to be worth. What matters most is what the market is actually doing: a rising price is treated as information, and so is a falling price. The price movement itself becomes the basis for the decisions.

Investment decisions are usually made with the aid of statistical and mathematical models that identify and follow market price movements. This does not mean that an algorithm can predict the future. On the contrary, most managers begin with the assumption that the future is uncertain. No one knows which markets will be next year's winners or losers. Capital is therefore often spread across a large number of positions at the same time. The goal is not to find a single perfect investment, but to create a portfolio capable of capturing the trends that actually emerge.

Trend following does not need to be right on a majority of trades to be profitable. A small number of large winners can offset many limited losses. Many small losses are accepted in exchange for the opportunity to benefit from the few occasions when markets move exceptionally far. Historically, such movements have occurred during economic crises, periods of inflation, currency realignments, commodity shortages, and other major changes in the global economy.

Another characteristic of trend following is that the method works in the same way regardless of the market. The same basic principle can be applied to stock indexes, government bonds, currencies, energy, metals, and agricultural products. The manager does not need to develop a separate investment philosophy for each market. Price movement remains the central focus.

Although trend following is the best-known strategy, it is far

from the only one. Many modern CTA managers also use models based on carry, relative-value strategies, seasonal patterns, and other quantitative methods. Trend following nevertheless remains the historical core of the industry and has contributed greatly to its development and popularity.

The Role of Managed Futures in a Portfolio

A Managed Futures manager rarely aims to beat the stock market every year. Nor does the manager try to forecast the economy, central-bank actions, or the next market crash. The task is to create a source of return that is less dependent on the factors driving traditional investments.

This is generally described as an absolute-return objective. Unlike relative management, in which performance is assessed against an index, the CTA manager seeks to generate positive returns over time regardless of whether traditional markets rise or fall. A well-constructed Managed Futures portfolio should be able to benefit from trends across different markets, whether those trends are positive or negative.

This is what makes Managed Futures a form of diversification that differs from traditional investments, whose returns often depend on the economy performing well over time. Stocks benefit from economic growth and bonds often benefit from falling interest rates, while Managed Futures can adapt to different market environments. Rising commodity prices, falling stock markets, sharp currency movements, or changing interest-rate levels can all create opportunities if sufficiently strong trends develop.

This does not mean that Managed Futures works in every situation. Like all investment strategies, CTA programs go

through periods of weak performance. Environments in which markets lack clear trends and move back and forth without a lasting direction are particularly challenging. The aim is not to profit every month or every year, but to create over time a distinct source of return that can be combined with traditional stock and bond investments.

For this reason, CTA managers rarely compare themselves with broad stock indexes such as the S&P 500. A stock index is designed to measure the performance of the equity market. A CTA has an entirely different mandate: to identify opportunities across a large number of markets and control portfolio risk. Comparing a trend-following CTA with the S&P 500 is rather like comparing a bond fund with a real estate index. Both are investments, but they are trying to accomplish different things.

Diversification is another area in which Managed Futures has attracted considerable attention. When stocks and bonds are affected by the same problems, trend-following strategies may respond in a very different way. Historically, this has been particularly evident during periods of financial stress, inflation, or sharp market change. This characteristic has made Managed Futures a recurring component of many institutional portfolios around the world.

For pension funds, foundations, and other long-term investors, the issue is rarely finding the investment that will deliver the highest return over the next year. The aim is to build robust portfolios that can function under many different economic conditions. Managed Futures has played an important role in that effort for several decades.

Understanding how the industry is structured is only the beginning. The next step is to examine how returns are generated and which characteristics may allow the strategies to continue

working in the future. This explains why Managed Futures has attracted capital from professional investors around the world, although past performance is never a guarantee of future results.

3

Why Trends Exist

The Commodity Cycle Creates Trends

To understand why trend following continues to work even though its underlying quantitative models are well known, widely studied, and used on a large scale, we must look beyond the method itself and focus on what drives price movements. Trend following is not about being a *rocket scientist* or discovering a unique strategy that no one else knows about and that would cease to work if it were published in a newspaper. At its core are the forces that create long-lasting trends. One of the most important is the commodity cycle—a fundamental yet underestimated force in financial markets.

In its simplest form, the commodity cycle is a recurring process in which price serves as the central signal coordinating supply and demand over time. When prices rise, the behavior of both producers and consumers changes. When prices fall, corresponding adjustments take place in the opposite direction. What makes this process particularly relevant is that these

adjustments do not occur immediately. They happen with a delay, creating the imbalances that give rise to prolonged upward and downward price movements, or trends.

When the price of a commodity rises sharply, whether it is oil, copper, or coffee, several things happen at once. Consumers begin to adjust their behavior and demand declines—not necessarily abruptly, but gradually. Companies seek substitutes, use the commodity more efficiently, or postpone consumption. Producers respond as well. Higher prices make it profitable to increase output, restart previously unprofitable projects, or invest in new capacity.

The problem is that increasing production takes time. In many commodity sectors, particularly energy and metals, several years may pass between an investment decision and the point when the resulting output actually reaches the market. During this period, the imbalance between supply and demand persists, which in turn drives the price still higher. This delay creates the conditions for persistent trends.

The delay varies considerably from one commodity to another. If a new orange tree is planted today, it takes two to four years to bear its first fruit, and roughly five to eight years before it reaches full production. If frost destroys a large share of the crop or many trees die, growers therefore have no way to restore production quickly. It may take many years for supply to return to its previous level.

The same principle applies to cattle. From the time a producer decides to expand a herd until the new animals are ready for slaughter, the process normally takes about two to three years. These are the animals traded on the futures market under the name *Live Cattle*. When production cannot adjust quickly to changes in demand, prolonged periods of shortage or surplus

emerge, which in turn give rise to the long-lasting price trends that trend-following strategies seek to exploit.

Once the new production reaches the market, the situation changes. Supply increases while the previously high price level has already dampened demand. The market moves from shortage to balance and then to surplus. The price stops rising, stabilizes, and eventually falls. This decline sets the next phase of the cycle in motion.

In some cases, however, the advance develops into a speculative phase. After prices have risen for a prolonged period, an increasing number of participants are drawn into the market, eventually including the general public. Demand is then driven not only by the underlying shortage but also by expectations of further price increases. The result is often a price far above what supply-and-demand conditions justify. Once the market realizes that the shortage is disappearing, the speculative rise may quickly be followed by an equally sharp decline.

At low prices, the opposite process occurs. Consumers increase their purchases and demand rises because the commodity has become cheaper. Producers, by contrast, reduce their activity. Projects are shut down, investments are postponed, and output declines. As during the upswing, adjustment occurs with a delay. It takes time to close capacity and even longer to restart it once the price begins to rise again.

This interaction—in which price first affects behavior and changes in behavior then drive the price in the opposite direction—lies at the heart of both the commodity cycle and trend following. The strategy does not depend on a particular market or period; it is a consequence of the way real economic systems operate. As long as price, production, and consumption remain connected, this cyclical dynamic will persist.

What makes this sequence especially relevant to trend following is its tendency to create persistent price movements rather than random fluctuations. Because supply and demand adjust with a delay, price movements often become both longer and more powerful than initially appears reasonable. Commodity markets therefore sometimes develop pronounced overreactions, with prices rising or falling far beyond what long-term fundamentals justify. A commodity can move from severe shortage to surplus, or vice versa, in a way that creates clear trends lasting months or years.

This recurring adjustment makes trend following a robust strategy. It rests on economic conditions that have repeatedly appeared in commodity markets over very long periods and are likely to continue in the future. At their core, trend-following models are a way of responding systematically to these cycles. They do not try to predict when a cycle will begin or end; they adapt once it is underway. This explains why they work over time—not because the models are inherently sophisticated, but because they are linked to an underlying process that recurs in markets.

One important aspect that is often misunderstood is how these movements appear to investors. To the average consumer, commodity prices seem relatively stable. The price of coffee in a store rarely changes dramatically from one week to the next, and even over longer periods adjustments tend to be gradual. This is because retail prices filter out much of the short-term volatility through inventories, pricing strategies, and contracts.

The futures market works differently from retail markets. The price of coffee in a store includes not only the raw commodity but also processing, packaging, transportation, taxes, and retail margins. Inventories, supply agreements, and corporate

pricing also mean that changes in commodity prices are not always passed on immediately. A futures price, by contrast, refers to a specified quantity of the commodity for delivery at a particular time and responds continuously to new information and changing expectations about future supply and demand. The movements of the commodity cycle therefore become clearer and directly tradable.

Both the physical commodity and the futures contract can experience large price movements. The relatively small margin requirement does not make the futures contract itself more volatile, but it does allow a large exposure to be controlled with less committed capital. Leverage therefore magnifies the gain or loss relative to the capital in the account, not the market's percentage price movement itself. Consequently, the magnitude of fluctuations in a CTA program is determined not only by the contracts traded but, above all, by position sizes and the portfolio's total risk. The same price movement may have a limited effect at low exposure and a much greater effect when positions are larger.

This brings us back to why trend following does not stop working even though it is well known. The answer is that it does not exploit a temporary inefficiency, but a structural feature of markets. The commodity cycle does not arise because market participants make mistakes in the traditional sense, but because production and consumption cannot adjust immediately to price changes. This inertia creates the movements captured by the models. As long as the world depends on physical commodities and production takes time to increase or reduce, these cycles will continue to exist.

Trend following does not depend on being unique or secret, but on the CTA manager responding consistently to these

movements. The strategy therefore often performs best during periods of greater imbalance. When markets are stable and supply and demand are in balance, there are few clear trends to exploit. But the balance is never permanent. Changes in supply, demand, production, and investor behavior sooner or later create new imbalances and, with them, the movements the strategy is designed to capture.

The combined positions of CTA managers account for only a small share of global trading in energy, metals, grains, and other commodities. Long-term price trends are therefore created primarily by changes in production, consumption, and inventories, not by the trading of trend followers. Even if the CTA industry continues to grow, it remains small relative to the global commodity economy.

At the same time, commodity cycles are only one part of a broader economic pattern. Economists have long tried to identify cyclical processes in the economy as a whole.

In the 1920s, the Russian economist Nikolai Kondratiev argued that economic cycles could extend over several decades. By studying commodity prices, interest rates, trade, and production, among other things, he identified waves of expansion and contraction lasting roughly 40–60 years. The theory is disputed, but it shows that the idea of long-term economic cycles has a long history. His conclusion that capitalist economies could undergo deep crises and subsequently recover conflicted with Soviet ideology. He was arrested in 1930 and executed during Stalin's purges in 1938.

But if long-lasting trends recur, another question arises. How can they be exploited systematically when the efficient market hypothesis says that known price patterns should not give investors a lasting edge?

The Efficient Market Hypothesis

If all investors had access to the same information and always acted rationally, market prices should adjust immediately to new information. As soon as news was published, its significance would be analyzed, valued, and instantly reflected in the price. In such a world, long-lasting trends could hardly exist. Prices would move from one equilibrium point to the next without extended periods of persistent advances or declines. Broadly speaking, this is the essence of the so-called efficient market hypothesis, a theory that came to dominate large parts of academic finance for several decades.

The researcher most closely associated with the theory was Eugene Fama of the University of Chicago. During the 1960s, he developed the efficient market hypothesis in a series of influential academic papers, thereby laying the foundation for much of modern financial economics. Fama's research had an enormous impact and shaped both university teaching and professional asset management around the world for several decades.

The efficient market hypothesis is based on the idea that market prices incorporate all available information. The implication is that no investor should be able to systematically earn a higher return than the market as a whole by analyzing publicly available information, because that information is already embedded in prices. Historical price movements should therefore provide no useful information about future price performance. From this perspective, trend following seems like a strange idea.

For a long time, trend following was therefore viewed with considerable skepticism in academic circles. If historical price movements contained useful information, other investors

should quickly discover and arbitrage away the opportunity.

In 2013, Fama was awarded the Nobel Prize in Economic Sciences for his empirical research on how asset prices behave, not for the theory itself. He shared the prize with Lars Peter Hansen and Robert Shiller, whose research pointed in a different direction in several respects. While Fama's work emphasized how quickly markets can absorb information, Shiller documented how psychology, herd behavior, and recurring bubbles can cause prices to deviate from fundamental values for long periods. The fact that both were honored at the same time illustrates how financial research has developed toward a more nuanced view of markets. They are neither perfectly efficient nor completely inefficient. Information spreads quickly and competition is intense, but human behavior, institutional constraints, and structural imbalances cause long-lasting trends and other anomalies to recur.

The problem with the efficient market hypothesis was that empirical reality did not fully correspond with the theory's predictions. In the decades after the hypothesis was established, both academic research and practical experience revealed the same phenomenon: markets tended to continue moving in the same direction longer than many models predicted. This did not apply only to isolated markets or specific periods. The phenomenon recurred in equities, commodities, currencies, and interest rates; in the United States, Europe, and Asia; and during periods of inflation, deflation, economic growth, and recession. The more data accumulated, the harder the observation became to ignore.

An important breakthrough came in 1993, when Narasimhan Jegadeesh and Sheridan Titman showed in the now-classic study *Returns to Buying Winners and Selling Losers* that stocks

with strong historical performance tended to continue outperforming stocks with weak performance. In 2012, Tobias Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen showed in *Time Series Momentum* that the same effect also existed in futures markets across commodities, currencies, interest rates, and equity indices.

Momentum thus became established as one of the most robust yet difficult-to-explain anomalies in financial economics. Put simply, momentum means that assets that have performed strongly over a period tend to continue performing strongly for some time, while assets that have performed poorly often continue to perform poorly. For many academics, this was an uncomfortable result because it conflicted with a strict interpretation of the efficient market hypothesis.

Professional investors nevertheless continued to observe the same thing: commodities could develop trends lasting several years, currencies could rise or fall for extended periods, and interest-rate markets could move in the same direction for an entire decade. Equity markets could remain in bull or bear markets far longer than most forecasters anticipated.

Consider the US bond market. From the early 1980s until the beginning of the 2020s, interest rates fell almost continuously. In commodities, we have seen multiyear advances in gold, oil, and agricultural products. In the foreign exchange market, the dollar has on several occasions developed strong movements extending over many years. If markets were perfectly efficient, such movements should be considerably harder to explain. This does not mean that the efficient market hypothesis has no value. It captures important features of the market: competition is intense, information spreads rapidly, and lasting advantages are difficult to create. But the theory does not provide the complete

picture.

A Market with No Answer Key

Think of a billiard ball. If the ball's exact position, speed, and direction are known, along with the characteristics of the table and the forces acting on it, it is possible in principle to calculate where it will be one second later. The future then becomes a mathematical consequence of the starting conditions. The French mathematician Pierre-Simon Laplace took this idea much further. He imagined an intellect that knew every force and the exact position of every object. With complete knowledge of the initial state, it could calculate both what had happened and what would happen.

The financial market, however, is not such a closed system. It consists of people and institutions that react to new information, changing prices, and one another's actions. Every decision therefore changes the market that everyone else is trying to analyze.

The market's future is therefore not hidden like a completed answer key in the information already available. It is partly created through decisions that have not yet been made. The British economist John Maynard Keynes, one of the most influential economists of the twentieth century, used the price of copper and the interest rate twenty years into the future as examples of questions that could not be calculated on any secure basis. His conclusion was simple: "We simply do not know."

Understanding in hindsight why something happened is not the same as being able to say in advance when and how it would happen.

Homo Sapiens and the Herd

Behind every buying and selling decision is a human being. Despite the high technological sophistication of financial markets, it is still *Homo sapiens* making the decisions—the same species that evolved over hundreds of thousands of years to survive in an uncertain and often dangerous environment. Our brains were not shaped to deal with stock markets, central banks, or derivative contracts. They were shaped to detect threats, seek safety in the group, and react quickly to changes in the surroundings. These very traits helped humanity become the dominant species on Earth. In financial markets, however, the same traits can amplify price movements and contribute to the emergence of long-lasting trends.

Herd behavior is one of the clearest examples: on the savanna, leaving the herd was often dangerous. If a lion pursued the herd, the individual that ran in the opposite direction was rarely the one most likely to survive. Following the group was a rational strategy in many situations. If a hundred individuals reacted in the same way, there was probably a good reason. This instinct remains with us.

The problem is that the world our brains evolved to navigate differs dramatically from the world we live in today. Lions have largely been replaced by market movements, economic news, and financial decisions. Even so, our brains often react in similar ways. When a market rises, a process therefore begins that is more complex than many economic models assume. Initially, the increase may have a fundamental cause. A company reports better-than-expected results, a central bank cuts interest rates, or demand for a commodity rises. Some event sets the movement in motion, but other forces then enter

the picture.

Commodity markets also contain major differences in information and experience. Mining companies, energy companies, agricultural businesses, and industrial buyers operate closer to the physical market than most financial investors. When conditions change, information therefore does not reach all market participants at the same time. Producers and consumers begin adjusting their decisions, while analysts and investors gradually try to understand the significance of the change.

Suppose a mining company discovers a new silver deposit with the potential to increase global supply substantially over time. The first people to recognize its economic significance may choose to sell silver futures to hedge future production or reduce their exposure. As the information gradually spreads to analysts, competitors, commodity traders, and then other investors, more participants begin making similar decisions. The price therefore rarely adjusts in a single step. Instead, the trend develops as new information is interpreted, confirmed, and disseminated through the market.

As prices rise, investors become more optimistic. They see the value of their holdings increase and feel that their decisions have been validated. Analysts raise their forecasts, banks publish more positive market assessments, and journalists write about the strong performance. Success stories receive greater media attention, and asset managers are interviewed about their results. All of this helps create a positive narrative. Once the narrative becomes strong enough, more investors begin to buy—not necessarily because they have analyzed the market more deeply than before, but because they see prices rising and others apparently making money.

The result is a self-reinforcing process. What began as a

fundamental price movement gradually comes to be driven by investors' own behavior. The more people follow the trend, the stronger it becomes, and the price movement can continue far beyond what the original fundamental changes alone would justify. Sooner or later, the market often moves back toward its fundamental value, but the return may take longer than most investors expect. The market can remain irrational longer than many can afford to wait.

Technology stocks in the late 1990s provide a clear example. At first, genuine innovations underlay the development: the internet changed the world, new companies were created, and productivity increased. Over time, however, rising prices attracted new investors and more and more people followed the herd. The further the market rose, the more people wanted to participate. By the time optimism peaked, many investors had stopped asking what the companies were actually worth. The focus shifted instead to how much prices had risen and how much further they were expected to climb. In such a situation, people debate whether the market is in a bubble. But the question is irrelevant to trend-following models, which regard the trend as their friend until the end.

Bubbles have recurred in tulip bulbs, railway shares, real estate, commodities, cryptocurrencies, and equities. The markets have differed, but the underlying psychology has remained the same. Herd behavior operates not only on the way up but just as effectively on the way down. When markets fall, the same psychological mechanisms operate in reverse. Investors become more pessimistic, analysts lower their forecasts, and negative news dominates the media. Losses increase uncertainty, prompting more investors to sell. The selling pushes prices down further and thereby reinforces the pessimism. The result

is a self-reinforcing process that, just as during an advance, can drive the market far beyond what the underlying fundamentals alone would justify.

Because markets consist of people, shifts in sentiment tend to occur gradually. New ideas are accepted step by step, optimism grows over time, and pessimism develops in the same way. Capital does not move instantly from one place to another. This is why trends can often continue even as more investors recognize what is happening. Simply because a trend is visible does not mean it is over. On the contrary, its visibility may attract additional participants and thereby strengthen the movement.

Herd behavior also does not necessarily imply irrationality. If everyone else is buying an asset that is rising sharply, wanting to participate may appear rational. It is impossible to predict in advance when optimism will turn into pessimism or determine when the market has become excessively optimistic or pessimistic. Once the market begins to move in a particular direction, the CTA manager tries to identify the trend and follow it. Trends can break, markets can surprise, and price movements can change character quickly. The difficulty of trend following should therefore not be underestimated. But as long as human psychology influences investment decisions, herd behavior is likely to continue amplifying price movements and thus help create the long-lasting trends that trend-following strategies seek to exploit.

Central Banks Create Imbalances

Another force has also exerted a decisive influence on global financial markets. It produces no goods and sells no services, yet its decisions affect the prices of equities, bonds, currencies,

and commodities worldwide. That force is the central bank. For much of modern financial history, the role of central banks was regarded as relatively limited. Their main tasks were to maintain price stability, ensure a functioning payment system, and adjust policy rates to influence the economy. The market did not regard the central bank as the dominant force it would later become. That view changed, particularly after the financial crisis of 2008.

When the global financial system was shaken by bank failures, credit losses, and rapidly falling asset prices, central banks around the world acted in ways without precedent in modern times. Interest rates were cut to levels previously regarded as unlikely. In several countries they fell all the way to zero. In some cases, rates even became negative.

But central banks did not stop there. They argued that further measures were needed to stabilize the economy and prevent a deep depression, and therefore began buying enormous quantities of financial assets, especially government bonds. The programs had different names in different parts of the world but collectively became known as quantitative easing, or QE. In the years after the financial crisis, the US Federal Reserve, the European Central Bank, and the Bank of Japan became some of the world's largest bond buyers. Through these extensive purchases, central-bank balance sheets expanded by trillions of dollars, euros, and yen. From a trend perspective, this is extremely important.

When a central bank buys bonds on a large scale, it affects more than a single market. It changes the conditions for the entire financial system. If the purchases are sufficiently large, bond prices rise and interest rates are pushed down. Lower rates, in turn, affect the valuation of equities, real estate, and

many other assets. Investors who had previously been able to earn a reasonable return in the bond market are forced to look elsewhere. Capital begins to move.

The effects did not stop at the bond market. As interest rates were pushed down, equities became more attractive. Discount rates fell and financing costs declined. Investors who had previously allocated capital to bonds began seeking higher returns in the stock market. Central banks also signaled that they were prepared to do a great deal to support the economy and the financial system.

Foreign exchange markets were affected as well. When a central bank cuts interest rates or expands the money supply, differences in returns between countries change. Capital tends to flow toward markets where returns are higher. Major shifts in monetary policy can therefore create currency trends lasting several years. At times, investors have observed currencies gradually strengthen or weaken as central banks' policies diverged.

The important point is that central banks influence not only market levels but also market direction. When a central bank makes a significant change, the adjustment is rarely immediate. Banks, pension funds, insurance companies, businesses, and private investors need time to reposition. Capital is gradually reallocated through the system. New investments are made. Old positions are unwound. This gradual repositioning can therefore continue for a long time.

This applies not only when interest rates are cut. The same occurs when they are raised. For many years after the financial crisis, investors became accustomed to a world of extremely low interest rates. When inflation returned after the pandemic, the landscape changed quickly. The Federal Reserve, the ECB, and several other central banks began their most aggressive

rate-hiking cycles in decades, and the effects were dramatic.

Bonds fell sharply in value, currencies began moving in new directions, and equity markets faced higher interest rates. The movements were not short-lived, but continued as investors adjusted their portfolios to the new conditions. For trend followers, whether central-bank decisions were regarded as good or bad mattered less. What mattered was that the decisions created movement. When sufficiently large amounts of capital are forced to adjust to a new economic reality, the very long-lasting trends that the Managed Futures industry seeks to capture often emerge.

This leads to an important insight. Central banks' stated objective is to stabilize the economy and the financial system. From the perspective of the Austrian School of Economics, however, their interventions risk distorting market price signals and causing resources to be misallocated. Through rate cuts, rate increases, bond purchases, and other monetary-policy measures, central banks influence capital flows on a scale that few other participants can match. When large amounts of capital are redirected, powerful and long-lasting trends can arise.

The Forces Behind Trends

There are further explanations for why markets develop long-lasting trends. One is the business cycle, which is closely related to the commodity cycle because both are driven by changes in production, demand, and investment. During economic expansions, production, employment, and business investment increase. Demand for energy, metals, and transportation rises while corporate profits improve. When the economy later slows, investment declines and consumers become more cautious. New

downward trends can then take shape. For example, the global expansion from 2003 to 2008 contributed to sharp increases in the prices of oil, copper, and several industrial metals before the financial crisis triggered a clear movement in the opposite direction.

Technological innovation is another source of powerful trends. When a new technology changes productivity or creates entirely new markets, the prospects of whole industries are reassessed. The emergence of railways in the nineteenth century transformed transportation, electrification created new industries, and the internet revolutionized communication and commerce. In recent years, advances in artificial intelligence have led to sharply increased demand for semiconductors, data centers, and electricity. Technological change comes first. Investors then begin to reassess the future, capital flows toward the winners, and price movements may develop over many years.

Political decisions and geopolitical conflicts can also mark the beginning of major trends. New laws, taxes, tariffs, and subsidies change the economic conditions facing businesses and can shift capital between entire sectors. War also affects trade routes, production, and energy supplies. Russia's invasion of Ukraine in 2022, for example, caused sharp movements in natural gas, oil, wheat, and fertilizer. The initial price reaction was swift, but the adjustment of companies, governments, and investors to the new geopolitical situation continued for much longer.

Natural disasters operate in a similar way. A hurricane can disrupt oil production, and a severe drought can reduce wheat or corn harvests. Frost in Brazil's coffee-growing regions has on several occasions triggered prolonged increases in coffee prices. Even when the initiating event is brief, production often takes a

long time to return to normal levels.

What all these examples have in common is that they change the market's underlying conditions. They set a process in motion, but they do not determine how far the price will move.

Two factors deserve particular attention because they do not necessarily create new trends, but often amplify them once they have begun.

The first is leverage. Borrowed capital makes both advances and declines more powerful. When the market rises, the value of collateral increases, creating room to buy more assets. This can produce a positive feedback loop: higher prices permit more borrowing, which in turn drives prices higher still. The reverse occurs when the market falls. Collateral loses value, lenders demand more capital, and investors are forced to sell to reduce their debt. Forced selling increases the pressure and can cause an established decline to accelerate.

The second factor is the options market. During certain periods, options trading can create exceptionally powerful price movements through what is known as a *gamma squeeze*. To understand the phenomenon, one must first recognize that the banks and *market makers* that sell options normally try to protect themselves against price movements by buying or selling the underlying asset. If many investors buy call options at the same time and the market begins to rise, *market makers* are gradually forced to buy increasing quantities of shares or futures contracts to maintain a neutral risk position. These purchases push the price still higher, which in turn requires further buying. The result is a self-reinforcing process in which hedging in the options market amplifies the trend already underway.

It is important, however, to distinguish between what starts a trend and what drives it further. Leverage and gamma squeezes

rarely create long-lasting market trends on their own. They can, however, make an existing movement both faster and more dramatic. The original force is usually found elsewhere—in the business cycle, technology, politics, geopolitics, nature, or some other fundamental change.

As long as prices are primarily determined by market supply and demand, commodity cycles will arise. And as long as economies develop, new technologies are introduced, governments make decisions, wars break out, natural disasters occur, and people continue to revise their expectations, new trends will also be created. This is precisely why Managed Futures strategies focused on trend following have survived so many different market environments and remain relevant from one generation to the next.

Trends Through the Centuries

The economic and psychological explanations for trends are persuasive, but one crucial question remains: are trends a modern phenomenon, or have they recurred under entirely different economic and institutional conditions? If trend following had worked only during recent decades, the results might be attributable to an unusually favorable period, changes in market structure, or mere statistical chance. Answering this question requires a much longer history than is available for today's CTA programs.

In 2017, Brian Hurst and Yao Hua Ooi of AQR Capital Management, together with finance professor Lasse Heje Pedersen, sought to answer the question in the study *A Century of Evidence on Trend-Following Investing*, published in *The Journal of Portfolio Management*. AQR is a US asset-management firm that has

employed quantitative and systematic investment strategies, including Managed Futures, since its founding in 1998.

The study built on *Time Series Momentum* from 2012, in which Ooi and Pedersen, together with finance professor Tobias Moskowitz, had documented momentum in commodities, currencies, bonds, and equity indices from 1985 onward. The next question was whether the results were primarily a phenomenon of modern financial markets or whether the same pattern could be traced much further back.

To investigate this, Hurst, Ooi, and Pedersen constructed a systematic trend-following strategy using historical data from 1880 to 2016. The strategy took long positions in markets that had risen and short positions in markets that had fallen. They reduced dependence on any single model by combining signals based on price performance over one, three, and twelve months.

The study covered a total of 67 markets across four asset classes: 29 commodities, 11 equity indices, 15 bond markets, and 12 currency pairs. Because modern futures contracts did not exist throughout the entire period, the researchers used futures data where available and supplemented it with historical price indices for earlier periods. Their calculations also included estimated transaction costs and a traditional fee model with both management and performance fees.

The results showed that the strategy would have been profitable over the 137-year study period and generated positive returns under widely differing economic conditions. The history encompassed depressions, expansions and recessions, wars, stagflation, financial crises, and periods of both rising and falling interest rates. The strategy worked across all four asset classes and over several time horizons. The study therefore provided no support for the idea that trend following's historical

returns were solely a consequence of the market environment of recent decades or a statistical coincidence.

The strategy's returns also showed low correlation with equities and bonds. Trend following performed particularly strongly during extreme years in the stock market and produced positive results during several of the largest declines in a traditional portfolio consisting of 60 percent equities and 40 percent bonds. This does not mean that the strategy always rises when the stock market falls. The results do show, however, that trend following does not depend on the same market conditions that normally favor equities and bonds, and can therefore provide diversification when a traditional portfolio comes under severe pressure.

In 2014, researchers Alex Greyserman and Kathryn Kaminski extended the historical perspective even further in the book *Trend Following with Managed Futures*. Both combined academic research with extensive experience in the CTA industry. Greyserman had worked for several decades with trend follower Larry Hite at Mint Investment Management. Kaminski had analyzed CTA managers at Sweden's RPM and also coined the term Crisis Alpha to describe trend following's potential to benefit from prolonged price movements during market crises.

To investigate whether trends also occurred long before today's financial system emerged, they compiled price series that in some cases extended back to 1223. Using this material, they studied a simple form of trend following over roughly 800 years and across widely different historical eras, long before modern exchanges, central banks, and futures contracts.

Trends recurred in this material as well, across different centuries and economic systems. The constructed strategy generated positive returns, limited correlation with traditional

assets, and favorable characteristics during several historical periods of crisis. Trends are therefore not a product of today's exchanges, trading technology, or financial system. The study's main significance lies in showing how long-lasting price trends have recurred throughout history and under widely differing economic conditions.

Hurst, Ooi, and Pedersen show that a systematically defined trend-following strategy has worked for more than a century, while Greyserman and Kaminski trace the underlying price movements much further back. Together, the results suggest that trends are not a temporary anomaly of recent decades, but a recurring feature of market history. Herd behavior, slow adjustment to new information, economic imbalances, and crises are older than today's financial markets. As long as these forces persist, there is also reason to expect new trends to emerge.

Trend following, however, is only one part of Managed Futures. Although it is the best-known and historically dominant strategy, CTA managers have several other ways of approaching markets and making investment decisions.

Strategies in Managed Futures

More Than Trend Following

Trend following has been the dominant strategy in Managed Futures for most of the industry's modern history and remains the method most investors associate with CTA management. Media coverage of Managed Futures often focuses on trend-following models. Analyses of the CTA industry also tend to revolve around trend-following strategies.

But the CTA industry is much broader. The term CTA does not describe a single strategy, but a form of investment management that can employ several different methods. This chapter covers the most common strategy groups: trend following, *carry*, *mean reversion*, and *spread* and relative-value strategies. They do not provide a complete picture, but they illustrate the breadth of the industry.

The CTA industry has undergone extensive development in recent decades. Computing power has increased dramatically. The availability of data has exploded. Academic research has

identified new market effects and new ways of analysing risk. Competition within asset management has also intensified. As a result, many managers have gradually expanded their toolkits.

Technological advances have also made quantitative methods more common. An investment programme based primarily on mathematical and statistical models can be described as quantitative. Many CTA programmes are quantitative, but the terms are not synonymous. Not all CTA managers are systematic, and not all quantitative funds trade futures. What many CTA programmes share instead is the use of liquid futures contracts, decisions made according to defined rules, and a central role for risk management.

Today, it is not unusual for a modern CTA to use several models simultaneously. Some strategies seek to capture long-term trends. Others focus on short-term price movements. Still others look for opportunities in the relationships between markets or exploit structural phenomena arising from the design of futures markets.

Two managers can therefore operate within the same industry while working in entirely different ways. One may be almost exclusively trend following, building the entire business around identifying long-term movements in equity indices, currencies, interest rates, and commodities. Another may combine trend following with models that seek temporary overreactions or attempt to benefit from differences between related markets. On the surface, both can be described as CTA managers, but their investment processes may differ substantially.

This is also an important reason why returns can vary so widely among CTA firms. It is not enough to know that a manager belongs to the Managed Futures industry. Investors must also understand which models are used, how risk is

managed, and how the portfolio is constructed. Even managers trading the same markets can produce entirely different results depending on the strategies behind their decisions.

At the same time, Managed Futures is one of the few investments for which investors cannot determine whether it has risen or fallen merely by following the economic news. Strong performance by Managed Futures as a group in a given month does not mean that every CTA programme performed well. The differences between individual programmes can be substantial.

The Backbone of the CTA Industry

Trend following remains the industry's historical core. As we saw in the previous chapter, the strategy does not try to determine what a market should be worth. Instead, it identifies and follows the direction that actually develops.

Trend following can be described as a systematic attempt to exploit momentum—the tendency for a price movement, once under way, to continue in the same direction. The strategy uses price-based rules to identify and follow these movements as they emerge.

Trend following can be implemented with several types of price-based signals. Common examples include breakout models, in which positions are taken when the price moves beyond previous highs or lows, and moving averages used to identify the market's overall direction.

Many modern CTA systems use considerably more sophisticated models, but the underlying logic is the same. The objective is to distinguish genuine trends from short-term market noise. The models do not attempt to capture exact tops or bottoms. They often enter a move late and exit late. Trend followers accept

this because the goal is not to achieve the impossible—buying at the bottom and selling at the top—but to capture the middle portion of the trend.

The models can operate across different time horizons. Some respond to movements that develop over a few weeks, while others seek to capture trends lasting several months or longer. The chosen time horizon affects how quickly the model responds and how sensitive it is to short-term fluctuations.

During some prolonged market crises, trend following has generated positive returns while traditional portfolios have struggled. The reason is that the strategy can take both long and short positions and follow major movements across several asset classes. This does not guarantee protection against sudden market crashes, but it can become highly valuable when a crisis develops into clear and persistent trends.

Trend following seeks to profit from market movements, whereas carry is based in many respects on the opposite principle. Carry strategies look for markets that offer an embedded return for holding a position over time.

Carry Strategies

Many investors are accustomed to the idea that returns arise when an asset increases in value. If a share is bought for 100 kronor and later sold for 120 kronor, the investor has made money because the price rose. Carry follows a different logic. The objective is not necessarily to benefit from large price movements, but from the ongoing return the market offers for holding a particular position.

Carry varies across markets. In currencies, it comes from interest-rate differentials between countries. In bonds, it arises

from yields and the slope of the yield curve. The common feature is that the investor receives a risk premium for holding the position over time.

It can be viewed as the market paying the investor for patience in certain situations. Currencies are perhaps the easiest market in which to understand carry. Not all countries have the same interest-rate level. During some periods, Japan has had very low rates, while Australia, for example, has had considerably higher rates.

Suppose an investor borrows Japanese yen at an interest rate of 0.5 per cent and invests the capital in Australian dollars yielding 4.5 per cent. The difference between the two rates is four percentage points per year. If the exchange rate does not change, the investor retains this difference. This is the classic *carry trade*, which has been an important part of currency markets for several decades.

It is, of course, not risk-free. If the Australian dollar falls sharply against the yen, the currency loss may exceed the interest income. But as long as the exchange rate remains relatively stable or moves favourably, the investor can earn money even if the market does not move very much.

Similar phenomena occur in the bond market. The interest-rate market does not consist of a single rate, but of an entire spectrum of rates with different maturities. A bond maturing in two years normally trades at a different yield from one maturing in ten years. These differences create opportunities for carry strategies.

In practice, many fixed-income strategies are based on different forms of carry. The manager seeks to identify parts of the yield curve where the compensation for taking risk appears particularly attractive.

Carry can therefore be regarded as a risk premium: the investor receives an ongoing return for bearing a particular risk for as long as the position is held.

This does not mean, however, that carry is free money. If it were, the opportunity would disappear. These are not risk-free arbitrage profits. Like every other investment strategy, carry requires the investor to accept certain risks in exchange for a potential return. During periods of market stress, carry strategies can suffer substantial losses. Currencies can move sharply, and fixed-income markets can change quickly.

Long-term returns are therefore not a reward, but compensation for accepting these risks. Carry tends to produce steadier, lower returns over time, interrupted by short but deep periods of loss. Trend following, by contrast, often experiences longer periods of weak performance, while a large share of its total return is generated during relatively brief periods. Trend following makes money when markets move; carry makes money while the market is waiting to move.

Mean Reversion – When the Market Goes Too Far

Trend following is based on the idea that strong price movements sometimes continue and develop into trends. Mean reversion, by contrast, rests on the observation that markets often overreact and reverse without establishing a trend, or that an existing trend sometimes goes too far over a short period. When prices are temporarily driven far from a normal equilibrium level, the probability of a move back increases. The strategy seeks to exploit this reversion. The reversion, however, is neither immediate nor certain. The phenomenon has been observed in many financial markets and forms the basis of a

large group of quantitative investment strategies.

A simple analogy is a rubber band. The farther it is stretched, the greater the force pulling it back towards its original position.

Mean reversion occurs across many time horizons. Some strategies seek to capture movements that develop over months or years, while others operate much faster and look for small price movements arising over days, hours, or even minutes. Trend following and mean reversion often interpret exactly the same price movement in different ways. When a market rises sharply, trend-following models may see a sign that a new trend is beginning. Mean-reversion models may instead conclude that buyers have become overly optimistic and that a temporary pullback is approaching.

One strategy does not necessarily have to be right and the other wrong. On the contrary, both can make money in the same market.

Suppose, for example, that both a trend-following strategy and a mean-reversion model act when the price is 10 dollars. The trend follower buys because the model has identified the beginning of a possible upswing. The mean-reversion strategy sells short because the price is judged to have risen too quickly. If the market then falls to 9 dollars, the mean-reversion manager takes a quick profit and closes the trade. The trend-following model, however, remains in the position because smaller counter-trend moves are accepted as a natural part of a larger trend. If the market later turns higher and continues to 20 dollars, the trend follower can still close the trade with a considerably larger profit.

Both strategies can therefore be profitable even though they initially took opposite positions at exactly the same price level. The main difference is not the direction, but how long the

position is held, how large a loss is accepted, and how profits compare with losses.

What matters, then, is not who was right about the initial price movement, but how each strategy is constructed. Mean reversion tends to take many smaller profits with a relatively high hit rate, much like carry.

A trend-following strategy with a 35 per cent hit rate can nevertheless be considerably more profitable than a strategy with an 80 per cent hit rate if its average winners are sufficiently large. Likewise, a strategy with many winning trades can produce high returns by steadily accumulating small profits, provided that the few larger losses are kept under control. Ultimately, success is not about being right as often as possible, but about the relationship between the hit rate, the average profit, and the average loss.

The market environment often determines which strategy performs best. During periods of strong and persistent trends, trend following tends to perform well, while mean reversion struggles. During calmer periods, when prices fluctuate back and forth without a clear direction, the situation is reversed.

Relative-Value Strategies

Trend following, carry, and mean reversion can be applied to individual markets. Spread and relative-value strategies instead focus on the relationship between two markets or contracts.

Here, the focus is not on the absolute price level, but on the relationship between two markets. The manager does not focus on whether oil prices will rise or fall, but on whether one type of crude oil is expensive or cheap relative to another. Similarly, the model does not focus on whether interest rates will rise or

fall, but on whether the difference between two parts of the fixed-income market is reasonable. This may sound like a small distinction, but in practice it represents an entirely different way of thinking.

One of the best-known examples is the relationship between Brent and WTI crude oil. A trend follower may buy Brent because its price is rising, while a relative-value strategy instead goes long Brent and short WTI because the price difference between them is expected to change.

Both are crude oils affected by global economic activity, geopolitics, OPEC production, energy demand, and several other common factors. Even so, they do not always trade at the same price. At times, Brent becomes considerably more expensive than WTI. At other times, the difference narrows or almost disappears. These variations may result from transport capacity, inventory conditions, regional imbalances, or changes in trade flows.

A relative-value strategy may therefore be based on the assumption that the difference between the two markets will eventually move back towards more normal levels. The manager buys the market that appears relatively cheap and sells the one that appears relatively expensive. If the relationship develops as expected, the strategy can generate a profit even if the overall price of oil does not move very much.

Similar relative-value positions can be constructed between closely related agricultural commodities, for example by buying wheat and selling corn when the price difference between them is considered unusually wide.

Perhaps the most sophisticated form of relative-value trading is often found in fixed-income markets. Here, the focus is not merely on individual bonds, but on entire yield curves. There

is no single interest rate in the economy, but a large number of rates with different maturities. Two-year government bonds trade at one yield, five-year bonds at another, and ten-year bonds at a third. The differences between these yields change constantly.

During some periods, the yield curve steepens. During others, it flattens or even inverts. These changes reflect the market's view of economic growth, inflation, and central-bank policy.

The manager's focus is not on whether interest rates in general will rise or fall, but on how the relationship between different parts of the fixed-income market will develop.

Spread Strategies

Another common form of spread trading is the calendar spread. Here, the manager does not trade two different commodities, but two futures contracts on the same commodity with different delivery months. If the longer-dated contracts are more expensive than the shorter-dated contracts, the market is in contango. If the longer-dated contracts are cheaper, the market is in backwardation.

Suppose, for example, that cotton is trading in unusually steep backwardation. The nearest futures contract is then considerably more expensive than the contract for later delivery. If the model determines that this price difference is unusually wide, the strategy may sell the nearby contract while simultaneously buying the later-dated contract. The aim is to benefit as the price difference between the contracts gradually returns towards more normal levels.

The cotton price itself is therefore not the central issue. The price may rise, fall, or remain unchanged, and the strategy can

still generate a positive return if the relationship between the two contracts develops as the model expects. Like other relative-value strategies, calendar spreads are based on the tendency of temporary imbalances between nearby and longer-dated futures contracts to eventually normalise.

CTAs Combine Several Strategies

Today, many CTA managers combine several different strategies in the same portfolio, while others focus on a single strategy group, such as trend following or spread trading. No strategy works optimally in every market environment.

By combining trend following with carry, mean reversion, and relative-value strategies, the manager creates more independent sources of return. Diversification across both markets and strategy types makes the portfolio less dependent on any single market environment. One possible trade-off is that a smaller risk allocation to trend following reduces returns during strong trends. At the same time, combining strategies may allow risk to be distributed across more independent sources of return.

There are good reasons to make trend following the dominant building block in a Managed Futures portfolio. The other strategies can create substantial value, but they often depend on more specific market conditions. Carry depends on how interest-rate differentials and other risk premia are priced. Mean reversion depends on temporary overreactions actually being followed by a return towards more normal levels. Relative-value and spread strategies depend on the relationship between two markets or two futures contracts deviating from its historical pattern and subsequently normalising.

What these strategies have in common is that they seek more

narrowly defined opportunities in which the potential profit is often limited by the size of the deviation.

II

The History of Managed Futuresxx

When Prices Were Set Free

Everything Begins with a Price

The modern Managed Futures industry is built on markets in which prices can move freely enough to create trends. Whether the market is crude oil, gold, US Treasury bonds, the euro or wheat, the starting point is the same: a price that can change relatively freely.

It is easy to take free market prices for granted. For today's generation of investors, it is difficult to imagine a world in which exchange rates do not move, interest rates do not respond to economic change and commodity prices are kept within politically determined limits.

We live in an age in which prices change every second, stock markets react to new information in real time and currencies trade around the clock. The market no longer waits a couple of days for news from New York to reach Stockholm. Commodity prices rise and fall with changes in weather, geopolitics, supply and demand. Yet from a historical perspective, this world is

surprisingly new.

Throughout history, kings, governments and central banks have repeatedly tried to control the prices of food, energy, currencies and other strategically important assets. The authorities have sometimes been able to set the official price, but not to eliminate the economic forces that affect supply and demand. The result has often been shortages, surpluses and malinvestment.

Prices carry information and coordinate decisions about production, consumption and investment. In his 1920 essay *Economic Calculation in the Socialist Commonwealth*, the Austrian economist Ludwig von Mises argued that economic calculation requires market prices. Without trade and genuine market prices, it is impossible to compare different uses of raw materials and capital or to determine how society's scarce resources can be put to their most productive use.

Friedrich Hayek, who was awarded the Nobel Prize in Economics in 1974, developed this argument further. He emphasised that the knowledge affecting an economy is dispersed among a great many people and can never be fully assembled by a single decision-maker. Market prices convey this knowledge and make it possible to coordinate decisions that no central authority could oversee on its own.

An artificially low price stimulates demand while weakening producers' incentive to increase supply. Conversely, an artificially high price can create overproduction.

A clear example is the United States' petrol price controls in the 1970s. When the government held the petrol price below the level at which supply and demand were in balance, the result was long queues, empty petrol stations and rationing. The low official price did not create more petrol. On the contrary,

demand increased while the incentives to produce and distribute petrol weakened.

When the price was no longer allowed to signal that the product had become scarce, petrol instead had to be allocated through queues, rules and personal connections. The shortage did not disappear – it merely took another form.

For much of the twentieth century, many of the markets that now form the foundation of the Managed Futures industry were far from free. Exchange rates did not change continuously but were often fixed through international agreements. Interest rates were heavily influenced by political decisions and direct central-bank control. Agricultural markets were regulated through price supports, production quotas and various forms of government intervention.

A currency fixed against another currency cannot develop a clear trend. An interest rate held within a narrow, politically determined range offers limited scope for substantial market movements.

The true origins of the CTA industry lie in the transition from controlled markets to freer ones. As more markets began to be priced more freely in the early 1970s, the investment landscape changed fundamentally: currencies could rise and fall, interest rates could respond to inflation and economic growth, and commodities could develop lasting trends based on genuine changes in supply and demand. An environment emerged in which investors could seek systematic opportunities across many different markets.

Futures markets and speculators already existed before then. Several investors were already using methods that can retrospectively be described as trend following. The opportunities were considerably more limited, however, because there were

few mutually uncorrelated markets and the strategies were mainly applied to individual markets, especially the stock market.

Only when financial markets gradually opened and the range of futures contracts expanded did it become possible to apply the same basic idea across a large number of markets: commodities, currencies, interest rates and, later, stock indices.

But this change did not arise out of nowhere. To understand why markets opened and prices began to move more freely, we must first return to the international monetary system created after the Second World War: the Bretton Woods system.

Bretton Woods – the World of Fixed Exchange Rates

As the Second World War drew to a close, the world's political and economic leaders faced a formidable challenge. Europe lay in ruins. International trade had collapsed. Currency chaos, competitive devaluations, high tariffs and economic nationalism had contributed to instability during the interwar years and deepened the global depression of the 1930s.

Behind these developments also lay government credit expansion, the abandonment of the classical gold standard and repeated attempts to manipulate currency values to favour domestic economies. If the world was to be rebuilt, a new economic framework was therefore needed to create stability and predictability.

Against this background, representatives of 44 countries gathered in the small town of Bretton Woods, New Hampshire, in the summer of 1944. The British economist John Maynard Keynes was one of the conference's most influential participants and led the British delegation. Together with the American economist

Harry Dexter White, Keynes formulated the principles that would underpin the post-war international monetary system. The result became known as the Bretton Woods system.

The basic idea was simple. To promote trade, investment and economic recovery, countries needed stable exchange rates. Businesses importing and exporting goods would be able to plan without constantly worrying about sharp currency movements. International investment would become less risky and trade between countries could grow. To achieve this, a system of fixed but adjustable exchange rates was created. At its centre stood the US dollar.

At the time, the United States had the world's strongest economy and controlled most of the world's gold reserves. The dollar, which had been tied to gold at \$35 an ounce since 1934, now assumed the role of the world's anchor. Most other major currencies were in turn pegged to the dollar at fixed but adjustable exchange rates. The result was a kind of hierarchical system, with gold at the top, the dollar anchored to gold and the other currencies tied to the dollar.

As long as the United States guaranteed that foreign central banks could exchange dollars for gold, the entire system possessed a fundamental stability. To the average citizen, this appeared positive: exchange rates rarely changed and international trade became easier. Economies could be rebuilt after the war. During the 1950s and 1960s, many Western countries experienced one of the strongest periods of growth in modern history.

From a societal perspective, Bretton Woods was therefore long regarded as a success. From a market perspective, however, the system created a very different reality because currency markets were not, in practice, free.

If the German mark, French franc or British pound moved too far against the dollar, central banks were obliged to intervene. They bought or sold currencies to defend the established rates. The objective was not to let the market determine the price, but to keep it within a predetermined range. Under Bretton Woods, currency markets were largely administrative constructions rather than free markets.

This does not mean that every market was completely static. Commodities could still move. Grain prices were affected by harvests, and metals responded to industrial demand.

The major global macro trends that would later become the CTA industry's most important raw material did not yet exist on the same scale. Currency trends were constrained, interest-rate markets were much more heavily regulated and cross-border capital flows were controlled. The system's imbalances grew ever larger.

During the 1950s and 1960s, Bretton Woods worked relatively well because the US economy was dominant and the country's gold reserves lent credibility to the dollar. Over time, the balance changed. Europe and Japan recovered, international trade grew and the supply of dollars in circulation increased faster than US gold reserves. The cracks in the system became increasingly visible. The decisive turning point came in August 1971.

The Nixon Shock of 1971 – the Great Turning Point

On 15 August 1971, US President Richard Nixon delivered a televised address to the American nation. For most viewers, the speech concerned inflation, unemployment and economic problems. For economists, central banks and financial markets, however, it would prove to be one of the most consequential

speeches of the entire twentieth century. Nixon announced that the United States would temporarily suspend the conversion of dollars into gold. The word “temporarily” proved misleading. In practice, the decision marked the end of the Bretton Woods system and the beginning of an entirely new financial era.

What later became known as the Nixon Shock changed not only the world’s monetary system. It also transformed the conditions for trade, investment and asset management. The decision became a crucial precondition for the emergence of the modern Managed Futures industry. The background lay in the problem Nixon faced.

During the 1950s and 1960s, the dollar was the world’s anchor, and foreign central banks could exchange their dollar reserves for gold at the official rate of \$35 an ounce at any time. The system worked as long as the rest of the world trusted this guarantee, but its weaknesses became increasingly apparent over time.

The United States financed the Vietnam War, expanded welfare programmes and pursued an increasingly expansionary economic policy. The number of dollars increased while the gold held in US reserves did not grow at a corresponding rate. This created what became known as the Triffin dilemma.

A growing supply of dollars was needed to finance political ambitions, public spending and international commitments. Yet the more dollars that were created, the harder it became to uphold the promise that every dollar could ultimately be converted into gold. By the late 1960s, several countries had begun to question the system’s sustainability. Central banks and governments realised that the United States had issued far more dollars than its gold reserves could reasonably cover.

If everyone had demanded to exchange their dollars for gold

at the same time, the system would have collapsed. In practice, it was a classic bank-run problem on a global scale. French President Charles de Gaulle became one of the most outspoken critics. He argued openly that the United States benefited from its privileged position by being able to create dollars that the rest of the world accepted as a reserve currency.

France began systematically converting its dollar reserves into gold and repatriating the metal. The action became a powerful symbol of the growing distrust in the sustainability of the Bretton Woods system. Other countries followed, and pressure on US gold reserves increased.

Eventually, the Nixon administration faced a choice: continue defending the link to gold and risk depleting US gold reserves, or sever the link altogether. It chose the latter. When Nixon closed the gold window, the final link between the dollar and gold disappeared.

The world's most important reserve currency thereafter rested solely on confidence in the US government and central bank. The decision was presented as a temporary measure intended to calm markets and prevent panic, but the gold window never reopened. In practice, it marked the beginning of the end of the Bretton Woods system.

Nixon did not, therefore, free markets from political control. Closing the gold window removed the last external constraint on how much the US government and central bank could increase the supply of dollars. The monetary system did not become freer. Instead, the supply of the currency could be controlled politically to an even greater extent. The result was continued credit expansion, inflation and a sharp decline in the dollar's purchasing power.

Since 1971, the US dollar has lost about 88 per cent of its

purchasing power, as measured by the Consumer Price Index. Someone who placed \$100 in a safe in 1971 would still have the same banknotes, but their purchasing power today would correspond to only about \$12 in 1971 money. Freer exchange rates should therefore not be confused with freer money itself. The market gained greater influence over the relative prices of currencies, while their supply and long-term purchasing power remained dependent on central-bank decisions.

During the following years, maintaining fixed exchange rates became increasingly difficult. Market forces were simply stronger than the political agreements intended to keep currencies in place.

In 1973, the leading industrial nations abandoned fixed exchange rates and introduced floating rates. Currency values were thereby determined by supply and demand. For the first time in several decades, the market gained greater influence over exchange rates, which once again reflected changes in inflation, interest-rate differentials and capital flows. Suddenly, lasting currency trends became possible. The same development subsequently spread to other parts of the financial system.

As currency risks grew, new instruments and futures markets were needed. The economist Milton Friedman recognised early that an organised market was required to manage these risks. Even before the Bretton Woods system collapsed, he had argued that free exchange rates would create demand for futures contracts through which companies and investors could protect themselves against currency fluctuations. Many were initially sceptical and maintained that fixed exchange rates made such a market unnecessary. After the Nixon Shock, however, conditions changed rapidly.

In 1972, the Chicago Mercantile Exchange (CME) opened

the International Monetary Market (IMM), the world's first organised exchange for financial currency futures. Futures contracts could be traded there on currencies including the Japanese yen, British pound, Swiss franc and Canadian dollar. This marked the beginning of modern financial-futures trading. Trading in interest rates, currencies and other macroeconomic assets expanded rapidly, and over the following decades the markets that would later form the core of the Managed Futures industry developed.

More Markets Open

Currencies were not the only financial prices to move more sharply during the 1970s. High inflation, repeated changes in monetary policy and growing government debt also produced substantial fluctuations in interest-rate markets. Banks, pension funds and other institutions had a growing need to protect themselves against changing interest rates. Rising rates could quickly reduce the value of a bond portfolio or increase the cost of future financing.

In 1975, the first exchange-traded interest-rate futures contract was launched on the Chicago Board of Trade. The underlying asset was US government-guaranteed mortgage-backed securities. Two years later, futures on long-term US Treasury bonds were introduced. Market participants could now both hedge against and speculate on changes in long-term interest rates.

The next major step came in 1981, when the Chicago Mercantile Exchange launched the Eurodollar futures contract. The contract was based on short-term dollar interest rates outside the United States and developed into one of the world's largest

and most liquid futures markets. Together, these contracts enabled investors to trade interest-rate movements across different maturities, from short-term money-market rates to long-term government bonds.

The introduction of interest-rate futures did not mean that interest rates were set free in the same way as exchange rates. The Federal Reserve continued to set its policy rate and therefore exerted considerable influence over short-term market rates. Longer-term rates and bond prices, by contrast, were influenced to a greater extent by market trading. Futures contracts made interest-rate risk tradable.

For CTA managers, this development was decisive. They could now position themselves for both rising and falling bond prices and apply the same systematic rules to interest-rate markets as they did to currencies and commodities.

Interest-rate futures formed part of a much broader transformation. As more market prices began to move freely, new trends emerged across several asset classes.

Gold is a particularly interesting example. When the United States stopped redeeming dollars for gold in 1971, the official price link disappeared and the market gained greater influence over the metal's price. Gold became one of the great winners in the years that followed. Investors sought protection against the deterioration of money's purchasing power, and demand rose sharply. Between the beginning of the 1970s and the price peak in 1980, gold rose by several hundred per cent.

The energy crises of the 1970s also created a need for better instruments to manage large price movements in oil and fuels. Heating-oil futures were launched in 1978, followed in 1983 by the futures contract on US WTI crude oil. Energy markets subsequently became an important part of the markets available to

CTA managers. Weather, geopolitics, production decisions and changing demand could create long-lasting price movements and thus another source of trends.

Alongside the emergence of energy futures, another important market group was added. Futures contracts on the S&P 500 Index were launched in 1982, and stock-index futures subsequently became increasingly important. Entire stock markets could now be traded through standardised contracts. Instead of buying individual shares, investors could gain exposure to indices such as the S&P 500, Nikkei, FTSE and DAX.

For the CTA industry, this represented another step towards global diversification. In this way, the modern CTA industry took shape. Each time a new asset class became available for futures trading, the opportunities for diversification increased. Each time liquidity improved, the capacity to manage large volumes of capital grew.

Donchian, Dennis and the Turtles

David Ricardo and an early rule of speculation

Several of trend following's fundamental principles can be traced back to the early nineteenth century, long before the strategy acquired its modern name. One of the earliest formulations is associated with the British political economist David Ricardo. Working in Adam Smith's classical liberal tradition, he argued for free trade, competition and fewer trade barriers. His theory of comparative advantage showed how countries can benefit from trade even when one country is more efficient in almost every area of production.

Parts of Ricardo's economic theories were later challenged by Carl Menger and the Austrian school. For the development of trend following, however, it is primarily Ricardo's experience as an investor and speculator that is of interest.

Ricardo was not merely an economic theorist. Before becoming one of the most influential figures in economics, he had made a considerable fortune in the financial markets.

Ricardo began working with his father at the London Stock Exchange while still a teenager and later traded on his own account, primarily in British government securities. His success was not based solely on economic analysis. He also placed great emphasis on how losses and profits were managed once a position had been taken.

In *The Great Metropolis*, published in 1838—fifteen years after Ricardo's death—the author James Grant recounted three rules that Ricardo reportedly used to recommend to his friends. Two of them have lived on together as one of the best-known principles in financial markets:

“Cut short your losses and let your profits run on.”

The wording captures one of trend following's most important asymmetries. A losing position should be exited before the loss grows large, while a profitable position should be retained for as long as the market continues in the right direction. The result therefore depends less on being right in every trade than on allowing profits to become larger than losses.

Ricardo naturally used no computerised models and did not describe himself as a trend follower. His rule nevertheless shows that one of the strategy's fundamental principles was known long before modern futures markets, CTA programmes and systematic trading models emerged.

Jesse Livermore, before trend following had a name

David Ricardo had formulated the principle of cutting losses and letting profits run. Nearly a century later, Jesse Livermore made the same idea a central part of his trading. Through his experience in the markets, he developed several of the principles that would later characterise modern trend-following models.

Few people have acquired such a mythical status in the investment world. Livermore was born in 1877, began speculating in stocks as a teenager and became one of the world's best-known speculators during the first half of the twentieth century. Over the course of his career, he built and lost enormous fortunes several times. At its peak, his wealth is estimated to have been worth hundreds of millions of dollars in today's money.

He became particularly well known after the stock market panic of 1907 and later during the crash of 1929, when he made enormous sums from falling equity markets. Livermore's story is also a reminder of how quickly success can turn into adversity. After several financial setbacks, his final years were marked by considerable difficulties.

What has kept his name alive from one generation to the next is his ideas, not the size of his profits. In 1923, the journalist Edwin Lefèvre published *Reminiscences of a Stock Operator*, a fictionalised account of Livermore's life. To this day, it remains one of the most widely read investment books ever written.

What is fascinating is that it contains many of the principles that would become central to classical trend following and Managed Futures much later.

Livermore discovered that success did not come from constantly buying low and selling high. Time and again, he stressed the importance of identifying the dominant trend and following it for as long as it continued. One of his best-known quotations is:

"It never was my thinking that made the big money for me. It always was my sitting. Got that? My sitting tight!"

The quotation captures the part of Livermore's philosophy that later became central to trend following: the ability to hold on to a winning position while a larger move develops. What

matters in this context is not which technical tools he used, but that he gave greater weight to the direction of price than to his own view of what the market ought to do. In that respect, he can be seen as a forerunner of the systematic traders who later turned similar observations into fixed rules. The person who took the next major step was Richard Donchian.

The man who followed prices rather than forecasts

If Jesse Livermore was the legendary speculator who intuitively understood the power of trends, Richard Donchian was the man who attempted to systematise the same idea and establish fixed rules for entry and exit. He is less well known to the general public, but many in the Managed Futures industry regard him as the true father of trend following. Most CTA managers can trace at least part of their intellectual heritage back to Donchian's work.

To understand how radical his ideas were, one must appreciate how investors thought during the period in which he worked. In the 1940s and 1950s, the investment world was dominated by fundamental analysis and forecasts. Investors tried to understand corporate earnings, analyse financial reports and assess the future course of the economy. Market commentary focused on what the economy ought to do, what interest rates were likely to do or which stocks ought to be undervalued. Most professional investors saw it as their task to predict the future better than others.

It was in this environment that Richard Donchian chose to think differently.

Donchian focused on the price itself. He was less interested in economic explanations than in the market's actual behaviour.

If the price rose, it was a sign that buyers were dominant; if it fell, sellers had the upper hand. He did not need to explain why. What mattered was whether price movements could be captured and followed using objective rules. Today this may sound self-evident, but at the time it was an unusual perspective. Donchian regarded price as the most important source of information. This became the foundation of his life's work.

Donchian developed objective rules for entry and exit that could be followed independently of the news flow and the manager's own emotions. One of his best-known contributions was the Donchian Channel, a method that used previous highs and lows to identify potential breakouts. This work was carried out long before the advent of the personal computer, which meant that the models were calculated manually and charts drawn by hand.

Donchian was also among the first to put these ideas into practice in professional asset management. In 1949, he founded Futures, Inc., often described as one of the world's first funds in which capital from several investors was pooled and managed in a diversified portfolio of commodity futures. The fund was an early forerunner of today's Commodity Pools and Managed Futures funds. Although the industry was still small, Donchian demonstrated that systematic methods could be applied across many markets and used to manage capital according to these principles.

It was this philosophy that later inspired the next generation of trend followers. Among them was a young commodity trader from Chicago who would become far better known than his predecessor: Richard Dennis.

Richard Dennis – the boy from Chicago

If Richard Donchian formulated the fundamental ideas of trend following, Richard Dennis demonstrated that they could create enormous fortunes.

Richard Dennis remains one of the most legendary names in the Managed Futures industry. He grew up in Chicago, the city that developed into the global centre of commodity and futures trading during the post-war period. The Chicago Board of Trade and the Chicago Mercantile Exchange were places where farmers, trading firms, speculators and investors met to trade contracts for the future delivery of grain, livestock and other commodities.

As a young man, Dennis was fascinated by this world. He began working on the trading floor while still in his teens. For many others, it would have been a temporary job. For Dennis, it marked the beginning of a lifelong involvement in the markets.

Dennis borrowed money from his family to trade futures. He initially worked with relatively simple principles, but over time developed a more systematic approach, inspired in part by Richard Donchian. Price movements guided the decisions, and rules determined how positions were managed.

Dennis began his career during a period when new currency, interest-rate and energy markets produced sustained moves that suited his method well. From starting capital of a few thousand dollars, he gradually built a fortune that eventually amounted to hundreds of millions of dollars.

This development eventually led Dennis to move beyond trading only his own capital and begin managing money for outside investors. Through Richard J. Dennis & Company, he operated as a CTA (Commodity Trading Advisor) and ran

systematic trading programmes in the futures markets. He thus became not only a well-known speculator but also part of the emerging Managed Futures industry.

Richard Dennis did not attribute his results to any mysterious talent, nor did he regard himself as a market prophet capable of predicting the future better than others. In his view, the results largely came from following a rule-based process. The discipline to follow those rules was decisive.

This perspective distinguished him from many other prominent speculators throughout history. Market legends are often explained in retrospect by intuition, instinct or exceptional market feel. Dennis began to wonder whether the results could be replicated.

Can success be taught?

After the strong trends of the 1970s, Richard Dennis was regarded as one of the world's most prominent commodity traders. His career raised a question that continued to fascinate him: why did some speculators succeed so much more than others? The traditional explanation was that success required a special intuition or an innate feel for the market. Many viewed trading as an art rather than a craft. Richard Dennis was not convinced.

The more he reflected on his own career, the more doubtful he became that strong results were primarily a matter of innate ability. When he analysed his best trades, he saw no magical intuition. Instead, he saw rules, discipline and risk management within a process repeated time and again.

This led him to suspect that the market might not reward talent above all else. Perhaps it rewarded discipline instead. Perhaps it rewarded the ability to follow the system when

emotions said otherwise. Perhaps it rewarded those who could accept uncertainty and still act consistently.

This conviction led to a long-running debate with his friend and business partner William Eckhardt. In many ways, Eckhardt was Dennis's opposite. He had a strong mathematical background, was known for his analytical thinking and later operated as a CTA himself through Eckhardt Trading Company.

Although he had been successful in the markets himself, Eckhardt doubted that the same ability could be taught to others. Their debate came to revolve around a question that is as relevant today as it was then: can successful trading be taught, or does it require some special quality that some possess and others do not?

Dennis argued that the most important components could be taught just like the skills required in any other profession. If a person received the right training, clear rules and sufficient discipline, there should be a good chance of succeeding in the markets.

Eckhardt disagreed. He believed that some people possessed qualities that made them naturally better suited to trading. In his view, the ability to handle stress, make decisions under uncertainty and resist herd behaviour was not something that could easily be taught in a classroom.

For most people, the discussion would have ended there, as yet another interesting philosophical debate without a definitive answer.

But Richard Dennis was not satisfied with a theoretical discussion. He had built his career by testing ideas in practice. Instead of continuing to argue over who was right, he could conduct an experiment. If he selected participants with no previous experience, taught them his rules and gave them capital to trade,

he would have his answer. If they failed, Eckhardt was right. If they succeeded, Dennis was right.

The Turtle experiment

The story of the Turtle Traders sounds almost like a Hollywood film. A prominent commodity trader claims that successful trading can be taught. His partner says it cannot. To determine who is right, they recruit participants with no previous experience, teach them their methods and give them money to trade. Had this been a novel, many readers would probably have found the plot exaggerated. But that is exactly what happened.

In 1983, Richard Dennis placed advertisements in *Barron's*, *The Wall Street Journal* and *The New York Times* seeking people who wanted to be trained as speculators. The application process was unusual because candidates needed neither extensive experience in finance nor a background in economics or mathematics. What mattered was their ability to think in a disciplined manner and follow rules.

The response was enormous, with several thousand applications. The applicants included everyone from students and accountants to gamblers, engineers and people with no experience of the financial markets. Dennis and Eckhardt conducted interviews and gradually selected those they believed had the greatest potential.

Dennis trained a total of around twenty participants in two groups, the first in 1983 and the second in 1984. Over the course of a few weeks, they were taught a complete set of rules for trend-following trading. The training covered breakout signals, entry and exit rules, stop-losses, position sizing and pyramiding. The methods were not particularly complicated, and several of the

principles were already widely known. The precise rule set and the way the participants applied it, however, were kept secret from the outside world. The essence of the experiment was their ability to follow the rules consistently.

Risk management was just as important as the trading signals themselves. Positions were adjusted to market volatility, and losses were limited according to predetermined rules. The participants were judged primarily on their ability to follow the system, not on who performed best in each individual trade.

After only a couple of weeks of training, the decisive moment arrived: the participants were given real capital to manage. This was not a theoretical exercise; it was Dennis's own money that was at stake.

The name Turtle Traders came from a visit to a turtle farm in Singapore. Dennis argued that he could train skilled speculators in the same way that turtles were raised on the farm—quickly, systematically and according to a defined method.

The result that shocked Wall Street

According to the most widely cited figures, the Turtle participants generated a combined profit of around \$175 million for Richard Dennis during the five years of the experiment. Dennis had never claimed that just anyone would automatically succeed as a speculator. His hypothesis was that profitable trading could, to a large extent, be taught through rules, discipline and risk management. The results provided strong support for his view.

The outcome, however, also proved Eckhardt partly right. All participants received the same training and the same rules, but their results varied widely. Some followed the system with discipline and achieved very strong results, while others

struggled to manage losses and deviated from the rules.

The experiment proved Dennis right that the method could be taught, but also Eckhardt right that not everyone could apply it with the same discipline.

Many of the participants became professional asset managers and later established their own businesses as CTA managers. Among them was Jerry Parker, who founded Chesapeake Capital after completing the training.

For several decades, the firm became one of the best-known names in the Managed Futures industry and based its management on the same fundamental trend-following philosophy. But Parker was far from alone.

Paul Rabar, Liz Cheval and Tom Shanks also built careers in asset management and are still mentioned in discussions of systematic trading and trend following. Others went on to start their own firms or work elsewhere in the industry.

By establishing their own asset-management firms, several of the participants carried the method forward. The Turtle experiment thus became more than an attempt to determine whether trading could be taught. It also helped move trend following from individual speculators into a growing professional asset-management industry.

From niche to institution

When Wall Street began to listen

Despite its success through much of the 1970s and 1980s, trend following remained on the fringes of the financial world. Most institutional investors focused almost exclusively on equities and bonds. Pension funds, insurance companies and endowments built their portfolios around traditional asset classes and assumed that long-term success depended on finding the right allocation between the equity and fixed-income markets.

Alternative investments existed, but they were often regarded as speculative or marginal holdings rather than serious components of an institutional portfolio.

In this environment, CTA managers appeared to be outsiders. They traded futures markets, followed mathematical models and spoke about trends rather than valuations. Many investors found it difficult to understand what they actually did.

To traditional asset managers, their activities often looked more like a form of advanced speculation than professional in-

vestment management. They did not analyse corporate balance sheets, meet company executives or try to determine whether a stock was cheap or expensive. Their attention was directed towards price movements in markets that many institutions barely followed.

The results, however, were difficult to ignore. During periods when traditional investments performed well, there was little reason to take an interest in these unusual managers. Trend followers began to stand out primarily during crises and other extended market moves because their results did not depend on a strong economy or rising equity prices.

Institutions began to recognise that CTA managers could provide a source of return that did not depend directly on rising equity and bond markets.

At the same time, the markets themselves were evolving. Futures exchanges expanded and liquidity improved. New contracts were launched and technology advanced, making it easier to trade globally and build diversified portfolios across many markets. This created the conditions for larger inflows and a growing share of capital from institutional investors.

The real shift came only when pension funds, endowments and other large asset owners began to regard the sector as a legitimate part of a professional portfolio. The industry was then transformed at a fundamental level. What had previously been a niche for commodity traders and entrepreneurs began to develop into a global asset-management industry.

The CTA wave of the 1980s

At the beginning of the 1980s, when Richard Dennis conducted his Turtle experiment, the Managed Futures industry was still small. Interest began to grow during the decade, however, and an activity built around individual pioneers gradually developed into a professional asset-management industry.

The expanding range of financial futures and the arrival of computers made it possible to process larger volumes of price data and apply the same rules across more markets. Systematic trading could therefore be conducted on a scale that had previously been difficult to achieve.

The organisational infrastructure also improved. When the National Futures Association began operating in 1982, it assumed a central role in registration and oversight within the US futures industry. Clearer rules helped make the market more formalised and easier for investors to examine.

John W. Henry and Larry Hite played important roles in the industry's development during the 1980s. Through John W. Henry & Company and Mint Investment Management, they demonstrated that systematic futures trading could attract substantial capital and be conducted on a global scale. Their success helped bring Managed Futures to the attention of investors far beyond the traditional commodity exchanges.

At the same time, established CTA managers such as DUNN Capital and Campbell & Company continued to build increasingly long, documented track records. Towards the end of the decade, Chesapeake Capital also emerged, carrying the ideas of the Turtle generation into professional asset management.

At first, the capital came mainly from wealthy private investors, entrepreneurs and family offices. By the end of the

decade, advisers and smaller institutions had also begun to follow the sector more closely. Broad participation by the large pension funds still lay further ahead.

The crises that established credibility

The credibility of the CTA industry was built not only through long track records and growing assets. It was strengthened above all when managers were tested during genuine market crises. Investors could then see how the programmes behaved when established relationships broke down, liquidity deteriorated and prices moved far beyond their normal ranges.

An early test for the sector came during the stock market crash of 1987, the event that later became known as Black Monday.

On 19 October, the Dow Jones Industrial Average fell by more than 22 per cent in a single day, the largest one-day percentage decline in the index's history. The crash exposed weaknesses in many portfolios and risk models. It also brought greater attention to CTA managers because their results were not directly tied to a continuously rising equity market.

The crash nevertheless revealed an important limitation. Trend following normally needs time to identify a move and adjust positions. A sudden decline within a single trading day is therefore not necessarily favourable. Managers that already held short positions could benefit from the fall, while others were able to react only after much of the move had already occurred. Protection during a crisis is greater when price movements last long enough for the models to adjust. There is no guarantee against every unexpected shock.

The next major test came with Iraq's invasion of Kuwait in 1990 and the subsequent Gulf War. Oil prices rose sharply as

concerns about energy supplies increased, while the movements spread to currencies, interest rates and equity markets. As the course of the war became clearer, several markets reversed quickly. The period therefore illustrated both the opportunities and the difficulties of trend following. Large moves created the potential for substantial profits, but sudden reversals could also erase part of them.

A more prolonged test followed the bursting of the dot-com bubble. Technology stocks had risen to extreme valuations during the late 1990s, but a sustained decline began in the spring of 2000. The Nasdaq lost around 80 per cent of its value before reaching its low in 2002. Unlike the crash of 1987, the decline unfolded over several years and gave the models time to adjust their positions.

Moves also developed in other markets. Interest rates fell as central banks tried to support the economy, bond prices rose and currency markets were affected by changing capital flows and growth expectations. Several CTA programmes therefore reported positive results during a period in which large parts of the equity market declined. The dot-com crash provided important evidence that Managed Futures could perform during a prolonged bear market, even when the crisis originated in a single asset class.

The financial crisis of 2008 caused a broad collapse across equity and credit markets. For the CTA industry, the crucial factor was that the crisis generated sustained trends across several asset classes. Equity indices fell, government bonds rose, commodity prices turned lower and currency markets underwent major shifts. Several trend-following programmes had therefore had time to adjust their positions before the crisis reached its most dramatic phase.

Results in 2008 varied between managers, but several large CTA programmes delivered some of their strongest years while global equity markets fell sharply. These events were more important to the industry than the earlier crises. They showed that the strategies could generate substantial returns during a prolonged and wide-ranging crisis.

Even a result close to zero could be valuable during a deep equity-market decline because the capital did not fall with the stock market.

Trend following is not, however, a strategy that simply waits for the next crisis. Such periods made its distinguishing characteristics especially clear, but a large share of its returns was generated in other market environments.

The crash of 1987 attracted attention, the dot-com collapse provided further evidence and the financial crisis of 2008 brought the major breakthrough. Together, these periods showed that CTA programmes could continue to follow market movements even when traditional portfolios were under severe pressure. This ability—combined with the returns generated between crises—gave the industry its growing credibility.

From private investors to pension funds

The transition from private investors to institutional capital involved far more than pension funds beginning to allocate larger sums. It changed the conditions for the entire CTA industry.

To gain acceptance from large asset owners, it was no longer enough to present strong results. A manager also had to demonstrate that the business could be examined and controlled, and could continue to operate if a key person disappeared or the

markets came under extreme stress.

The early CTA firms were often built around one individual or a small team. Investors' decisions largely centred on the person behind the strategy. Contact was direct, decision-making paths were short and the organisations were small. For a wealthy private investor, convincing results and strong personal trust could be sufficient grounds for investing.

Pension funds, insurance companies and university endowments operated differently. They managed capital on behalf of pensioners, policyholders and future generations and had to justify every investment to boards, risk committees and regulators. They were therefore not investing solely in a trading programme. They were investing in an organisation expected to manage capital for many years.

The track record remained important, but it formed only the first part of the review. Institutions wanted to understand how the results had been produced, what risks the programme assumed and how the business would operate under extreme market conditions or if a key person disappeared. Trading systems, risk limits, business-continuity planning, independent audits, external administrators and transparent reporting therefore became increasingly important.

For many CTA managers, this required extensive adjustment. A smaller firm had previously been able to concentrate almost all of its resources on trading itself. Institutional capital also required staff in risk control, legal affairs, administration, technology and investor relations. Processes that had previously been handled informally had to be documented, while the systems had to support a significantly larger operation.

Managed accounts became an important part of this development. Instead of placing capital in a pooled fund, an institution

could open a separate account in which the CTA manager received authority to execute trades. The institution retained ownership and gained greater transparency into positions, risks and transactions.

A new group of intermediaries also emerged. Investment consultants, fund platforms and specialist advisers began analysing CTA programmes, while databases and industry indices made it easier to compare managers and study longer track records.

Institutional participation affected the way capital was allocated. The largest mandates usually went to firms with long track records, stable organisations and the capacity to manage substantial allocations. The distinction between being a skilled manager and running an institutional asset-management firm became increasingly clear.

The large capital flows also created new demands. Managers had to assess how much capital each model and market could absorb without increasing costs or influencing prices. At the same time, large investors could negotiate fees, reporting and risk levels, making Managed Futures more adaptable to the needs of different asset owners.

The development continued through the 1990s and early 2000s, but gained further momentum after the financial crisis of 2008. By then, several CTA firms had built long track records and organisations capable of passing institutional due diligence. When demand increased, an established infrastructure was therefore in place to absorb significantly more capital.

The trading idea itself did not change when institutions entered the industry. The models continued to follow market movements according to established rules. The change took place around the trading. What had once been operated by a handful of people developed into firms with advanced

technology, independent control functions and international distribution.

The CTA industry had thus moved from personal trust to institutional structure.

Sweden, Brummer and Lynx

Accounts of the Managed Futures industry almost always focus on Chicago, New York and London. This is natural. Most of the pioneers worked there, and the largest futures exchanges were located in the United States and the United Kingdom. Sweden, however, also came to play an important role in the development of alternative asset management.

For much of the twentieth century, Swedish asset management was dominated by traditional equity and fixed-income investments. Alternative investments had limited scope, and hedge funds were often regarded as unfamiliar and difficult to understand. It was in this environment that Patrik Brummer founded Brummer & Partners. The firm helped establish hedge-fund management as a professional part of the Swedish capital market. The business grew and, over time, attracted both Swedish and international institutions as investors.

Significantly, Brummer & Partners was never a pure CTA manager. Its importance to the Managed Futures industry lies instead in the way it helped change institutional attitudes towards alternative investments as a whole and created a platform for several managers with different approaches.

One of them was Lynx, founded in Stockholm in 1999 by Svante Bergström, Jonas Bengtsson and Martin Sandquist. In 2002, Brummer & Partners acquired 40 per cent of the management company, giving Lynx access to an established in-

stitutional environment and a broader investor base. Lynx subsequently developed into one of the larger international participants in the CTA industry.

The emergence of Brummer & Partners and Lynx demonstrates how alternative and systematic strategies also became established outside the industry's traditional centres. It was not necessary to be located in Chicago, New York or London.

The CTA industry becomes global

The CTA industry grew from a handful of American pioneers into an international segment of asset management.

Several US firms laid the foundations for this development. Millburn, Campbell & Company and DUNN Capital began operating during the 1970s and belonged to the first generation of CTA managers to use quantitative, rule-based methods across multiple futures markets. Their long, documented track records helped establish Managed Futures as a professional form of asset management.

A later US participant was Crabel Capital Management, founded by Toby Crabel in 1987. The firm differed from traditional trend followers by focusing primarily on short-term patterns in global futures and currency markets.

A similar development gathered pace in the United Kingdom during the 1980s. One of the most influential firms was AHL, founded in London in 1987 by Michael Adam, David Harding and Martin Lueck. AHL developed systematic models for trading global futures markets and became a pioneer in large-scale quantitative asset management.

Man Group acquired a majority stake in the company in 1989 and purchased the remaining shares in 1994. Funds based on

AHL's models became early examples of large-scale European quantitative management.

Under the name Man AHL, the business expanded into more markets, trading models and time horizons. The firm thus became a clear example of how CTA management moved from individual personalities to larger organisations.

AHL also became a training ground for the next generation of British managers. After the sale to Man Group, David Harding founded Winton in 1997, while Martin Lueck co-founded Aspect Capital that same year.

Winton built an operation centred on statistical analysis and developed from a small start-up into an international asset-management organisation. Over time, its approach also broadened beyond traditional trend following.

Aspect Capital built on parts of the same systematic tradition and developed several different systematic strategies.

A significant CTA business also became established in the Netherlands. Transtrend began trading in the early 1990s and based its operations on systematic analysis of global futures markets. Its Diversified Trend Program came to be applied across futures, currency and swap markets.

The industry's geographical centre of gravity remained in the United States, where the futures exchanges, regulatory framework and first large CTA firms had emerged.

At the same time, London developed into an important European centre for systematic asset management. As the industry spread further, Singapore and Hong Kong assumed growing roles as links between international managers, Asian investors and the region's markets. Managed Futures had thus developed from an American speciality into a global asset-management industry, although its structure remained strongly shaped by

the United States.

The CTA winter

The period after the financial crisis was considerably calmer for trend-following CTA programmes than the dramatic years surrounding 2008. In retrospect, parts of the 2010s have therefore been described as the CTA winter. The expression is used somewhat loosely and does not describe an uninterrupted decline throughout the decade. Rather, the period consisted of two weaker phases separated by a clear recovery.

The years 2010–2013 were demanding for many trend followers. Results were characterised by short-lived price moves and repeated trend reversals that never developed into sustained trends.

Market conditions subsequently improved, and strong moves emerged during 2014 in currency, interest-rate and energy markets, among others. The dollar rose sharply while oil prices fell, creating opportunities. For many CTA programmes, 2014 was a strong year, and by 2015 several had recovered earlier losses and reached new highs.

A new period of weakness followed from the second half of 2015 until the end of 2018. The general performance of many trend-following programmes was flat or negative, although results varied between managers.

After the financial crisis, markets were characterised by extremely low interest rates, extensive central-bank intervention and periods of low volatility. Trends that appeared to be taking hold were often interrupted by new monetary-policy announcements, making it difficult for trend-following models to generate strong returns. To produce strong results, at least a

few markets need to develop larger and more persistent trends.

Global equity markets rose during much of the decade, making the comparison particularly demanding. As CTA programme returns weakened and some drawdowns became prolonged, many investors reduced their allocations. Fees were questioned, and the media began asking whether trend following had stopped working.

The CTA winter was especially difficult for smaller managers. Many had limited financial resources and depended on a small number of investors, making it difficult to cover fixed operating costs when assets declined. Several were therefore forced to close their programmes even though their models had not necessarily stopped working.

Larger and more established managers also experienced substantial redemptions, but their scale, broader client bases and stronger organisations gave them better prospects of surviving the prolonged period of weakness.

When the trends returned

Market conditions began to change during 2019. Gold broke out of a consolidation that had lasted several years and began a sustained advance, providing an early sign that larger trends were once again starting to form.

The development accelerated when the Covid-19 pandemic spread around the world in early 2020. Lockdowns, supply-chain disruptions and sharply changing consumption patterns generated large movements in financial markets.

After the initial turbulence, strong trends emerged. Equity markets recovered, interest rates were pushed lower and several currencies and commodities began to display more sustained

trends. The type of price movement that trend-following models are designed to capture had therefore returned.

During the inflation shock of 2022, both equities and bonds fell. The shift in monetary policy and disruption in commodity markets created clear and sustained trends across interest-rate, currency, energy and agricultural markets.

Many CTA managers were able to make use of their ability to take both long and short positions and delivered some of their strongest results in many years. The diversifying characteristics of trend following therefore became particularly clear.

The sector subsequently entered a weaker period in 2023 as several of the major moves from the previous year lost momentum. Conditions later improved, particularly in 2025 as strong trends developed in gold and other precious metals.

III

Management in Practice

The Participants in the CTA Industry

An investment in Managed Futures involves several participants with different responsibilities. The CTA manager makes the trading decisions, the broker facilitates the relationship and sometimes the order, the FCM holds the client's capital and maintains the account, the exchange matches the trades, and the clearing organization manages counterparty risk. The CFTC and NFA, in turn, are responsible for regulation and oversight. The structure may initially appear complicated, but it creates independent control across management, custody of capital, and reporting.

The Broker

The broker is often the investor's first point of contact with the U.S. futures industry. In the United States, the term Introducing Broker, or IB, is used for a firm that introduces clients to an FCM and helps them gain access to the accounts and infrastructure required for trading. An IB may accept or transmit trading orders, but it may not hold clients' money or other assets as

collateral for trades. The capital must therefore be placed in an account with an FCM, a *Futures Commission Merchant*.

The IB's work begins before the account has even been opened. The broker can help the investor select a suitable FCM, explain the different account options, and ensure that applications, powers of attorney, and regulatory documents are completed correctly.

When an investor engages a CTA, the relationship among the investor, the manager, and the FCM must be documented. The broker can coordinate the account-opening process and ensure that the CTA manager receives the trading authorization required to execute trades. This authorization normally does not give the manager the right to withdraw capital from the account.

After the account has been opened, the IB generally serves as the investor's main point of contact. Questions concerning account statements, deposits, withdrawals, trading platforms, and technical issues can be handled through the broker, who in turn communicates with the FCM or CTA manager. This service can be particularly valuable when the investor is located in another country or when several firms are involved in the management arrangement.

The term broker covers several functions that need to be distinguished. An Introducing Broker is primarily responsible for the client relationship, the introduction to the FCM, and ongoing service. An executing broker is responsible for routing the order to the market and executing it.

The FCM maintains the futures account itself, holds the client's margin funds, and is responsible for recordkeeping, risk monitoring, and clearing. The same firm may perform several of these functions, but the roles and legal responsibilities are

fundamentally different.

How the order reaches the market depends on how the trading program is organized. It may be sent through the broker's system, through a separate *executing broker*, or directly from the CTA manager's order-management system. Even when the IB does not participate in every order, it may continue to be responsible for the client relationship, administration, and contact with the FCM.

The IB is usually compensated through commissions or an agreed share of the transaction revenue generated in the accounts of the clients it has introduced. The charge may be assessed as an amount per contract and divided among the broker, the FCM, the exchange, and the clearing organization. Some arrangements also include fixed account or service fees. The account agreement and fee disclosures should explain how the costs are allocated.

Futures Commission Merchant

FCM stands for *Futures Commission Merchant* and refers to the firm that maintains the client's futures account and administers the financial side of the trading. The CTA manager determines which positions to take, but the FCM holds the client's collateral, records the trades, and verifies that the account meets its financial requirements. The FCM therefore serves as the financial hub of the futures account.

When an account is opened, the FCM reviews the application, identifies the client, and assesses whether the account is suitable for the intended trading. The firm also handles the agreements and powers of attorney governing the relationship between the investor and the CTA manager. If the CTA is to trade on the

client's behalf, a limited trading authorization is registered, giving the manager the right to place orders but normally not to remove money or other assets from the account.

The client's capital is deposited directly with the FCM and used as collateral for the positions. The funds are not ordinary bank deposits, although the FCM's function may in some respects resemble that of a bank. The capital is held in a futures account and is subject to the rules governing customer funds in the derivatives markets. The FCM is responsible for handling and accounting for the money in accordance with those rules.

The account's total equity must be distinguished from the margin in use. Account equity represents the client's total balance after profits, losses, fees, deposits, and withdrawals have been recorded. Margin is the portion of the capital that must remain available as collateral for open positions. It is used to cover price movements that may occur while the positions remain open.

If an account contains one million dollars, for example, the entire amount need not be tied up as margin. A portion may be reserved for current positions while the remainder constitutes available capital. This surplus provides room to withstand normal market movements, take new positions, or meet higher margin requirements. The amount of available capacity depends on the contracts traded, the size of the positions, and prevailing volatility.

The FCM continuously calculates the required margin. The exchange and clearing organization establish basic requirements for each contract, but the firm may impose higher levels if market risk or the client's positions warrant it. Margin requirements may therefore change even when the CTA manager has not entered any new trades. During periods of sharp price

movements or deteriorating liquidity, requirements can rise quickly.

All open futures positions are valued daily. At the end of each trading day, the value of the positions is compared with the previous settlement price. Profits are credited to the account and losses are deducted. This daily settlement means that changes in market value have an immediate effect on account equity and on the capital available to support the positions.

The FCM records every completed trade, open position, commission charge, exchange fee, and other account entry. The client's account statement generally shows the account balance, open positions, completed trades, realized results, and the market value of positions that remain open. Although the reporting may distinguish between realized and unrealized results, the daily settlement of futures positions affects the capital actually available in the account.

Profits increase available capital and may create room for new positions. Losses, by contrast, reduce both the account's value and its margin capacity. An account may therefore encounter margin problems even if the long-term trading strategy itself remains unchanged. The FCM assesses the account's financial condition based on the actual positions and the capital available to support them.

If the capital falls below the required level, the FCM may require additional funds to be deposited or positions to be reduced. The client or CTA manager may then need to close certain trades to free up margin capacity. If this does not happen quickly enough, the FCM generally has the right to reduce or close positions itself. Under extreme market conditions, such action may be taken without first giving the client an opportunity to add capital.

This right also entails an obligation to the clearing organization and the market. The FCM is responsible for its clients' positions within the clearing chain and cannot allow an account to continue accumulating uncovered losses. It therefore monitors, among other things, concentrated positions, changes in volatility, approaching delivery dates, and how quickly positions could be liquidated. These controls are conducted independently of the CTA manager's own risk models.

The CTA and the FCM therefore view risk from different perspectives. The CTA manager determines how risk should be allocated in order to follow the trading program's rules and achieve its objectives. The FCM instead assesses whether the account can meet its financial obligations. A position that the CTA considers reasonable within the overall portfolio may still be restricted by the FCM if it requires excessive margin, is difficult to liquidate, or creates an overly concentrated exposure.

The FCM is also responsible for the connection with the clearing organization. Some FCMs are themselves clearing members of the relevant exchanges, while others use an external clearing member for certain markets. For the client, this does not alter the basic division of responsibilities: the FCM records the trade in the client's account and ensures that the necessary collateral is present in the clearing chain.

An important part of the structure is that the client receives trade confirmations and account statements directly from the FCM. The information therefore comes from a party other than the CTA manager. The investor can compare the manager's performance reports with the positions, account activity, and values reported by the firm maintaining the account. This independent reporting creates a control function that would not exist if the manager made the trading decisions, held the

capital, and reported the results alone.

Segregated Customer Funds

A fundamental principle of the U.S. futures industry is that customer money must be kept separate from the FCM's own assets. This is described by the term *segregated customer funds*. The purpose is to create a clear legal and accounting boundary between capital belonging to customers and capital belonging to the FCM.

When an investor deposits money in a futures account, the capital therefore does not become part of the FCM's unrestricted working capital. The firm may not use customer funds to finance its own operations, pay its debts, or speculate for its own account. The money may be used for purposes connected with customers' trading, such as collateral for their positions and payments in the clearing process.

Customer funds must be held in accounts clearly identified as belonging to the FCM's customers. They may be held with banks, clearing organizations, and other approved depositories. The FCM may also invest portions of the capital in certain permitted and regulated assets, but the rules restrict the investments that may be used and impose requirements concerning safety and liquidity. The capital must be available when needed for customers' margins, withdrawals, and daily settlements.

A segregated account does not, however, mean that each customer's money is held in a separate bank account in that customer's name. The FCM may hold the assets of several customers together in designated customer-funds accounts, sometimes called omnibus accounts. Its own records must nevertheless show how much of the pooled capital belongs to

each individual customer.

In its internal records, the FCM must be able to determine each customer's balance and continuously verify that the aggregate customer funds match the firm's obligations to its customers. The capital may be distributed among banks, clearing organizations, and other approved depositories, but it must be identifiable and reported as customer property.

Segregation becomes particularly important if the FCM encounters financial difficulties. Because customer funds must be kept separate from the firm's own assets, they should in principle not be used to pay its general creditors. Customer capital therefore receives stronger legal protection than it would if the money formed part of the firm's ordinary balance sheet.

The bankruptcy of MF Global in 2011 became a clear test of the rules governing segregated customer funds. MF Global was one of the largest FCMs in the United States, but it improperly used customer funds to meet the firm's own liquidity needs. When the business collapsed, approximately \$1.2 billion was missing from customer accounts.

Some account holders recovered their capital within a couple of weeks, while others had to wait considerably longer. The event contributed to stricter requirements for the reporting, reconciliation, and control of customer funds held by FCMs.

MF Global demonstrated that the rules governing segregated customer funds do not always guarantee immediate access to capital when an FCM fails. Even so, all customers ultimately recovered their capital in full.

Futures Exchanges

The FCM holds the client's collateral and maintains the futures account, but the trading itself takes place on a futures exchange. Each contract is tied to a particular exchange whose rules specify, among other things, the contract size, quality, maturity, delivery terms, and trading hours. When the CTA manager sends an order, it must therefore be directed to the marketplace on which the relevant contract is listed.

The fundamental role of a futures exchange is to create an organized and standardized market by designing contracts and setting the terms of trading. A gold contract, for example, specifies the number of troy ounces covered and the required quality of the metal, while an agricultural contract specifies quantity, quality, and approved delivery locations. Standardization ensures that everyone trading the same contract knows exactly what rights and obligations it represents.

Buy and sell orders meet in the electronic marketplace. They are received from market participants, entered into the order book, and matched when price and other conditions coincide. The exchange also publishes prices, trading volume, and *open interest*, meaning the total number of contracts that remain outstanding. This information gives market participants a continuous view of trading activity and price formation.

At the end of each trading day, the exchange establishes official settlement prices used to value open positions. It also sets trading hours, order rules, and the requirements applicable to each contract. Under special circumstances, it may use price limits, position limits, and other control mechanisms to maintain an orderly market. Trading activity is monitored to detect prohibited conduct and other violations of exchange

rules.

The exchange's role is to design the contracts, organize trading, and supervise the marketplace.

An important distinction exists between an exchange group and the individual exchanges within it. CME Group, for example, is not a single futures exchange but a group comprising four U.S. exchanges: the Chicago Mercantile Exchange, CME; the Chicago Board of Trade, CBOT; the New York Mercantile Exchange, NYMEX; and the Commodity Exchange, COMEX. They share much of the same technology and clearing infrastructure, but their contracts remain listed under the rules of the respective exchanges.

The exchanges have different areas of specialization. Within CME Group, contracts are traded on U.S. equity indices, interest rates, energy, metals, and agricultural products. ICE has a strong position in areas including Brent crude oil, European natural gas, and so-called *soft commodities*. Other important marketplaces include Eurex, the London Metal Exchange, and several Asian futures exchanges.

A global CTA therefore needs access to several exchanges. This infrastructure makes it possible to build a portfolio across many markets and geographic regions.

Two exchanges may offer futures contracts based on similar underlying markets, but trading is often concentrated on one of them. The CTA manager then selects the contract that best meets the program's requirements for liquidity, transaction costs, contract design, and reliable price data.

Historically, much futures trading took place on physical trading floors through a system known as *open outcry*. Today, most trading is electronic. CTA models can monitor markets in real time, send orders automatically, and manage positions on

multiple exchanges through the same technical system. Many of the largest futures contracts can be traded for almost twenty-four hours a day on weekdays.

The electronic infrastructure allows an order to travel from the CTA manager's system to an exchange in a very short time. Once buy and sell orders have been matched, the financial processing of the trade remains. At this point, the clearing organization takes over and manages the obligations arising between buyer and seller.

From Order to Cleared Trade

When a position appears on an investor's account statement, it has already passed through several technical and institutional stages. The trading decision comes from the CTA manager, but the order is executed, cleared, and recorded by other parties. The exchange, clearing organization, and FCM perform different functions, even though modern electronic trading often completes the process in fractions of a second.

The process begins when the CTA manager makes a trading decision. The decision may, for example, involve buying a specified number of futures contracts, reducing an existing position, or replacing a long position with a short one.

The order is then created in the CTA manager's order-management system. It contains information about the contract to be traded, whether the order is to buy or sell, the number of contracts involved, and the conditions governing execution. The system must also identify the client accounts or programs to which the order belongs. If the CTA trades the same program for many clients, a larger block order may be sent to the market and subsequently allocated among the relevant accounts according

to predetermined rules.

The order may be routed through an executing broker or sent directly from the CTA manager's electronic order-management system to the marketplace. In either case, it must pass controls ensuring that the contract is permitted, the order size is within approved limits, and an identified FCM and account stand behind the trade. Incorrect or unauthorized orders may be stopped before they reach the exchange.

When the order reaches the futures exchange, it enters the exchange's order book, where it meets buy and sell orders from other market participants. Orders are matched according to published principles and the rules of the contract. The exchange does not, however, determine whether the CTA manager's trade is suitable, nor does it hold the client's money.

If the CTA sends a buy order, a corresponding seller must be present in the market for the trade to occur. When the buy and sell terms meet, the exchange's trading system matches the orders and a trade is created. Both parties receive an electronic confirmation showing the contract, quantity, price, and time of the trade. An order may be filled in full at a single price or divided into several executions, depending on its terms and the opposing orders available.

After execution, the trade is forwarded to the clearing organization. The exchange and clearing organization may belong to the same corporate group, but they perform different functions: the exchange matches the trade, while the clearing organization is responsible for ensuring that the financial obligations can be met.

This occurs through a process known as *novation*. The original agreement between buyer and seller is replaced by two separate obligations. The clearing organization steps in as seller to the

buyer and buyer to the seller. The buyer therefore does not have to rely on the original seller's ability to perform, and the seller does not have to assess the buyer's ability to pay. The clearing organization thus becomes the central counterparty to both.

The clearing organization does not have a direct relationship with every individual investor. Contact is handled through its clearing members, which are responsible to the clearing organization for positions and collateral.

An FCM may be a direct clearing member of certain exchanges and clear its clients' trades in its own name. It does not, however, need to be a member of every exchange on which its clients trade. For other markets, the FCM may need to use another clearing member that carries the trade further through the clearing chain.

Once the trade has been accepted for clearing, the position is recorded in the client's account with the FCM. The FCM records the contract, quantity, direction, execution price, and fees associated with the trade. If the order was executed as a block order for several clients, the contracts are allocated among the client accounts according to the CTA manager's documented allocation method. The allocation may not be changed afterward to give certain clients better execution prices than others.

The account's margin requirement is then updated. The FCM calculates how the new position affects the client's total collateral requirement and verifies that sufficient capital is available. The clearing organization performs a corresponding calculation at the clearing-member level. Margin control thus operates through several layers, from the client account at the FCM to the FCM's aggregate obligations to the clearing organization.

In some arrangements, the trade is executed by an *executing*

broker and then transferred to the FCM where the client's account is maintained. This is known as a *give-up*. The arrangement allows the CTA manager to use different brokers for order execution while consolidating positions with one or a small number of FCMs for clearing, margin management, and account administration.

After the position has been recorded, it is valued at the end of each trading day. The exchange establishes an official settlement price, and the clearing organization calculates the change in value from the previous settlement. Profits and losses pass through the clearing chain among the clearing organization, clearing members, and customer accounts at the respective FCMs. This daily settlement continues for as long as the position remains open.

The client can follow the process through the trade confirmations and account statements provided by the FCM. These show the executed trade, the open position, account performance, available capital, and fees charged. When the CTA manager later decides to close the position, an opposing order is sent through the same chain. The exchange matches the order, the clearing organization terminates the obligation, and the FCM removes the position from the client's account.

What appears to the investor as a single trade is the result of a carefully divided process. The CTA makes the trading decision, the broker or order system transmits the order, the exchange matches it, the clearing organization manages counterparty risk, and the FCM maintains the account. This division of responsibilities allows each stage to be controlled and the trade to be completed without buyer and seller needing to know or assess one another.

During the *open-outcry* era, an order followed a considerably

longer route. It was phoned to the trading floor, written on an order ticket, and carried by a *runner* to the pit broker, who executed the trade using shouts and hand signals. The confirmation then traveled back through the same chain. Today's electronic trading has made order flow both faster and easier to follow.

How the Participants Are Paid

An investment in a CTA program involves several types of fees and costs. Some go directly to the CTA manager, while others compensate the broker, FCM, exchange, clearing organization, and the firms administering the investment structure. To understand the total cost, the investor must therefore look beyond the management fee stated in the program's marketing materials.

The CTA manager is generally compensated through a management fee, a performance fee, or a combination of the two. The management fee is normally calculated as an annual percentage of assets under management and charged monthly or quarterly. It provides the manager with recurring revenue intended to cover staff, development, trading systems, market data, regulatory compliance, and administration, among other expenses.

The performance fee depends on the program's results and is calculated as a share of the profit generated for the investor. Many CTA programs use a high-water mark. This means that a performance fee is charged only on profits above the highest level at which such a fee was previously calculated. If the account first declines in value, the loss must therefore be recovered before a new performance fee can be charged.

A CTA program may, for example, charge a management

fee of two percent per year and a performance fee of twenty percent. The management fee is calculated on assets under management, while the performance fee is calculated only on fee-eligible profits. The precise calculation methods may vary among programs and must therefore be disclosed in the management agreement or offering memorandum.

An Introducing Broker is usually compensated through commissions, such as an amount for each futures contract bought or sold. The compensation may be included in the transaction fee shown in the client's account or charged as a separate service fee. How it is divided among the IB, FCM, exchange, and clearing organization depends on the relevant agreement.

The FCM also earns revenue from commissions and clearing-related fees. A portion of the amounts charged to the client covers the FCM's work in account administration, risk monitoring, reporting, and maintaining the financial connection with the clearing organization. The firm may also charge for trading platforms, market data, account administration, withdrawals, or special reporting.

Customers' cash collateral may also generate interest income. Within the regulatory framework, the FCM may invest customer funds in certain permitted assets or hold them in interest-bearing accounts.

Depending on the account agreement, all of the interest may accrue to the client or the FCM, or it may be shared between them. During periods of high interest rates, this can represent a significant source of revenue for FCMs. This means that the cost of the relationship is not always apparent solely from the commission shown on the account statement.

The exchange charges fees for providing the marketplace and processing the order. The clearing organization, in turn, charges

for registering, guaranteeing, and settling the trade. These costs are generally passed on to the client through the FCM. Exchanges also sell market data, and the cost may vary depending on whether the user is classified as retail or professional and on the information required.

If the investment is made through a Commodity Pool, additional costs arise from the pooled structure. The pool may need to pay for administration, accounting, auditing, legal advice, regulatory reporting, and valuation of investors' interests. If a CPO is responsible for operating the pool, a separate CPO fee may also apply.

In some structures, these costs are covered by the standard management fee. In others, they are charged separately to the pool and thereby reduce investors' net returns.

Although the number of fee items may appear large, most trading-related fees are individually very small. Exchange and clearing fees, along with other minor transaction charges, normally represent only a limited share of the total cost. The larger items are usually the management fee, the performance fee, and, where applicable, the costs of a Commodity Pool. What matters is therefore not the number of different fees, but their combined size relative to the capital under management and the program's trading activity.

CFTC – The Federal Regulator

The Commodity Futures Trading Commission, or CFTC, is the federal agency that oversees the U.S. market for futures, options, and large parts of the swaps market. Its mandate covers both the marketplaces and the firms operating in them, including futures exchanges, clearing organizations, FCMs, Introducing

Brokers, CTA managers, and CPOs.

Oversight is structured in several layers. Exchanges are responsible for the marketplaces on which trading takes place, and clearing organizations handle the financial obligations arising after trades. FCMs hold customer funds and maintain customer accounts, while CTA and CPO firms manage capital or operate pooled investment structures. The CFTC establishes the overarching rules these participants must follow and verifies that the different parts of the markets operate safely and properly.

A central part of the mandate is protecting customer assets. The CFTC sets rules governing how FCMs must hold, report, and reconcile segregated customer funds. It also establishes financial requirements for registered firms and rules for record-keeping, reporting, and documentation. This reporting enables regulators to detect deficiencies before they develop into larger problems and to intervene when a firm fails to meet its obligations.

The CFTC also oversees market integrity. It collects and analyzes trading data to detect suspicious behavior, excessive concentrations, and attempts to influence market prices improperly. Its work includes investigations of fraud, manipulation, misleading marketing, improper use of customer funds, and trading that violates market rules.

If the CFTC suspects a violation, it may open an investigation and request documents, trading records, and communications from the firms involved. The agency may then bring an action in federal court or pursue an administrative proceeding.

These actions may result in fines, restitution to customers, trading bans, revocation of registration, or other restrictions on the business. When criminal conduct is suspected, information

may also be referred to U.S. law-enforcement authorities.

Not all ongoing supervision is performed by the CFTC. The U.S. futures industry is partly based on self-regulation, under which exchanges and the National Futures Association, among others, oversee their members. The CFTC, in turn, verifies that these self-regulatory organizations maintain effective rules, control systems, and disciplinary processes. The CFTC stands above the self-regulatory system and can intervene if oversight is inadequate.

Registration should not be interpreted as a seal of quality for an investment. The CFTC does not determine whether a CTA model will be profitable and does not guarantee future results. A registered manager may therefore comply with every rule and still experience significant *drawdowns*. The agency's role is to enforce the regulatory framework and act against fraud, manipulation, improper use of customer funds, and other violations. Assessing the strategy's financial risk remains the investor's responsibility.

NFA

The National Futures Association, or NFA, is the self-regulatory membership organization for the U.S. derivatives industry. The NFA is not a government agency but operates under CFTC oversight and is responsible for much of the ongoing supervision of the industry's registered firms and individuals.

The NFA is funded by membership dues and transaction assessments, not by federal tax revenue. A small NFA assessment is charged on futures transactions, reported and remitted to the organization by the FCM.

The division of responsibilities between the CFTC and NFA

can be described in simplified terms as follows: the CFTC is the federal regulator and establishes the overarching regulatory framework, while the NFA conducts much of the practical supervision of members. At the same time, the CFTC oversees the NFA to ensure that it fulfills its responsibilities and that its rules and control systems are consistent with federal law.

NFA members include FCMs, Introducing Brokers, CTA managers, and CPOs. The NFA administers large parts of the registration system and reviews firms' records, customer agreements, trading records, marketing, and internal control procedures. For CTA firms, reviews may also cover how historical performance is calculated, presented, and supported by underlying trading data.

Marketing is another important area. Members may not present historical results or future opportunities in a misleading manner. Claims must be substantiated, risks must be described, and hypothetical results must be clearly distinguished from actual trading. The NFA may request marketing materials for review and require inaccurate or unbalanced wording to be changed.

The organization also monitors how members conduct themselves toward customers. This includes sales practices, information provided before an investment, handling of customer complaints, and supervision of employees and representatives. Firms must maintain written procedures and be able to demonstrate that the business is supervised by responsible individuals.

If a member violates the rules, the NFA may initiate disciplinary proceedings. Measures may include fines, warnings, temporary restrictions, or expulsion from the organization. Because membership is normally a prerequisite for conducting registered business, expulsion may in practice mean that the

firm or individual can no longer operate in the U.S. futures industry.

One of the NFA's most useful tools for investors is its BASIC database, the Background Affiliation Status Information Center. It allows anyone to check free of charge whether a firm or individual is registered, which individuals are associated with the business, and whether there are disciplinary decisions or other regulatory history. The absence of such decisions does not guarantee that an investment is problem-free, but it allows investors to verify a manager's information against an independent source.

Transparency and Independent Verification

The division of responsibilities among participants in the CTA industry may appear complicated, but it makes the investment easier to verify. Investors do not need to examine every technical detail themselves. It is enough to understand who manages the capital, where it is held, and how performance is reported.

In the United States, a CTA manager's registration and any disciplinary history can be checked free of charge through NFA BASIC. In a managed account, the investor can also verify the management through account statements received directly from the FCM. The CTA manager's limited power of attorney normally grants authority to trade, while the investor retains control over deposits and withdrawals.

As a general rule, a prospective client of a registered CTA program must receive a current *Disclosure Document* before entering into the management agreement. The document is filed with the NFA, which reviews it for compliance with regulatory requirements and to ensure that the information

is not misleading. The review does not, however, mean that the NFA approves the trading program, recommends the manager, or considers the investment suitable.

The document describes the CTA manager and responsible individuals, the trading program's approach, the markets that may be traded, the principal risks, fees, potential conflicts of interest, and relevant disciplinary or legal history.

It also presents the program's historical performance according to established rules, together with drawdowns and an explanation of how returns have been calculated. The investor therefore receives the most important information in a single regulated document and does not need to map the entire institutional chain independently.

In a Commodity Pool, the futures account is owned by the pool rather than by the individual investor. The CPO is responsible for the legal and administrative structure, while the CTA manager is responsible for the trading program. An external administrator may calculate the net asset value, and an auditor may review the accounts. The investor therefore receives reporting at the fund-interest level rather than an individual account statement directly from the FCM.

Regardless of the structure, the underlying principle is the same: trading decisions, custody of capital, and reporting do not need to reside with the same party. Investors can therefore verify key information through several independent sources. This division of responsibilities should primarily be seen as protection for the investor, not as an extensive control exercise that the investor must personally perform.

The Greatest Risk

The Problem Is Rarely the Strategy

The greatest risk to a trend-following model is rarely the strategy itself. The greatest risk is often the investor. Many CTA managers fail not because the model stops working, but because investors leave the program.

At first glance, this may sound strange. If a strategy has a documented track record and a clear investment philosophy, its outcome should reasonably be determined by how well the strategy works. In practice, however, investing is not only a matter of mathematics and markets, but also of psychology and human behavior.

An investment strategy can generate returns only for those who actually follow it. It does not matter how successful a strategy has been historically if the investor sells after a period of adversity. The returns belong only to those who remain invested long enough to experience both the weak and the strong periods. History is full of examples.

The same applies to index funds. Today, many regard index investing as the obvious solution for long-term saving. Yet index funds also require patience. Anyone who invested before the dot-com crash in 2000 experienced two severe market declines in less than a decade. For many, it was a period filled with doubt and frustration. Some chose to sell after heavy losses and consequently missed the recovery that followed.

Trend following and Managed Futures are no exception. Because the strategy is often characterized by periods of underperformance and many small losses, investors' patience is repeatedly tested. When markets lack clear trends, results may remain weak for several years. That is when the most difficult questions arise. Has the strategy stopped working? Have the markets changed? Should the capital be moved elsewhere?

These questions recur in almost every form of investing. The difference is that trend following makes them particularly acute because the strategy's success depends to a large extent on the investor's willingness to accept periods when the results look poor.

This creates a paradox. Investors often devote enormous energy to choosing the right investment, but far less time to considering whether they will be able to live with the strategy when adversity arrives. In reality, the two questions are inseparable.

Success in investing therefore often depends more on patience than on accurate forecasts. Markets change, strategies pass through different phases, and underperformance is inevitable. The question is not whether setbacks will occur, but how the investor responds during difficult periods. A strategy may survive its losing periods, but the question is whether the investor will.

The Reality of Returns

Trend following rarely produces its long-term return through a steady succession of profitable months and years. Results are unevenly distributed across trades, markets, and points in time. Flat or negative performance may be followed by sharp gains when clear trends emerge in one or more markets.

This return pattern follows directly from the strategy's construction. A trend-following CTA does not try to determine what a market ought to be worth. The models take positions when prices begin moving in a clear direction and exit them when the move breaks down. Most signals never develop into major trends and therefore result in small losses. The win rate can consequently be low without making the strategy unprofitable. A limited number of positions, however, may grow into large winners and account for a significant share of the long-term return.

The average annual return tells only part of the story. Two strategies may reach the same final result while following entirely different paths. One may progress relatively steadily, while the other alternates between extended periods of flat performance and shorter periods of powerful gains. To understand Managed Futures, one must therefore examine both the result and how returns are distributed over time.

In asset management, the term *drawdown* describes how far a portfolio falls from a previous high before reaching a new peak. If a fund rises from 100 to 150 and then falls to 120, it has experienced a drawdown of 20 percent. The drawdown period does not end until the fund's value rises above 150 again.

Both the size and the duration of a drawdown matter. A rapid decline of 20 percent that is recovered within a few months

differs from a decline of 10 percent followed by several years of flat performance. In the latter case, the percentage loss is smaller, but the capital remains below its previous high for considerably longer.

A smooth and attractive equity curve may create an impression of robustness, but it may also mean that the strategy has not yet encountered the market environment in which its weaknesses become apparent. A system that has survived high volatility, deep drawdowns, and several different market regimes over a long period has, by contrast, been tested in the real world.

This does not mean that large losses are desirable or constitute proof of quality. But the ability to remain within the risk framework and recover after difficult periods provides stronger evidence of robustness than a short and almost uninterrupted rise.

A drawdown does not automatically mean that the model has stopped working. It may occur even when the program follows its rules and operates as intended. The investor must therefore distinguish between a normal setback and signs that something has genuinely changed. The manager needs sufficient conviction to follow the strategy, but also enough humility to limit risk when reality diverges from the model.

When Rational Decisions Become Difficult

Investors do not always make decisions as rationally as traditional economic theory long assumed. Uncertainty, losses, and recent performance influence decisions in ways that may cause an investor to abandon a long-term plan at the wrong time.

Success therefore depends not only on which strategy is selected, but also on the ability to remain invested when its

performance creates doubt and frustration.

During the 1970s and 1980s, psychologists Daniel Kahneman and Amos Tversky studied how people actually make decisions under uncertainty. Their research laid the foundation for what is now called behavioral economics and fundamentally changed the way investment decisions are understood. Rather than treating investors as rational decision-making machines, they showed that human choices are influenced by a range of systematic psychological biases.

One of the best-known psychological mechanisms is *loss aversion*. Kahneman and Tversky showed that people generally feel the pain of a loss more strongly than the pleasure of an equivalent gain. A loss of 10 percent therefore often feels more powerful than an equal gain.

From an evolutionary perspective, this is easy to understand. For our ancestors, a mistake could mean lost food, injury, or, in the worst case, death. Reacting strongly to losses was therefore often advantageous. The problem is that the same mechanisms still govern our decisions in financial markets.

For an investor, this means that a drawdown of 20 percent often feels far worse than a previous gain of 20 percent felt good. Although the percentages are identical, they are experienced very differently on an emotional level.

This is particularly relevant to Managed Futures. Even successful CTA strategies may experience significant drawdowns, the size of which varies with the program's risk level. From a statistical perspective, a decline may fall within the strategy's normal return pattern. From a human perspective, it is nevertheless often experienced as evidence that something has gone wrong.

Another psychological bias is what is known as *recency bias*.

People tend to place disproportionate weight on what has happened recently. If a strategy has produced strong returns over the past twelve months, it feels natural to assume that it will continue to perform well.

If the same strategy has instead had several weak years, it is easy to conclude that it has stopped working. The problem is that markets rarely behave that way.

The best future investments are usually not those that have performed best most recently. Likewise, the worst future investments are not always those that have recently performed poorly. Investors nevertheless continue to make decisions as though the past will automatically repeat itself.

For trend following, this creates a particular challenge. The strategy's strongest results may arrive after prolonged frustration and weak performance. Just as investors begin to lose patience, the market environment the strategy was designed to handle may emerge.

Herd behavior also affects how investors evaluate their managers. When the stock market rises while trend following performs poorly, owning a strategy that diverges from the majority can feel uncomfortable. The psychological pressure increases when colleagues, neighbors, and the financial media highlight investments that have recently performed better.

The risk is that the investor leaves the CTA program because it feels wrong at that particular moment, even though the strategy's role in the portfolio remains unchanged. The loss is visible in the account, while the missed gain becomes apparent as other markets rise.

Many of the greatest challenges in investing have very little to do with the markets themselves. Instead, they concern how people respond to uncertainty, losses, and social signals. The

investor's greatest opponent is therefore often not the market, but the investor's own behavior.

Chasing the Latest Winner

When loss aversion, recency bias, and herd behavior influence investors' decisions, capital tends to flow toward whatever has performed best recently. It is one of the best-documented and most costly behavioral patterns in asset management.

The process often begins when a fund, market, or investment strategy produces strong returns. The financial media take notice of the success, and the manager is interviewed. After such a period, the strategy almost sells itself.

The herd moves toward what has recently worked and interprets the strong historical return as a sign of low risk. By the time the capital is invested, however, a significant part of the advance has usually already occurred.

If performance then weakens, doubts arise. A drawdown, a period of flat performance, or several years of lower returns may cause the investor to reconsider the decision. Eventually, the holding is sold and the capital is moved to something that currently looks more attractive. If the former investment subsequently recovers, the investor is no longer there.

A frequently cited example is Fidelity Magellan under Peter Lynch. Lynch managed the fund from 1977 to 1990 and presided over one of the most successful periods in the history of the fund industry. During his tenure, the average annual return was approximately 29 percent.

The fund was opened to a broader public in 1981, after Lynch had already established several years of strong performance. Between 1981 and 1990, Magellan generated an average annual

return of approximately 21.8 percent. One frequently cited calculation, however, has estimated the average investor's money-weighted return over the same period at approximately 13.4 percent per year.

The difference arose because investors added and withdrew capital at different times. Those who joined after a strong advance did not participate in the earlier return. Those who left after a weak period risked missing the subsequent recovery. The fund's reported performance and the return actually earned by investors' capital thus became two different things.

For several decades, the American research firm Dalbar has attempted to measure the same broader phenomenon. The firm compares market performance with estimates of the returns earned by investors' capital after actual contributions and withdrawals. Its findings have repeatedly shown that the average investor has earned a lower return than the market.

The phenomenon is particularly evident in strategies with unevenly distributed returns. A strategy may generate mediocre results for several years and consequently appear uninteresting.

A limited number of favorable months may then account for a significant share of several years' cumulative return. The strong results attract new capital, but by the time investors join, the market environment may already be changing.

This does not mean that an investor should buy a strategy solely because it has recently performed poorly. The results may reflect either normal market conditions or genuine problems in the management of the program. The point is that recent performance alone is rarely sufficient to determine which investment will perform best in the future.

The same behavioral pattern appears in equities, value investing, small-cap stocks, commodities, and other strategies that

periodically deviate from market expectations. Investors buy after success and sell when the investment feels most uncomfortable. For a CTA manager, helping investors understand the program's return pattern before adversity occurs is therefore an important responsibility.

Trend Following Feels Wrong

If there is one common reason why many investors struggle to remain committed to trend following, it is that the strategy often conflicts directly with human intuition. For most people, it feels natural to want to buy something that has become cheaper. If a product is offered at a discount, it is perceived as an opportunity. If a stock has fallen sharply, the thought often arises that it should now be more attractive than before.

Likewise, people naturally want to sell something that has become expensive. The logic is simple: buy low and sell high.

The problem is that financial markets do not always behave as intuition suggests. When a market establishes an upward trend, the trend follower buys, which often means buying after the price has already risen. When a market establishes a downward trend, the trend follower sells or goes short to profit from the decline. Trend followers therefore act in ways that many investors find strange.

Consider, for example, a market that has risen sharply for several months. After such an advance, it is easy to feel that the move has already gone too far. One wants to wait for a pullback or hope for a better buying opportunity.

The trend follower reasons differently: if the market continues to demonstrate strength, that strength is precisely the signal. The fact that the price has risen is not an argument

against buying. It is the very reason the position is taken. The same principle applies in declining markets.

When a market falls sharply, there is often a temptation to buy because the price appears low relative to previous levels. The trend follower instead observes that the market is in a downward trend and therefore chooses to sell or go short.

Most people dislike admitting mistakes. It is a natural human trait. If an investment performs poorly, there is often a desire to give it a little more time. Perhaps the market will reverse. Perhaps the analysis was correct after all. Perhaps one simply needs to wait a little longer.

But intellectual accuracy and financial results are not the same thing. It is not enough to be proved right eventually about the direction. The position must also be taken at a time and in a size that make it possible to preserve the capital until the analysis is potentially confirmed. The forecast-driven investor is inclined to ask, ‘How confident am I that the analysis is correct?’ A more important question is, ‘How much will I lose if it is wrong?’

If the market develops in a way that contradicts the position, the trade is closed. The loss is accepted and the capital is moved elsewhere. The objective is not to defend the original analysis, but to protect the capital. From a psychological perspective, this is much harder than it sounds.

Every loss is, in practice, an acknowledgment that the market did not develop as expected. For many investors, this feels like failure. For the trend follower, however, it is a normal part of the process. In fact, the ability to accept being wrong quickly is one of the strategy’s most important characteristics.

A low win rate means that the investor must repeatedly accept small setbacks without knowing when the next major trend will emerge. Mathematically, the logic may be simple.

Psychologically, it is far more difficult to live with.

Human instinct wants to buy low – trend following buys strength.

Human instinct wants to sell high – trend following sells weakness.

Human instinct wants to be right often – trend following accepts being wrong often.

Human instinct wants to avoid losses – trend following accepts small losses as part of the game.

This is why many investors feel that the strategy is wrong even when it is working exactly as intended. The difficulty is not that the basic rules are complicated, but that they often conflict directly with our natural instincts.

This also helps explain why trend following has endured across so many different market environments. The strategy's basic principles have been presented in books, academic studies, and industry publications for decades. The rules are well known and, in many cases, relatively simple, but knowing them is not the same as being able to apply them consistently.

Understanding the strategy is therefore only the first step. The investor must also be able to live with its return pattern.

CTA Management

The CTA as Risk Manager

A CTA's work involves far more than a model for buy and sell decisions. The strategy's signals indicate where opportunities exist, but risk management determines the significance of each position for the portfolio as a whole.

A manager's day-to-day work therefore revolves largely around aggregate risk. How much risk does the portfolio carry, and which markets dominate its exposure? Are several positions effectively expressions of the same underlying risk? The manager must also assess how the portfolio would be affected if volatility increased or markets that had previously moved independently suddenly began moving together.

In this respect, the CTA manager has more in common with an airline captain monitoring instruments, risks and safety margins than with the stereotypical market guru portrayed in the financial press. Risk management is not added after the signals have been generated; it is an integral part of the

investment process.

A central part of the CTA manager's work is to construct a portfolio in which an incorrect signal or an unexpected market move cannot have an unreasonably large impact on performance.

Entry Signals

Systematic models can be technically sophisticated, but the basic principle of trend following is simple. When a market displays sufficient strength, the model may take a long position. When it displays weakness, the model instead takes a short position. The challenge lies in defining strength and weakness in a way that can be applied consistently across many markets.

A signal is based on a predetermined set of rules: an algorithm that specifies when a position should be entered, the direction it should take and when it should be exited.

CTA programs often use several price-based signals, such as moving averages, breakout models and momentum. Signals can be combined across different time horizons. Faster models respond earlier to changes in direction but are more affected by short-term noise, while slower models respond later but filter out more temporary price movements.

The signals are rarely based on a secret formula. Differences in the performance of CTA programs often depend more on market selection, time horizons, position sizing and risk management than on the entry signal itself.

Position Sizing

Once a model has identified a buy or sell signal, a crucial decision remains: how large should the position be? The signal indicates which market to trade and in which direction, but the size of the position determines how much the trade matters to the result.

Two managers may use the same signals and take positions in the same markets yet achieve entirely different results because of their position sizes. A more cautious manager applies the model with low risk, while a more aggressive manager takes larger positions. As exposure increases, both profits and losses are magnified.

Position sizes cannot, however, be determined solely by the number of contracts or the notional exposure. Futures contracts differ greatly in price level, contract size, point value and normal volatility.

A natural gas contract, for example, may change far more in value from one day to the next than a gold or government bond contract. If the same number of contracts were used in every market, a handful of volatile contracts would dominate the portfolio's performance.

To calculate position size, CTA managers commonly use market volatility. The principle is that a market with large price changes receives a smaller position, while a more stable market may receive a larger one. The aim is not to allocate the same amount to every market, but to prevent certain positions from having a disproportionate influence on performance.

This comparison requires converting the market's price movement into a monetary value. Every futures contract has a specified point value that determines how much a price change is worth. A one-point move represents one dollar amount in

a gold contract and an entirely different amount in a bond or energy contract. The manager must therefore calculate how much the contract normally changes in value before deciding how many contracts to trade.

A common tool for this purpose is ATR, Average True Range. ATR measures volatility—that is, the size of a market's price movements over a selected period. The calculation takes account of both the day's trading range and any price gaps from the previous day's close. A rising ATR indicates that market volatility is increasing, while a lower reading indicates that price movements are subsiding.

Suppose, for example, that a move equal to one ATR in a gold contract represents \$2,000, while the corresponding move in a natural gas contract represents \$4,000. If both positions are to contribute approximately the same amount of risk, the model could, in simplified terms, trade twice as many gold contracts as natural gas contracts. In practice, the calculation is also affected by available capital, the contracts' minimum trading size, the program's risk target and any stop levels.

Volatility thus makes position sizing adaptive. It is never constant, but changes with market movements. When price movements increase, the position is reduced; when volatility falls, it may instead be increased. The same market and the same signal can therefore result in different position sizes at different times.

Correlations among positions are also part of the calculation. Every position must fit within the program's overall risk profile, which specifies how much total risk the portfolio may carry and how that risk may be distributed among individual markets.

Suppose the model simultaneously receives buy signals in U.S., European, Japanese and Australian stock indices. These may

appear to be four separate positions, but all of them may be affected by the same macroeconomic event.

Similarly, long positions in several government bond markets may represent a shared exposure to falling interest rates. Positions in different currency pairs may sometimes be several expressions of the same exposure to the U.S. dollar.

Because futures contracts are traded on margin rather than paid for in full, position size should not be determined by the contract's notional value. Instead, the manager focuses on the potential loss and the portfolio's aggregate risk.

The difference between notional exposure and actual risk was illustrated by George Soros in a 1998 interview on the CBS program *60 Minutes*. The Quantum Fund's position against the British pound in 1992 amounted to approximately \$10 billion, but according to Soros, the potential loss represented only about two to three percent of the fund's capital. A position can therefore be very large in notional terms while still presenting limited risk, provided that the loss scenario is clearly defined and the position can be closed as planned.

Large losses also have disproportionate consequences because the recovery begins from a lower capital base. After a decline of 20 percent, a gain of 25 percent is required to restore the capital. After a decline of 50 percent, a gain of 100 percent is required. This is an important reason why position sizing and the limitation of large losses matter so much.

Unrealized Profits and Growing Positions

Trend-following models can build a position gradually through a technique known as pyramiding. The initial position is taken when the entry signal occurs. If the market then continues in

the direction of the trend and the original position shows a profit, additional contracts are added. The manager is therefore not adding to a losing position in the hope of a reversal, but increasing a position in which the market has already confirmed the signal.

Pyramiding means that the largest positions are normally built in the markets that develop the strongest trends. A position that was initially small and conservatively sized may eventually become one of the portfolio's most important exposures. In some cases, a single market can therefore account for a large share of the program's total return over an entire year.

As the position and its unrealized profit grow, the portfolio's short-term fluctuations often increase sharply. A reversal can then have a substantial effect, particularly after the market has already moved a long way in the direction of the trend.

It is important to distinguish between the position's profit and the account's performance. A position may still be profitable while the account is in a drawdown. This happens when part of the unrealized profit has disappeared. The account value has then declined from its previous high even though the position remains profitable.

This is what it means to ride a trend: the manager must accept reversals against the prevailing direction without exiting the position when the profit declines. If every move against the trend led to an exit, the model would fail to follow the large trends that generate a significant share of its long-term return.

The price is that part of the unrealized profit sometimes disappears before the exit signal arrives. A position may at one point show a profit of 80 percent but ultimately be closed with a gain of 50 percent. The final result remains very strong, yet it may still feel negative to someone who saw the higher account

value.

The manager can never know in real time whether a reversal is merely a temporary interruption or the beginning of the end of the trend. The position must therefore be retained until the predetermined exit rule indicates that the market's behavior has changed. Only after the position has been closed is it possible to determine how much of the trend the model actually captured.

This volatility is often not apparent in a long-term performance record. If results are reported only monthly, large intramonth movements may disappear between two reporting dates. The fluctuations can, however, become very visible from one month to the next.

A position may, for example, have contributed a 30 percent return at month-end and then reverse so sharply that the program's capital falls by 10 percent during the following month. The decline is then caused by the reduction of a previously unrealized profit, but it still lowers the account's market value. Unrealized profits are real money, but in a trend-following model they must be allowed to fluctuate while the manager waits for the trend either to continue or to generate a clear exit signal.

Fluctuations often intensify the longer a trend continues. The initial position is usually taken under relatively calm conditions, before the market has attracted much attention. As the price move continues, more participants are drawn in, speculation increases and volatility rises. The combination of a larger position and a more volatile market then causes the unrealized profit to vary substantially.

A modest position can thus develop into a roller coaster and ultimately account for a large share of the year's result. Accepting the fluctuations and surrendering part of an unrealized profit is not a failure of the model, but the price of following a trend

long enough for it to make a meaningful contribution.

Risk and Volatility

In finance, volatility is often used as a measure of risk. The greater the fluctuations in an investment, the higher the risk is considered to be. But volatility does not tell the whole story. It describes how much the value has varied, not why the fluctuations occurred or what losses may arise in the future.

A trend-following position may display high volatility after a substantial unrealized profit has accumulated. A sharp reversal can then reduce the account's value without the trade necessarily threatening the original capital. The main risk at that point is that part of the unrealized profit will be surrendered before the exit signal arrives. This is a real decline in account value, but it is different in character from a loss incurred on the capital base.

There is a more fundamental risk that volatility cannot measure: the risk that the model will not work in the future. Historical standard deviation describes how large past fluctuations have been, but it does not determine whether the strategy's underlying logic will continue to generate returns. This model risk is difficult to express as a single percentage. Assessing it requires judgment, common sense and an understanding of what the model is actually trying to capture.

The assessment must begin with the philosophy behind the strategy. Why should the model work at all? The premise of trend following is that markets repeatedly develop prolonged moves. As we have seen, these moves arise in part from economic imbalances, the slow adjustment to new information and herd behavior—not because a particular combination of

parameters produced the best historical result.

High volatility therefore does not automatically mean that a trend-following model is at greater risk of ceasing to work. Conversely, low volatility does not mean that the model is robust or that the risk is small. Some strategies may produce smooth, apparently reassuring returns for long periods while substantial risk builds beneath the surface.

A clear example is provided by strategies that regularly write options and collect premiums. Under normal market conditions, many of the options expire worthless, allowing the manager to report a long sequence of small profits. The equity curve is smooth, the win rate is high and measured volatility is low. The strategy may therefore appear considerably safer than a trend-following program whose results vary more from month to month.

But the premium is not free income. It is compensation for the option writer assuming the risk of a major market move. When an unusual but powerful move occurs, the loss may be many times larger than the premiums previously collected. Several years of small profits can then disappear in a matter of days or hours.

In such a case, low volatility gave a misleading picture of risk. It showed how the strategy had performed under normal conditions, but not what could happen when the market regime changed. Author and former options trader Nassim Nicholas Taleb describes this as picking up pennies in front of a steam-roller. Profits arrive frequently and regularly, while losses occur rarely but can be very large.

Volatility shows how much performance fluctuates. Model risk concerns whether there is still a sustainable reason for the return to arise. The first can be measured relatively easily; the

second must be assessed from the logic of the strategy.

Simulations

When a trading model contains fixed rules, it can be tested on historical market data. The rules must specify when long and short positions are taken, when losses are limited, when winning positions are exited, how many contracts are traded and which markets are included. This type of historical simulation is commonly called a backtest.

A simulation cannot predict the model's future return. It shows what would have happened if today's rules had been applied to yesterday's markets. This is an important limitation, but it does not mean that the simulation has no value. It gives the manager an indication of which results have historically been normal for the model and which future events would constitute a clear deviation.

It can show, for example, how often the model has historically produced winning and losing trades, how large the losses have been and how long positions have normally been held. It also shows the drawdowns that have occurred, both in depth and duration. The manager is therefore better prepared for prolonged periods of weak performance if these form part of the model's normal behavior.

Historical data can also be used to size positions and thereby set the program's risk level. By simulating different position sizes, it is possible to estimate how each choice would have affected returns, volatility and drawdowns. The same model can therefore be used to construct a more conservative program with lower expected fluctuations or a more aggressive program with a higher return target and larger expected drawdowns.

The test result does not mean that the future will follow exactly the same pattern. The largest future drawdown may be both deeper and longer than any previous decline.

The simulation provides a reference point, not an absolute limit. If the model has historically experienced drawdowns of 20 percent, the manager must not conclude that 25 percent is impossible. On the contrary, there should be a margin of safety to account for the possibility that the future will be worse than the past.

The simulation's most important purpose is therefore not to show how much return the model will generate. It should show how the model behaves under different market conditions. If live trading later deviates sharply from the historical behavior, there is reason to investigate why.

A historical simulation may also show whether the rules have previously contained a statistical advantage—an edge. The result, however, is only as credible as the assumptions on which the simulation is based.

One problem is that the risk of the strategy failing in the future increases when the rules are fitted too closely to historical data. By testing enough entry levels, time periods, filters and exceptions, it is almost always possible to find a combination that would have performed unusually well. The result may look convincing in the simulation while depending on chance patterns that never recur. This is commonly known as curve fitting, or overfitting.

A robust model should therefore not depend on one exact parameter value. If a breakout over 100 days produces a fantastic result while 90 and 110 days perform poorly, there is reason to be suspicious. The model may have been fitted to a handful of historical events that happened to favor that particular param-

ter. If several neighboring values instead produce reasonable results across many markets and different time periods, there is stronger support for the conclusion that the model is capturing recurring behavior.

In a simulation, it may be tempting to include only the markets in which the model historically performed best and exclude those that produced weak results. But such a selection is based on information that was not known when the trades would have been made. A market that was unprofitable for trend following for several decades may suddenly develop one of the program's most important trends.

Cocoa provides a clear example. For decades, cocoa may have produced weak results for certain trend-following models, but it later developed exceptionally powerful price moves. If cocoa had been excluded solely because older simulations looked poor, the manager would have been unable to participate when the major trends finally emerged. Retaining markets despite weak individual simulations can therefore contribute to a more robust and better diversified portfolio.

Trend following does not require every market to be profitable at all times. By following many different markets, the model has more opportunities to capture price movements. If only historical winners are included, the program risks being designed for yesterday's trends rather than tomorrow's.

The distinction becomes particularly relevant when more advanced models are used. Neural networks and other AI-based methods can analyze a very large number of variables and detect relationships that simpler models miss. The same flexibility, however, increases the risk of overfitting. A model with enough parameters may learn the noise in the historical data and produce an impressive simulation that deteriorates

quickly when the market environment changes.

Simple trend-following models begin from a different premise. They often contain few parameters and are based on rules. There is therefore less to fine-tune and fewer opportunities to create a perfect historical curve through optimization. The performance history of trend-following programs is far from optimal, but perhaps that is precisely why the concept continues to work—as long as no one tries to make it perfect.

The purpose of a simulation is not to find the combination of rules that would have made the most money historically. It should help both the manager and the investor understand the model's characteristics, weaknesses and possible behavior.

IV

Allocating to Managed Futures

Crisis Alpha

The Insurance No One Wants to Buy

Kathryn Kaminski, who together with Alex Greyserman conducted the extensive study of trend following over 800 years, also coined the term Crisis Alpha. The term described why Managed Futures had performed well during the 2008 financial crisis, when traditional investments suffered extensive losses. In the report *In Search of Crisis Alpha: A Short Guide to Investing in Managed Futures*, she analyzed how trend-following strategies had performed during earlier market crises.

The results confirmed something that many CTA managers had long observed in practice: during several of the largest financial crises, trend-following strategies had been able to generate positive returns while traditional investments suffered heavy losses.

Crisis Alpha does not, however, mean that Managed Futures always rises when the stock market falls. The strategy has no permanent inverse relationship with equities, which makes the

term misleading if it is interpreted as a promise of immediate returns during every crisis. The opportunity for large profits arises only when the crisis produces price movements that are sufficiently broad and persistent for the models to adjust their positions.

To benefit from the returns that Managed Futures can generate during a crisis, the investment must be in place before the need for protection becomes obvious. By the time the trend is clear enough to attract the attention of a wider group of investors, trend-following programs have generally already established their positions. It may not be too late to invest, but a significant part of the move may already have occurred.

During favorable periods when equity markets are rising, interest in risk management is often limited. After several years of advancing markets and strong portfolio performance, protection may feel like an unnecessary expense. Interest is often awakened only after the market environment has changed and prices have already begun to move sharply.

Few people would try to buy fire insurance after their house had caught fire or install a backup generator in the middle of a power outage. Protection feels most valuable when the need is obvious, but by then it is often too late to obtain it on reasonable terms.

The CTA manager does not sit waiting for the next disaster, but periods of turmoil are one of several market environments in which trend following can perform well. Trend following does not make money from disasters, but from price movements.

Crises often happen to set very large capital flows in motion. Investors leave risky assets, central banks change monetary policy, currencies are repriced and commodity markets adjust to new conditions. When these changes persist for long enough,

they create the environment that trend-following strategies are designed to exploit.

Several of the CTA industry's most successful periods have coincided with some of the most turbulent episodes in modern economic history. This is the relationship described by the term Crisis Alpha.

For a traditional investor, Managed Futures can serve as a diversifying component of a portfolio when stocks and bonds encounter difficulties. For professional CTA managers and active speculators, however, it may be a stand-alone investment and the principal source of long-term returns.

Crisis Alpha therefore describes an important feature of trend following, but not its only purpose. The strategy is not designed for the world to fall apart. It is designed to adapt when the world changes.

The 2008 Financial Crisis

When Lehman Brothers filed for bankruptcy in September 2008, the event became the symbol of a financial crisis that shook the entire global economy. For Managed Futures, the year became a decisive test of the strategies' ability to navigate a prolonged crisis.

The crisis did not, however, begin with Lehman Brothers. Like most major crises, the problems had been building for several years. Low interest rates, rising house prices and generous lending had fueled a powerful expansion in the U.S. housing market. Banks lent money to increasing numbers of households, while mortgages were packaged into complex securities and sold to investors around the world.

As long as house prices continued to rise, the system appeared

to work. When prices began to fall, it became clear that the risks were far greater and more widely dispersed than most people had understood. Credit losses increased, and banks became increasingly reluctant to lend to one another. During 2007 the problems became ever more apparent, yet few people still understood how far the crisis would spread.

When Lehman Brothers collapsed, it was no longer clear which banks, securities or counterparties could be trusted. No one knew which institutions carried the largest losses or how far the problems would spread. Credit markets froze, companies struggled to finance their operations and investors sought to reduce risk at almost any price.

Equity markets fell sharply around the world. From peak to trough, several major indices lost approximately 40–50 percent of their value. Credit strategies suffered substantial losses, and a number of hedge fund strategies that had been marketed as diversifiers proved dependent on rising asset prices and functioning credit markets.

Many investors discovered that seemingly different holdings were in practice exposed to the same underlying risk.

It was in this environment that many CTA managers began to stand out. This was not because they had predicted the financial crisis or possessed unique knowledge of the U.S. housing market, but because their models were designed to respond to the powerful and increasingly persistent price movements created by the crisis.

As equity markets developed clear downward trends, many trend-following models gradually established short positions in stock indices. The models did not need to understand the cause of the decline or determine which banks would survive. It was enough for prices to continue falling and for the move to

become sufficiently clear to generate signals.

At the same time, capital fled risky assets and moved into government bonds. Demand for U.S. and German government securities, among others, rose, yields fell and bond prices developed strong upward trends. Many CTA programs were therefore able to combine short positions in stock indices with long positions in government bonds.

Currency markets provided additional opportunities. As banks, companies and investors sought liquidity, large amounts of capital moved between countries and currencies. The U.S. dollar strengthened during important phases of the crisis, while other currencies developed clear moves in the opposite direction. For systematic models, these flows created more powerful trends to follow.

Commodity markets underwent dramatic changes. During the first half of 2008, oil prices had risen to record levels. As the global economy slowed, the trend reversed and energy prices fell sharply. Similar movements occurred in several other commodities as markets began to price in weaker growth and lower demand.

The crisis therefore created large and persistent trends across several asset classes at the same time. It was close to an ideal environment for a diversified trend-following portfolio. The opportunities did not depend on a single trade or market, but appeared in stock indices, interest rates, currencies and commodities.

Not every CTA program made money, however, and results varied. The industry is not homogeneous, and models respond at different speeds.

As a broad indicator of performance, the SG Trend Index gained 20.9 percent in 2008. The index tracks ten of the

largest trend-following CTA programs and is widely used as a benchmark for this part of the Managed Futures industry.

Behind the average were substantially stronger results. Winton gained approximately 21 percent, AHL more than 30 percent and BlueTrend 43.4 percent. Some smaller CTA programs achieved even higher returns. The year therefore demonstrated that the CTA industry could generate positive performance during a financial crisis, while also revealing large differences among managers.

The crisis showed that trend-following strategies could generate positive returns when traditional portfolios were suffering severe losses.

Covid 2020

The next major test for the CTA industry showed that not all crises look alike and that Crisis Alpha is not activated automatically as soon as markets become unsettled.

During the first months of 2020, a local outbreak of a virus in China developed into a global pandemic. Governments closed borders, air travel collapsed and companies were forced to suspend or restrict their operations. No one knew how long the shutdowns would last or what their economic consequences would be.

Between February 19 and March 23, the U.S. S&P 500 stock index fell by approximately 34 percent. The decline was not the largest in history, but it was one of the fastest. A market fall that in other crises might have unfolded over many months occurred in just a few weeks.

During the 2008 financial crisis, several of the most important market moves developed over a longer period and proved

relatively persistent. Trend-following models therefore had an opportunity to identify the direction, exit old positions and gradually establish new ones.

The first phase of the pandemic crisis was much faster. Markets fell almost vertically and, only a few weeks later, began an equally rapid recovery.

Many models therefore could not adjust their positions as quickly as the market changed. Programs positioned for the previous market environment could first be hurt by the decline and then by the rapid reversal.

Faster models had a greater ability to respond early, while slower models required more confirmation before changing direction. The results were consequently mixed.

This illustrates a central difference between conventional insurance and trend following. Insurance pays out when a predefined event occurs. Trend following works differently: the move must continue after the position has been taken.

If the decline occurs faster than the models can respond, or if the market immediately reverses, the strategy may struggle to benefit from it.

After the initial shock, new moves began to emerge. Developments in commodity markets were particularly dramatic as transportation and industrial production ground to a halt and demand for energy collapsed. On April 20, the U.S. WTI contract for May delivery fell as low as -\$40.32 per barrel and closed at -\$37.63.

This was not free oil. The buyer needed to be able to accept delivery in Cushing, Oklahoma, and arrange storage. As storage capacity began to run out and the contract's final trading day approached, many market participants were willing to pay to dispose of their contracts and avoid delivery. At the low, this

effectively meant that the buyer received 1,000 barrels of oil and just over \$40,320 for taking them.

When economies later began to reopen, the trend reversed. Demand recovered faster than production, supply chains remained disrupted and many commodities shifted from sharp declines to prolonged upward trends.

Equity markets also established a new direction. Once the advance had taken hold, trend-following models could gradually exit their short positions and follow the new trend upward.

The pandemic therefore contained two distinct phases. First came a rapid and chaotic shock in which many models struggled to adjust in time. This was followed by a longer period in which reopening economies, changing consumption patterns and large capital flows produced increasingly clear price movements.

The Inflation and Interest Rate Shock of 2022

Interest rate increases created prolonged declines in bond markets, while differences in monetary policy gave rise to major currency trends. The U.S. dollar strengthened during much of the year, and Russia's invasion of Ukraine intensified movements in energy, grains and other commodities.

Trend-following CTA programs were able to exploit movements across several asset classes simultaneously. Among other positions, they went short bond futures and stock indices, long the U.S. dollar, and followed the powerful trends in energy and commodity markets. It was this breadth, rather than the development of any one market, that made the period particularly favorable for diversified trend following.

The SG Trend Index, which tracks ten of the largest trend-

following CTA programs, gained 27.3 percent in 2022. It was the index's best calendar year since its inception in 2000. After a decade in which the relevance of trend following had repeatedly been questioned, the result demonstrated how quickly conditions could change.

A comparison of the crises shows why Crisis Alpha does not arise during every market decline. In 2008, prolonged and broad trends emerged. In the spring of 2020, an extremely rapid fall was followed by a sharp reversal. In 2022, prolonged moves again developed across several markets.

Anyone investing in Managed Futures must therefore accept that the strongest years are unevenly distributed and that the strategy may feel unnecessary for long periods.

Alternatives to Managed Futures

Managed Futures is not the only way to seek protection for a portfolio. Investors can hold cash, buy government bonds or gold, invest directly in commodities, or use options and specialized strategies designed to rise when equity markets fall. Each alternative, however, is exposed to different risks and works under different market conditions.

Cash does not normally fall in nominal value in the same way as stocks and gives the investor flexibility when attractive opportunities arise. If the interest rate is lower than inflation, however, the purchasing power of the capital declines over time. Cash can therefore serve as a safe haven during an individual crisis, but it does not generate a positive return from the price movement itself.

Government bonds are another traditional form of protection. During recessions and deflationary crises, central banks often

lower interest rates, which usually causes bond prices to rise. If inflation is driving the crisis, central banks may instead be forced to raise rates even as equity markets fall. Bonds and stocks may then lose value at the same time, as the events of 2022 demonstrated.

Gold is often regarded as protection against inflation, currency depreciation and declining confidence in the financial system. During certain crises, the metal has performed extremely well, but the relationship is neither stable nor immediate. Gold can fall when liquidity is needed, when real interest rates rise or when the U.S. dollar strengthens.

History also shows that the metal can expose its owner to very long periods of losses. After the 1980 peak, gold entered a decline that did not reach its bottom until more than 21 years later. Gold can therefore serve as a long-term store of value but provide weak protection for anyone who needs the capital during the wrong part of the cycle.

Directly owned commodities and other real assets have provided effective protection during certain inflationary periods, particularly when price increases were driven by shortages of energy, food or industrial metals. As general crisis protection, they are less reliable. When the economy weakens, demand often falls rapidly and commodity prices may decline alongside the stock market.

Put options on a stock index offer more direct protection. The option increases in value when the market falls and can provide a rapid payoff during a sudden crash. Specialized tail-risk strategies are often based on this principle and are designed to generate large profits during extreme market movements.

The protection, however, carries an ongoing cost. Options have a limited life and must be renewed regularly. If the

crisis fails to materialize, the premiums may be lost repeatedly. Protection also tends to be most expensive after the market has already become unsettled and demand for options has increased.

No form of protection works under all conditions. Cash offers stability but limited real returns. Government bonds may work well during recessions but less effectively during inflationary crises. Gold and commodities can protect against certain economic developments but involve permanent long exposure. Options provide more direct protection but must be paid for and renewed.

Managed Futures offers no guarantee either, but its ability to adjust exposure to the price movements that actually occur explains the strategy's role in diversified portfolios.

The question, therefore, is not which single form of protection is always best. No such protection exists. The question is which risks the investor wants to be prepared for and which combination of strategies can make the portfolio less dependent on a single economic outcome.

Managed Futures in the Portfolio

Why Managed Futures?

The practical question is how Managed Futures can be integrated into an investment portfolio and how large an allocation is required to affect the portfolio's characteristics.

When Managed Futures is integrated into a traditional portfolio, the classic 60/40 portfolio is a natural starting point. The purpose is not to dismiss the model, but to make the portfolio less dependent on rising equity and bond markets. Managed Futures can add an independent source of return and thereby another engine to the portfolio.

The objective is not to maximize returns in the prevailing market environment, but to build a portfolio that can handle different conditions for growth, inflation and interest rates. For this distinction to matter in practice, the allocation must be large enough.

Allocation size

As a general rule, the larger the share of a portfolio allocated to Managed Futures – or to any other single investment – the more difficult it becomes to endure periods of negative performance and retain confidence in both the strategy and the original allocation.

An investor who places all of their capital in trend following will feel every loss much more strongly than one who lets the strategy represent five percent of a broader portfolio. The models and results may be the same, but the consequences for the investor are entirely different.

If five percent of the portfolio is allocated to a CTA program and the program falls by 20 percent, the total portfolio declines by approximately one percentage point, assuming the remaining holdings are unchanged. If all the capital is invested in the same program, however, the decline is 20 percent. In the first case, it is relatively easy to view the loss as a normal part of the investment process. In the second, the same drawdown can create doubts about both the strategy and the decision to invest.

It is common for an investor who initially believes they can accept a 20 percent drawdown to begin having doubts when the decline reaches only 10 percent. This is especially true when the capital has been invested recently and the decline immediately represents a loss. A drawdown that begins after a gain, while the investment is still in positive territory, is often easier to endure.

The allocation that looks best in a historical calculation is therefore not necessarily the one that works best in practice. A historical portfolio analysis may show that a large allocation to Managed Futures would have improved the portfolio's risk-adjusted return. The calculation does not reveal whether the

investor would have endured several years of weak performance while the equity market rose and other investors made money.

If the allocation is too large, the risk increases that the investor will abandon the strategy during a drawdown and return only after the next strong period. The strategy may work over an entire market cycle while the investor's purchases and sales impair the result.

The allocation can also be so small that it has no practical significance. If Managed Futures represents one percent of the portfolio and the strategy rises by 30 percent during a market crash, it contributes only 0.3 percentage points to the total result. With an allocation of five percent, the contribution is 1.5 percentage points. The investor may therefore claim to be diversified without the allocation materially changing the portfolio's characteristics.

The allocation must therefore be large enough to affect the portfolio when the strategy performs differently from traditional holdings, yet limited enough for the investor to retain it through drawdowns and long periods of weak performance.

Optimal allocation

From a purely mathematical perspective, larger allocations to Managed Futures can appear attractive. When historical results are combined in a portfolio optimization with equities and bonds, an allocation of 20–40 percent to Managed Futures, or more, is not unusual. It is often at these higher allocations that the portfolio's Sharpe or Sortino ratio reaches its highest level.

The Sharpe ratio measures return above the risk-free rate relative to the portfolio's total volatility, while the Sortino

ratio considers only negative deviations. A high ratio therefore means that the portfolio has historically generated a high return relative to the risk taken.

The explanation is that the calculation rewards assets and strategies that have positive long-term returns but low correlation with equities and bonds. If Managed Futures has also performed strongly during some of the periods when traditional portfolios suffered their greatest losses, the calculated diversification benefit becomes even greater. This pattern appears in several published analyses.

An early example is a study published by CME Group in 2008. Using the efficient frontier, which identifies the combinations offering the best relationship between return and risk, the most efficient portfolio was calculated to consist of 20 percent Managed Futures, 40 percent equities and 40 percent bonds.

When RCM Alternatives later updated the analysis with data through the end of 2015, the Managed Futures allocation rose to approximately 35 percent. In calculations ending at the close of the years 2008–2012, it was around 40 percent.

Similar results have been presented in analyses by Litman Gregory, Wadhwani Asset Management and others. Despite using different periods, indices and optimization objectives, these calculations also pointed to higher CTA allocations than most investors use in practice.

Man Group's 2024 study Honey, I Shrunk the Trend-Following uses data from January 1995 through December 2023. The combination with the highest Sharpe ratio contained 40 percent trend following and 60 percent of a global 60/40 portfolio. When the objective was to minimize the largest historical decline, the trend-following allocation rose to 63 percent.

Some analyses also make use of the capital efficiency of fu-

tures. Because futures positions require only part of the contract value to be posted as collateral, trend-following exposure can in some portfolios be added on top of existing equity and bond holdings. A portfolio with 100 percent of its capital invested can therefore have total market exposure exceeding 100 percent.

In some examples, the theoretically optimal weight in Managed Futures can amount to 50 percent or more. Such a solution also involves leverage and requires sufficient liquidity to meet margin requirements. In practice, it is more common among wealthy private investors and institutional investors.

These figures were not produced by a single model. Analyses using different indices, periods and optimization objectives point in the same direction: historically, portfolios with high risk-adjusted returns have often contained substantially larger allocations to Managed Futures than most investors actually assign to the sector. The difference between the theoretically optimal weight and the allocation an investor can retain is decisive.

An optimization model does not account for the investor's reactions. The theoretical model feels no disappointment, anxiety or doubt. It rebalances the portfolio according to the specified rules without asking whether the strategy has stopped working.

An investor with extensive experience of trend following who accepts its drawdowns may be better equipped to retain a higher allocation than someone encountering the strategy for the first time. An allocation of approximately 5–10 percent does not change the character of the portfolio in the same way as 20–40 percent, but is often easier for the investor to manage.

A small allocation to a high-volatility CTA program can nevertheless have a greater impact than a much larger allocation

to a more conservative program. The program's risk level, not merely its share of capital, therefore determines its effect on the portfolio.

Sharpe and Sortino ratios may seem complicated. A simpler and more practical way to assess the size of an allocation is to calculate how a decline in Managed Futures would affect the entire portfolio. If the selected CTA program could reasonably fall by 20 or 25 percent, the investor can calculate the consequences for the total result.

With a CTA allocation of 10 percent, a decline of 25 percent reduces the total portfolio, all else being equal, by 2.5 percentage points. At an allocation of 40 percent, the corresponding decline is 10 percentage points. Using these figures as a starting point makes it easier to determine which allocation the investor can actually accept.

The best allocation is one that is large enough to improve the portfolio's characteristics, but not so large that the investor cannot retain it through several weak years. A portfolio that is optimal on paper but abandoned at the wrong time is inferior to a theoretically less efficient portfolio that the investor can hold over time.

Managed Futures is not tied to a particular stage of life. For someone who starts early, an independent source of return and the reinvestment of profits can have a considerable impact over several decades.

For someone approaching retirement, even a more limited allocation can reduce dependence on the equity market performing well during the specific years when the capital will be needed. An allocation of 10–15 percent need not change the portfolio's fundamental character, but can still have a noticeable effect.

Four portfolios

The significance and effect of the allocation become clearer when translated into actual portfolios. The traditional 60/40 portfolio serves as the starting point, together with three alternatives in which Managed Futures is assigned a weight of 10, 20 and 40 percent, respectively.

Table 12.1. Examples of four portfolios with different Managed Futures allocations.

Portfolio	Equities	Bonds	Managed Futures
Traditional portfolio	60%	40%	0%
10% Managed Futures	54%	36%	10%
20% Managed Futures	48%	32%	20%
40% Managed Futures	36%	24%	40%

The CTA allocation is funded by reducing the equity and bond allocations proportionally.

In the traditional portfolio, 60 percent is invested in equities and 40 percent in bonds. If 10 percent is allocated to Managed Futures, the equity allocation is reduced to 54 percent and the bond allocation to 36 percent. With a CTA allocation of 20 percent, the portfolio consists of 48 percent equities and 32 percent bonds. In the final alternative, 40 percent is invested in Managed Futures, while equities and bonds represent 36 and 24 percent, respectively.

The source of the capital used for the CTA allocation matters greatly. If the equity holding is reduced, the portfolio becomes less dependent on stock-market performance. This can limit losses during an equity decline, but can also dampen returns during a strong bull market.

If the capital is taken from bonds instead, the portfolio's interest-rate and duration exposure declines. This can be beneficial during an inflation and interest-rate shock, but less favorable in a deflationary crisis in which government bonds rise sharply. A proportional reduction in both equities and bonds, by contrast, preserves the relationship between the traditional asset classes and allows Managed Futures to form a third component. This is the method used in the following calculations.

Funding the allocation from cash has a different effect. Cash has low volatility, while the CTA program assumes active market risk. The change can increase the portfolio's expected return and improve diversification, but it does not necessarily reduce total risk.

There is therefore no neutral way to fund a CTA allocation. The choice determines which risks decrease, which increase and what function Managed Futures serves in the portfolio.

The comparison uses the S&P 500 including dividends as the equity index, the Bloomberg U.S. Aggregate Bond Index as the bond index and the SG Trend Index as the representative of large trend-following CTA programs.

The SG Trend Index is a non-investable, fee-adjusted index that tracks ten large trend-following CTA programs. It is an established international benchmark, but primarily represents large institutional managers whose programs often have lower volatility than more aggressive alternatives. An actual invest-

ment may deviate from the index.

The choice of index also affects the comparison. The IASG Trend Following Index, for example, rose by nearly 50 percent in 2008, compared with approximately 21 percent for the SG Trend Index. Nevertheless, the SG Trend Index is used here because it is clearly defined and well established. The difference illustrates why results must be interpreted in light of how each index is constructed.

Three years are of particular interest. In 2002, the S&P 500 fell by 22.1 percent, while the bond index rose by 10.3 percent and the SG Trend Index gained 26.1 percent. In 2008, the S&P 500 fell by 37.0 percent, while the bond index rose by 5.2 percent and the SG Trend Index gained 20.9 percent. In 2022, the equity index fell by 18.1 percent and the bond index by 13.0 percent, while the SG Trend Index rose by 27.3 percent.

Table 12.2. Calculated portfolio performance in 2002, 2008 and 2022.

	2002	2008	2022
60% equities / 40% bonds	-9.1%	-20.1%	-16.1%
54% equities / 36% bonds / 10% Managed Futures	-5.6%	-16.0%	-11.7%
48% equities / 32% bonds / 20% Managed Futures	-2.1%	-11.9%	-7.4%
36% equities / 24% bonds / 40% Managed Futures	+5.0%	-3.7%	+1.3%

The calculations are based on calendar-year results for the S&P 500 including dividends, the Bloomberg U.S. Aggregate Bond Index and the SG Trend Index. The portfolios are assumed to have been rebalanced to the specified weights at the beginning of each year. Sources: Bloomberg and Société Générale. Author's calculations.

Based on the index results, it is possible to calculate how the four portfolios would have performed in 2002, 2008 and 2022. The comparison shows the result for the traditional 60/40 portfolio and for portfolios with 10, 20 and 40 percent Managed Futures, respectively.

In 2002, the equity market was in the third year of its decline following the dot-com bubble. Bonds made a positive contribution, but the effect of Managed Futures was considerably greater. The traditional 60/40 portfolio would have fallen by approximately 9.1 percent. With 10 percent Managed Futures, the decline would have been limited to about 5.6 percent, and with 20 percent to 2.1 percent. An allocation of 40 percent would have turned the loss into a gain of approximately 5.0 percent.

In 2008, bonds worked as intended and limited part of the equity market's decline. Managed Futures provided additional protection. A CTA allocation of 10 percent would have improved the portfolio result by approximately 4.1 percentage points, and an allocation of 20 percent by about 8.2 percentage points. With 40 percent Managed Futures, the improvement would have amounted to approximately 16.4 percentage points and the portfolio's decline would have been limited to 3.7 percent.

In 2022, the difference became even clearer because both equities and bonds fell. The traditional portfolio then lacked a component that performed positively. With 10 percent Managed Futures, the decline would have been limited from 16.1 to 11.7 percent. With 20 percent, it would have stopped at approximately 7.4 percent. An allocation of 40 percent would have reversed the result entirely and produced a gain of approximately 1.3 percent.

Smaller losses shorten the road back. After a decline of 20 percent, a gain of 25 percent is required to restore the capital, whereas a decline of 10 percent requires a little over 11 percent.

The value of diversification therefore lies not only in a better result during the crisis year, but also in leaving the portfolio with a smaller loss to recover.

The calculation shows how the size of the allocation determines its significance for the total result. Even 10 percent Managed Futures would have reduced losses in all three years, although equities and bonds continued to dominate the portfolio. At 20 percent, the effect became clearer. With 40 percent, the portfolio's characteristics changed fundamentally: the loss in 2008 was limited, while the declines in 2002 and 2022 turned into modest gains.

Managed Futures and equities

For an investor seeking higher long-term returns, a combination of equities and Managed Futures may be an alternative to the traditional 60/40 portfolio. By omitting the bond component, a larger share of the capital can work in assets and strategies with higher return potential. This entails greater fluctuations and periods when both components may perform poorly, but also the possibility of substantially higher returns over time.

Trend-following CTA programs have demonstrated low correlation with equities and performed differently during several prolonged equity-market declines. A decline in the equity market often occurs during periods of economic slowdown, which in turn can create strong trends in both currencies and commodities as supply and demand adjust to the new economic conditions. Managed Futures does not provide direct protection against inflation, but can benefit from the price movements caused by inflation if they develop into persistent trends.

The strong results during crisis years show what a CTA allocation can contribute. It would, however, be misleading to report only periods when trend following performed well. The same independence from the equity market can work in the opposite direction and reduce portfolio returns during an extended bull market.

Table 12.3. Performance of the S&P 500 and the SG Trend Index during the CTA winter of 2011–2013.

Year	S&P 500 including dividends	SG Trend Index
2011	2.11%	–7.93%
2012	16.00%	–3.52%
2013	32.39%	2.67%
Total	56.81%	–8.80%

S&P 500 returns include dividends. The total represents the geometrically compounded return for the entire period. Sources: S&P Dow Jones Indices and Société Générale. Author's calculations.

Between 2011 and 2013, the S&P 500 including dividends rose by 2.11, 16.00 and 32.39 percent, respectively. The cumulative return was therefore 56.81 percent. By contrast, the SG Trend Index fell by 7.93 percent in 2011 and 3.52 percent in 2012, before rising by 2.67 percent in 2013. Over the full period, the result was a decline of 8.80 percent.

These years occurred during the prolonged slump that became known as the CTA winter, when the absence of persistent trends

made it difficult for many trend-following programs to generate returns.

An investment of 100 dollars in the S&P 500 would have grown to nearly 157 dollars during the period, while the equivalent investment in the SG Trend Index would have declined to just over 91 dollars. A CTA allocation would therefore have reduced the portfolio's return, and the effect would have become more pronounced as the share allocated to trend following increased.

It would have been easy for an investor to question the strategy as the gap relative to the equity market widened over several years. Three years can feel like a very long time, but it is a short period from a long-term investment perspective.

Diversification does not mean that every component of the portfolio should perform well at the same time. If all components always rose together, they would probably also move together on the way down and therefore provide less protection.

The results do not show that the relationship between equities and Managed Futures is always negative. They do show, however, that trend following can perform independently of the equity market and that the correlation during certain major market declines has been clearly negative. It is this independence that makes Managed Futures attractive as a diversifying component. Returns are generated through positions in multiple markets and do not depend on a rising equity market.

This distinguishes Managed Futures from investments described as protection that nevertheless retain significant exposure to the equity market.

The price of this independence is that the CTA allocation will sometimes lag during long equity bull markets. This disadvantage cannot be removed without changing the strategy's other characteristics. It is a consequence of owning something that

can genuinely deliver a different result when the equity market falls.

A more detailed year-by-year comparison could easily create false precision. The SG Trend Index cannot be invested in directly, and each CTA program has its own return profile depending on the markets, models, time horizons and risk levels employed. The statistics are therefore deliberately general. The purpose is not to show the exact result an investor would have obtained, but to illustrate the function and potential that Managed Futures can add to a portfolio.

A more extensive historical calculation could also create a misleading impression of what an actual investment can be expected to deliver. This is especially true if the capital is placed in a more aggressive CTA program whose return potential and fluctuations can be considerably greater than those of the institutional programs with billions under management included in the SG Trend Index.

The difference between the index and the selected program is therefore a risk that the investor must bear. The reported statistics show that the basic principles behind combining equities and Managed Futures are sound, but they cannot predict the performance of an individual investment. In reality, the result will always deviate from the index calculation – sometimes for the better and sometimes for the worse.

To rebalance or not

Portfolio weights change after an investment has been made. If Managed Futures rises while equities fall, the CTA allocation will automatically grow while the equity allocation declines. A portfolio that initially consisted of 50 percent equities, 30

percent bonds and 20 percent Managed Futures may therefore have an entirely different composition after a major market move.

Rebalancing restores the portfolio to its original weights. The investor then sells part of the assets that have performed most strongly and allocates the capital to those that have fallen. Positive CTA performance during an equity-market crisis can therefore serve two functions. It limits the portfolio's decline and creates capital that can be reallocated to equities or bonds after prices have fallen.

Rebalancing should not, however, be performed mechanically after every minor change. If the CTA program is in a strong trend, reducing the allocation too quickly may cause the investor to sell while the favorable market move is still under way. Frequent adjustments can also create costs.

The portfolio can be rebalanced at predetermined times, for example once a year. The investor then need not attempt to time the market or make decisions based on emotion, but simply follows the established strategy.

The rule should be determined when the investment is made, because otherwise the temptation is strong to rebalance only when it feels safe. If the principle is not established in advance, rebalancing risks being governed by the same fear and greed that the portfolio construction was intended to limit.

Another alternative is to act only when a weight deviates sufficiently from the chosen level. A target weight of 20 percent in Managed Futures may, for example, be allowed to move between 15 and 25 percent before the portfolio is restored to its intended allocation.

Many CTA programs and Commodity Pools have fixed redemption periods, notice periods or lockups. Rebalancing may

therefore require some advance planning. If the portfolio is rebalanced frequently, the investor risks reducing the components that are performing strongly before trends in equities or Managed Futures have reached their full potential. Together with transaction costs, rebalancing can impair the long-term result.

A simulation for the period 2000–2025, comparing annual rebalancing with allowing the components to develop freely, shows no decisive difference. With 40 percent allocated to Managed Futures, annual rebalancing produced a slightly better long-term result, but both methods led to similar performance. This is related to the fact that both equities and trend-following strategies performed positively during the test period.

There are two reasonable approaches to rebalancing. One is to restore the portfolio's original weights regularly. The other is to allow what is performing well to continue growing.

Just as a CTA manager lets profits run, the investor can refrain from reducing the component of the portfolio that is performing most strongly.

In practice, rebalancing means taking capital from the component of the portfolio that has performed best and moving it to the one that has performed worst. This can be a reasonable method when the portfolio consists of two well-diversified components that are both expected to generate positive returns over time.

Applying the same principle to individual investments, however, can be disastrous. Continually adding to losing positions and reducing winning ones allows losses to grow while cutting profits short.

The disadvantage of rebalancing is that a strongly performing investment is reduced before the entire advance has necessarily ended. The advantage is that the portfolio's risk allocation is re-

stored and no single component is allowed to become dominant. The choice depends on whether the investor primarily wishes to preserve the original risk composition or allow winning components to exert increasing influence.

It is therefore not wrong to allocate part of the capital to Managed Futures and then allow the investment to grow over time. Nor is it wrong to choose a more cautious approach and rebalance the portfolio regularly to preserve diversification and limit fluctuations.

There is therefore no allocation or rebalancing rule that suits everyone. The decision must be based on the function Managed Futures is intended to serve, the size of the decline the investor can accept, the source of the capital and how long the investment can be held. What matters is not finding a theoretically perfect portfolio, but building one whose components are significant enough to make a difference while remaining possible to retain when performance becomes uncomfortable.

The Market Does Not Require a Degree

Assessing a CTA Manager

The financial markets have always attracted some of the world's sharpest minds. Nobel laureates, professors, mathematicians, physicists, and economists have tried to understand their movements and turn their knowledge into profitable investment management. Yet there is no clear link between academic brilliance and long-term success in the market. An advanced education can be an asset, but it offers no protection against flawed assumptions, excessive leverage, or inadequate risk control.

The market takes no account of diplomas or intelligence. It does not reward the person who formulates the most complicated theory, but the one who makes decisions suited to the conditions that actually prevail. A manager does not need to understand every economic relationship or predict the next crisis. The business must, however, survive when the models are wrong and the markets change.

The trading floors of Chicago showed that success did not require an academic degree. In the noisy trading pits, people from widely different backgrounds came together. Some had started as runners and learned the profession from the ground up. Several successful floor traders have been described as surf bums—people without impressive educational credentials or a conventional career in finance. What mattered was the ability to read the market, make quick decisions, and control risk under pressure.

When trading later moved to electronic marketplaces, the conditions changed. Those who fared best were the ones able to abandon familiar methods and adapt to the new environment.

LTCM illustrates the same principle from an entirely different perspective. Long-Term Capital Management was founded in 1994 by John Meriwether, who had previously headed bond trading at Salomon Brothers. The circle around the fund included Myron Scholes and Robert Merton, two of the world's leading financial theorists and recipients of the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel. The team consisted of experienced traders, mathematicians, and academics with a deep understanding of financial models.

The strategy was based on identifying small price discrepancies between securities that had historically moved together and taking positions in the expectation that the relationships would return to normal. This produced many small and relatively steady profits. Because the return on each trade was limited, the fund used extensive leverage to increase returns.

The return profile was almost the opposite of trend following. Trend followers accept many small losses while waiting for a few large gains, whereas LTCM could accumulate many small profits but risk very large losses when the relationships continued to

diverge. Leverage amplified profits under normal conditions, but made the losses devastating when the market moved against the fund.

During the first few years, the strategy worked well, and its success contributed to the growth of the positions. When Russia defaulted on parts of its sovereign debt in 1998, market behavior changed. Investors moved toward the most liquid and safest assets and sold those perceived as risky. The relationships did not normalize as LTCM had expected, but instead continued to move in the opposite direction.

At the same time, the positions were so large that they could not be unwound without affecting the market. In September 1998, a group of 14 banks and brokerage firms provided \$3.6 billion so that the positions could be liquidated in an orderly manner.

In 1998, LTCM lost approximately \$4.6 billion, wiping out nearly 90 percent of the fund's capital. The collapse showed that even the most advanced models depend on risk control, liquidity, and humility before the market. Academic credentials can strengthen an investment process, but they can never replace these fundamental requirements.

Larry Hite is a clear example of a very different path. He grew up with severe dyslexia at a time when such difficulties were rarely recognized or met with support. He was also blind in one eye and had severely impaired vision in the other. School was not a natural path to success. Hite worked as an actor, screenwriter, and concert promoter, among other occupations, before becoming a stockbroker and later a co-founder of Mint Investment Management.

By the end of the 1980s, Mint was the world's largest CTA manager and the first hedge fund to surpass one billion dollars

in assets under management. During Hite's 13 years at the firm, its combined programs generated an average annual return of more than 30 percent before fees.

His philosophy was not based on being smarter than the market or predicting the future better than others. The emphasis was on probabilities, diversification, and, above all, limiting the loss when a trade went wrong. The traits that had made a traditional educational path difficult did not prevent him from building one of the largest CTA firms of his time. In the market, risk management carried more weight than formal education.

John W. Henry grew up in a farming family and encountered the futures market through grain trading. He attended several universities but never earned a degree. Through the family farm, he learned how futures could be used to lock in prices for soybeans, corn, and wheat. These experiences sparked an interest in speculation and eventually led to a trend-following system. In 1981, he founded John W. Henry & Company, which developed into one of the Managed Futures industry's most successful investment firms.

At the same time, there are plenty of examples of successful CTA managers with advanced scientific training. William Dunn held a doctorate and was a pioneer in the use of quantitative methods in the futures markets. Ed Seykota studied at MIT, while Jim Simons built Renaissance Technologies by recruiting mathematicians, physicists, statisticians, and computer scientists.

The market has rewarded and punished self-taught managers and distinguished scientists alike. What matters is not their background, but how their knowledge is applied in practice and combined with a sound investment process. For the investor, a résumé is therefore only the beginning. The assessment must

also cover how the firm manages risk, how the organization is structured, how much capital the strategy can accommodate, and which markets the program can actually trade.

Size, Capacity, and Flexibility

Historical indexes can show how Managed Futures has performed as a group, but investors cannot invest directly in the SG Trend Index. The actual result depends on the choice of CTA program. Different programs can perform very differently from both the index and one another, depending on the manager's size, market selection, and risk level.

Size does not determine which manager is best. Smaller and larger firms can serve different purposes in the same portfolio. An established manager can provide a long history and institutional infrastructure, while a smaller firm can offer greater flexibility and exposure to markets that the largest managers cannot trade efficiently.

Capital within the CTA industry is heavily concentrated among the largest and most established managers. In 2026, the Barclay BTOP50 Index consists of 20 large, investable CTA programs. Together, they represent at least 50 percent of the capital managed by the investable CTA programs in BarclayHedge's database.

The concentration is easy to explain. Institutional investors need managers that can accept large allocations without the program's trading having an excessive impact on the market. The largest CTA managers usually possess the market capacity, organization, and infrastructure required. A smaller program may be more flexible and able to trade markets that are too small for a large capital base. Yet this very advantage makes it harder

to accept a larger investment without changing the conditions under which the strategy operates.

Institutional investors do not restrict themselves to the largest firms. Many seek out smaller and growing CTA managers, known as *emerging managers*, because their flexibility and ability to trade smaller markets may offer greater return potential. The manager must, however, be able to demonstrate that its reporting, risk control, and organization can withstand the scrutiny that comes with a larger mandate.

Maximizing assets under management is not necessarily an obvious objective for a CTA manager. The demands of institutional investors can force changes in strategy, market selection, and risk level that reduce the ability to generate returns. Some smaller CTA programs therefore deliberately limit their size. Many successful programs eventually close to new investment in order to protect the strategy's flexibility and capacity.

Size and history are often connected. A long *track record* creates confidence and shows how the manager has handled losing periods and changing market cycles. An established manager with substantial capital may, however, have become more cautious over time. Once the business is profitable and the organization has grown, both the incentives and the risks that can be taken change.

As assets under management grow, the practical significance of risk changes. A 30 percent decline represents three million dollars in a program managing ten million, but three billion in a program with ten billion under management. The percentage loss is the same, but large positions may be harder to unwind, and substantial redemptions can have greater consequences for the business.

An established manager may therefore lower the program's risk level, reduce position sizes, or exclude less liquid markets. The result is often more limited drawdowns, but also lower returns during favorable periods. A large program designed for annual volatility of around ten percent should therefore not be compared directly with a smaller manager operating at twenty or thirty percent.

It is tempting to conclude that smaller and younger programs always have the greatest return potential. Yet programs that fail during their first few years often disappear from databases, while successful ones survive and build the history that can later be studied. The same thing happens in equity indexes: companies that fail disappear, while the winners remain and shape the historical record. The statistics may therefore present an overly favorable picture of young CTA programs.

Organization and Key-Person Risk

In a smaller organization, a larger share of time and resources can be devoted to investment management itself. Decision-making paths are shorter, and the person who develops the models may work closely with trading, risk management, and investors. Less time then needs to be spent managing large teams or dealing with multiple organizational layers.

A concentrated organization can also have a downside. A great deal of knowledge and responsibility may rest with one person or a small number of people. Investors therefore need to understand what happens if the lead manager falls ill or leaves the firm. Are the models and procedures documented? Can someone else oversee trading and risk? Are technical backup systems and an effective business continuity plan in place?

The manager should also be able to explain who oversees trading, how exceptions are handled, and which reports investors receive. The independent infrastructure surrounding FCMs, administrators, and auditors does not replace the investor's assessment of the organization, but it makes it possible to verify that the business operates as described.

Fees Must Be Assessed in Relation to Returns and Risk

Large institutional CTA programs often charge relatively low fees but usually require a higher minimum investment. A large capital base can make the business profitable even with a low fee, and institutional investors also have considerable negotiating power. When a program's expected return is relatively modest, fees have a significant effect on the result that remains for the investor.

Smaller CTA programs often need to charge higher fees to cover the costs of trading, technology, administration, and compliance. Many also operate at a higher risk level and can generate larger percentage moves. The fixed management fee therefore carries less relative weight in successful years. An annual fee of two percent, for example, is far more noticeable when the return is ten percent than when it is twenty percent. The higher risk level, however, also magnifies losses during weak periods.

Fees must therefore be assessed in relation to the program's risk level, net return, and what the investor actually receives. A more expensive program may be the better choice if it delivers a stronger net result, while a low fee does not in itself make a weak program more attractive. Likewise, a high fee is no proof of high quality or sought-after expertise.

A performance fee creates a shared interest in capital growth, but the incentives are not entirely symmetrical. The investor bears the entire loss, while the manager receives a share of the profit. Investors should therefore check whether a *high-water mark* is used, whether the manager has its own capital invested on the same terms, and whether the program's risk level can be changed without investor approval.

A baker does not get up at four in the morning and light the oven out of charity. Nor does a CTA manager work without financial incentives. A performance fee can encourage continued development of the models and reward good results, but it must form part of a structure in which the manager also has strong reasons to protect capital when the market moves against the program.

What Investors Can Examine

Investing in a CTA program need not be more complicated than investing in other funds. Above all, investors should understand how the program is expected to make money, which markets it trades, and under what conditions the strategy may struggle. The manager need not disclose every model or parameter, but should be able to explain the strategy in understandable terms.

The performance history should show whether the returns come from actual trading or a simulation and whether they are reported after fees. A long *track record* is valuable, but only if today's program remains broadly similar to the one that produced the historical returns. If market selection, models, or risk levels have changed substantially, older results may have limited relevance to the investment offered today.

Average returns are not sufficient as a basis for a decision.

Investors also need to know the program's largest drawdowns, how long they lasted, and the risk level at which the program operates. The crucial question is whether the investor can maintain the allocation when a normal losing period occurs.

For registered CTA managers, the NFA normally requires a current *Disclosure Document* to be provided to prospective clients before they invest. The document is reviewed by the NFA before use and must describe, among other things, the program, its risks, fees, conflicts of interest, and past performance.

Finally, communication should be assessed. A credible manager describes both the program's strengths and weaknesses, reports changes clearly, and remains available even after a weak period. The manager need not be able to explain every individual market move, but should be able to describe why the program has performed as it has and whether the result is within the expected range. Communication should be just as open when results are weak as when they are strong.

The most impressive education, the largest organization, or the smoothest historical return curve does not necessarily lead to the best choice. A CTA manager should be assessed according to how strategy, risk, capacity, organization, and incentives fit together. The objective is not to find a program without weaknesses, but one whose characteristics can be understood—and that the investor can retain even when results temporarily fall short of expectations.

V

Timeless Principles

New Times, Old Principles

New Times

When Richard Donchian developed his first trend-following models, charts were drawn by hand and orders were relayed by telephone to the exchange floor. Computers later made it possible to test ideas over longer periods, apply them across more markets, and execute trades faster and more consistently.

Artificial intelligence is the next step in this development. The question is whether it changes the foundations of investment management or primarily provides new tools for solving familiar problems.

Financial markets have a fondness for proclaiming the arrival of a new era. Whenever a new technology breaks through or a new market emerges, the idea soon follows that earlier experience has lost its relevance. Historical comparisons are dismissed, and the new landscape is said to require entirely different rules. New times!

Today, artificial intelligence and cryptocurrencies occupy

the center of this narrative. AI can process vast amounts of data and search for relationships that no human could uncover unaided. Cryptocurrencies have simultaneously created global markets that trade around the clock and in which new assets, trading venues, and derivatives can develop rapidly. Both will undoubtedly affect the CTA industry.

Yet new tools do not necessarily change the fundamental problems of investment management. More information does not automatically make the future easier to predict, and a faster model can draw the wrong conclusions just as efficiently as it can discover genuine relationships. The decisive question is therefore what the technology actually contributes. What changes when CTA managers gain access to AI, and what remains the same?

AI – Revolution or Evolution?

At its core, trend following is not a complicated strategy. Artificial intelligence is not required to observe that a market is rising and take a long position.

AI should therefore not be judged by its ability to eliminate every false signal and drawdown. These are not technical defects, but part of the price of participating when a major trend develops. The technology should instead be assessed by whether it can improve investment management without sacrificing the strategy's robustness.

The term artificial intelligence encompasses several methods. Put simply, machine learning allows a model to search for statistical patterns in data without requiring every rule to be specified in advance. Deep learning uses neural networks to handle more complex information. What these methods share

is the ability to process large datasets and test more potential relationships than a human could examine.

In quantitative funds, including many CTA programs, AI can be used to analyze information that is not already organized into clear price series, such as weather data, satellite imagery, and news flows.

From space, it is now possible to monitor the world's commodity flows almost in real time. Satellites can measure how deeply an oil tanker sits in the water and thereby estimate how much oil it is carrying. They can also read the shadows cast inside floating-roof storage tanks and calculate whether inventories are rising or falling. AI can organize these datasets, compare changes over time, and identify shifts in trade flows or crop conditions before they become visible in official statistics.

Not All Algorithms Think for Themselves

A very large share of trading in liquid equity, currency, and futures markets is conducted automatically. As early as 2015, the CFTC estimated that more than 90 percent of trades in the most liquid U.S. futures contracts involved some form of algorithm or automation. That does not mean, however, that 90 percent of capital is managed by fully automated funds.

Automation serves several different functions. An execution system can carry out an investment decision that has already been made by dividing a large order into smaller pieces, selecting an appropriate time, and seeking to reduce costs and market impact. Electronic market makers continuously place and adjust buy and sell orders, thereby contributing liquidity and market depth. Other algorithms trade short-term price differences between related markets.

At the other end of the spectrum are fully systematic funds and CTA programs in which the entire chain can be automated, from analysis and trading signals to position sizing, order execution, and exit. It is therefore important to distinguish between algorithms that make the trading decision itself and execution algorithms that merely carry out an order that has already been decided.

When robots are said to account for a large share of trading, the claim often concerns how orders are executed, not who or what makes the investment decision. An algorithm can be advanced without analyzing the economy, predicting the market, or independently managing a fund. Often, the machine has taken over order execution, not investment management.

Liquidity monitoring is a related area. A market that is normally easy to trade can quickly become thinner when uncertainty rises. AI can flag widening spreads and disappearing market depth, and help determine whether orders should be divided, postponed, or routed to another contract. It can also detect when seemingly independent positions suddenly begin moving together.

AI can make data validation, testing, and comparisons between models more efficient. At the same time, the risk of overfitting increases as more variables and model variants are tested. The technology can identify genuine patterns, but it can also give historical coincidences a convincing mathematical form.

When Smarter Does Not Mean Better

Trend following rarely needs more parameters or greater computing power. The underlying principle is simple and does not need to be optimized for every historical period. On the contrary, every additional refinement risks making the model better fitted to the past but less robust in the future.

The real challenge is not to engineer away the strategy's weak periods, but to accept them and continue following the model when markets lack clear trends. AI cannot make those periods favorable without simultaneously changing the strategy. A model adjusted to perform better in sideways and choppy markets may respond less effectively when major trends return.

AI is therefore better suited to short-term trading, relative-value strategies, and statistical models in which anomalies come and go. In those areas, models must be reassessed continuously and sometimes replaced when the relationships that generated returns weaken. Trend following presents a different problem: the difficulty is not finding a smarter model, but living with the simple model's disadvantages long enough to benefit from its strengths.

AI can process enormous datasets, test thousands of combinations, and continually search for new models to replace those that have stopped working. But this speed also creates a problem.

Many managers rely on the same publicly available price data – open, high, low, and close – and allow their models to search for similar anomalies. As more systems discover the same pattern, they also begin placing similar trades. Competition then compresses the statistical edge until it disappears.

If many models use similar data, respond to the same changes,

and reduce risk at the same time, their orders can push the market further in the direction that first triggered the signals. The statistical edge may therefore disappear just as the risk of crowded positions and rapid liquidation increases.

The more advanced models become, the harder they may be to interpret. When a simple trend model loses money, it is usually possible to see which positions caused the loss and why the signals arose. A neural network may indicate what it wants to do without the manager fully understanding why.

Taken together, AI can broaden analysis, improve order execution, monitor liquidity, and make model development more efficient.

AI Does Not Make Markets Predictable

Neural networks are not new to financial markets. They were already being used during the 1980s to search for relationships in market data. In 1988, economist Halbert White published a widely noted study in which a neural network was used to search for predictable patterns in IBM's daily stock returns. The model nevertheless failed to produce reliable short-term forecasts – an early example of how difficult it is to distinguish persistent patterns from market noise.

What now appears primitive was revolutionary at the time. Since then, computing power has increased enormously and models have become far more advanced. Even so, no generally available AI has made markets predictable.

One reason is that a model never operates entirely independently of humans. Someone must decide which data it should use, what it should try to predict, and how success should be defined. The manager also determines the time horizon, risk

level, transaction costs, and the conditions under which the model should be withdrawn. AI can find patterns in the material it receives, but the manager must still judge whether they have an economic explanation or are merely historical coincidences.

AI has clear advantages in fields where large quantities of comparable data can be collected and results shared. In medical diagnostics, for example, images, disease progression, and treatment outcomes can be compiled from many hospitals. The models improve as the common body of knowledge grows.

Financial markets work differently. Prices and economic statistics are largely public, but the most valuable knowledge is often kept secret. A manager who discovers a profitable relationship has strong reasons not to publish the model, because the edge may weaken when many participants begin exploiting it. Even in a market filled with intelligent machines, participants have limited information, protect their best discoveries, and react to one another's actions.

New Markets

When a new market emerges, the manager does not need to decide whether the technology will change the world or whether the asset is overvalued. What matters is whether the market offers price movements that can be exploited and instruments that make it possible to establish and unwind positions.

Bitcoin was launched in 2009 as a decentralized digital payment system with no central issuer. Unlike conventional currencies, its supply cannot be increased through political or monetary-policy decisions, but is limited to a maximum of 21 million bitcoin. This scarcity has led Bitcoin to be described as a form of digital gold.

In December 2017, CME Group launched a regulated U.S. bitcoin futures contract. The range of crypto derivatives has since expanded to include futures on Ether and Solana, among others. Other regulated exchanges now also offer cryptocurrency futures and broader crypto indexes. International crypto exchanges offer far more products, but many of these contracts fall outside the markets that a traditional CTA can or is permitted to trade.

The large and rapid price movements in cryptocurrencies make them attractive for trend following, but high volatility does not automatically produce profitable trends. Violent swings, false breakouts, and sudden reversals can generate many losses in a short period. At the same time, volatility creates opportunities, but requires smaller positions. If Bitcoin moves several times as much as a traditional currency, the exposure must be adjusted so that it does not dominate the risk of the entire portfolio.

The development of futures contracts has made cryptocurrencies more accessible to traditional CTA management. Regulated and centrally cleared contracts provide standardized terms, margin requirements, and a central counterparty. Many are cash-settled, like equity index futures. The manager therefore does not need to take delivery of or deliver the underlying cryptocurrency when the contract expires.

Crypto can now be incorporated into the same framework as commodities, currencies, interest rates, and equity indexes. The sector fits well with the CTA industry's tradition of seeking opportunities in new markets.

Cryptocurrencies are not the only area capable of broadening future CTA portfolios. Emissions allowances, electricity, freight, and other specialized futures markets can also develop large and

persistent price movements.

The freight market illustrates how a new investment area can gradually become accessible to systematic trading. The first exchange-traded ocean freight futures contract was launched in 1985 and was based on an index of the cost of transporting commodities along different trade routes. The contract later disappeared, but was replaced by more specialized freight agreements.

Today, derivatives exist on the cost of transportation by dry bulk vessels, oil tankers, LNG and LPG carriers, and container ships. Shipowners and buyers of transportation services can therefore hedge future freight costs, while other market participants can take positions on rising and falling prices in the global shipping market.

History shows, however, that far from every futures contract that is launched survives. Over the years, exchanges have introduced futures on everything from eggs and onions to frozen shrimp and cheddar cheese. There have even been plans for contracts whose value would be determined by the box-office receipts of individual Hollywood films.

Some of these markets disappeared when the need for hedging declined, while others never attracted sufficient trading. Onion futures were banned in the United States in 1958 after two traders manipulated the market and caused prices to collapse. Plans for futures on the box-office receipts of individual Hollywood films, meanwhile, were halted after opposition from the film industry, which warned of insider trading, manipulation, and outright gambling. The CFTC had already approved the contracts, but Congress intervened before trading could begin.

Such markets can also involve risks that are absent or less prominent in traditional futures contracts. The price of emis-

sions allowances can change as a result of political decisions, electricity markets have regional conditions, and weather-related contracts may depend on a single outcome rather than a persistent trend.

A futures contract can survive only if there is a genuine price risk to manage and enough participants on both sides of the market. It also requires clear terms, an appropriate contract size, reliable price formation, and sufficient trading activity for positions to be opened and closed without unreasonable costs.

Physical storage and delivery are not required, however. Freight futures, for example, can be cash-settled against an index of transportation costs even though no freight or shipload changes hands. What matters is that settlement is based on a transparent, representative, and manipulation-resistant price benchmark.

The Next Generation of CTA Managers

The CTA manager's role is now less about personally performing every calculation and more about framing the right problem, reviewing the result, and taking responsibility for risk. AI can produce more models and trading proposals, but someone must still decide which deserve real capital.

This requires a combination of market knowledge and technical understanding. A skilled programmer may discover a statistical relationship without understanding why it arose. An experienced manager may recognize a plausible market development but lack the tools to examine it systematically. The CTA manager of the future must combine both perspectives: understand how markets and contracts work, while also being able to assess how models were developed and what their results

actually show.

AI can suggest new variables, write computer code, and identify combinations that appear profitable in historical data. Responsibility for evaluating the model and allocating capital still rests with the manager.

Humans are therefore not disappearing from CTA management. Their responsibilities are shifting. The more analysis and trading are automated, the more important it becomes to understand the model's purpose, control how it is used, and set limits on the amount of risk it may take.

The unexpected can never be ruled out.

Successful investing is therefore not always about knowing what will happen. More often, it is about being prepared when it does.

Beyond Trend Following

The Man Who Beat Them All

Jim Simons was one of the most successful investment managers in financial history. He began his career in Managed Futures but went on to develop a form of quantitative management that lay far beyond traditional trend following.

Wall Street has never lacked famous names. George Soros made headlines with some of the boldest macro trades in financial history. Warren Buffett and Charlie Munger turned Berkshire Hathaway's annual meetings into international events, and millions of people followed their letters, statements, and investment decisions.

Jim Simons chose almost the opposite path. He rarely sought the spotlight, never disclosed his positions, and never revealed how the models worked. The business was built not on Wall Street but on Long Island, where mathematicians, statisticians, and computer scientists searched behind closed doors for recurring patterns in market noise.

It is difficult to say who is the greatest investor in history. The answer depends on whether the comparison is based on total profit, average return, risk, longevity, or the amount of capital that could be managed.

But if the comparison is based on long-term returns after fees, Simons is difficult to surpass. Over roughly three decades, his most successful fund is reported to have returned close to 40 percent a year after some of the highest fees in the financial industry.

Simons began managing currencies, commodities, and futures as a CTA manager. His earliest trades were based partly on his own judgment, but opinions were gradually replaced by mathematical models. The search shifted from major trends to a large number of small, short-lived statistical edges.

The Mathematician Who Left Academia

Jim Simons's path to the financial markets began far from Wall Street. He was born in Massachusetts in 1938 and grew up in a Jewish family whose father ran a shoe factory. Even as a child, he was fascinated by mathematical problems.

When he was three or four years old, his father said during a car journey that they needed to stop for fuel. Simons objected that the gasoline should never run out: the car could first use half, then half of what remained, and continue in the same way forever.

He had not yet considered that the car would also travel progressively shorter distances, but he had independently approached what is known as Zeno's paradox. The episode captures something that would follow him throughout his life: an ability to calculate, but above all a desire to question seemingly

obvious assumptions and approach problems from unexpected angles.

At seventeen, Simons began studying mathematics at the Massachusetts Institute of Technology, MIT. He earned his undergraduate degree at the age of twenty and then continued at the University of California, Berkeley. In 1961, at only twenty-three, he received his doctorate in mathematics. His dissertation concerned differential geometry, an abstract field in which geometry, algebra, and analysis are used to study curved surfaces and multidimensional spaces.

After completing his doctorate, Simons taught at MIT and Harvard, but his mathematical ability soon took him into an entirely different field.

During the Cold War, he was recruited by the Institute for Defense Analyses, IDA, an organization that conducted advanced research for the U.S. government. There he joined a group of mathematicians attempting to break encrypted messages and identify patterns in large quantities of seemingly disordered information.

The work had several similarities to what Simons would later do in financial markets. A codebreaker does not know the system that created the message. The task is to study the information, find recurring structures, and distinguish genuine relationships from random noise. This requires both mathematical skill and an ability to detect faint patterns that others miss.

Simons enjoyed the intellectual challenge and had considerable freedom to pursue his own research. His employment ended, however, after he publicly criticized the Vietnam War. In an interview, he openly stated his opposition to the war and his intention to devote part of his working time to campaigning against it. Management did not appreciate the statement, and

he was dismissed in 1968.

That same year, Simons became chairman of the mathematics department at Stony Brook University in New York State. He was only thirty years old. The university was still young and its mathematics department had not yet achieved prominence, but Simons recruited several leading researchers and helped turn Stony Brook into an international center for geometry.

Together with mathematician Shiing-Shen Chern, he developed what later became known as Chern–Simons theory. The work became highly influential in both mathematics and theoretical physics and helped establish Simons as a distinguished researcher.

He had thus reached a leading position in academia. Even so, he began looking for a new challenge.

For several years, Simons had been interested in trading currencies and other financial instruments. At first, it was something he pursued alongside his research.

The markets, however, presented a problem well suited to his way of thinking. Prices often appeared random, but perhaps recurring patterns were hidden within their movements. If such patterns could be identified mathematically, it might be possible to build models that made better and more consistent decisions than a human trader.

In 1978, Simons left his secure academic career to devote himself to financial markets full time. The decision was far from risk-free. He exchanged an established position in mathematics for a field in which he lacked traditional education and experience. He had never worked at an investment bank, analyzed companies on Wall Street, or built a career as a professional broker.

That outsider status would become one of his greatest assets.

Simons did not view the market through the eyes of a traditional investment manager. He saw it as a mathematical problem: an enormous body of data that could be examined for recurring relationships.

Monometrics and the Early Attempts

When Jim Simons left Stony Brook University in 1978, he was convinced that mathematics could be used to find recurring patterns in financial markets. Yet he was still far from an answer. The company he founded was named Monometrics, a combination of *money* and *econometrics*. The name expressed the ambition to unite money with the systematic analysis of economic data.

Simons first turned his attention to the currency market. After the collapse of the Bretton Woods system, exchange rates left their narrow, politically determined ranges and began to exhibit large and sometimes persistent movements. For a mathematician searching for patterns in price data, the currency market was a natural starting point.

There was, however, no ready-made model to use. Computers were slow, historical market data were difficult to obtain, and research in quantitative finance was still in its infancy. Simons therefore had no clear path from mathematical theory to a functioning trading system. He knew only that prices ought to contain more information than traditional analysis had managed to capture.

To help him, he recruited mathematician Leonard Baum, whom he had met while working as a codebreaker. Simons and Baum collected and analyzed historical currency prices. They searched for recurring behavior that could indicate what the

market was likely to do next.

Some of the early models were based on momentum and trends. If a currency had moved in a particular direction, they examined whether the movement tended to continue. The analysis also included possible relationships between currencies and, later, among commodities and other financial markets.

The first attempts therefore contained clear elements of trend following. Simons was not satisfied with a simple trading rule, however; he wanted to measure the strength of the relationship, when it worked, and the probability that the movement would continue.

At first, the currency trading performed well. The results encouraged Monemetrics to extend its models to markets including U.S. Treasury futures and commodities. It soon became clear that a system that worked in one market could not automatically be transferred to another. Losses grew, and the promising start was followed by significant setbacks.

The problem did not lie only in the models. Despite their mathematical backgrounds, Simons and his colleagues still made several decisions according to their own judgment. They might follow a signal when it felt convincing, ignore it when it seemed unreasonable, or intervene when a position developed in an uncomfortable way. The result was a mixture of systematic signals and discretionary decisions.

An objective model loses its purpose if the manager follows it only when its signals agree with his own view. Simons discovered that, despite his scientific ambitions, he was not immune to the very reactions the models were intended to restrain.

He then formulated a far more demanding objective: the models would not merely support trading decisions; they would

make the decisions themselves. The system would determine when to open and close a position, without a manager subsequently choosing which signals to follow.

Simons wanted to build models that could operate consistently even while he slept. That would free him from monitoring the market and making new discretionary decisions.

The path proved much longer than he had initially imagined. Markets changed, relationships weakened, and patterns that looked convincing in historical data did not always work in live trading. Moreover, the clean and continuous datasets required to test the ideas properly did not exist. Before any reliable patterns could be identified, the information had to be collected, checked, and made usable.

Monometrics' early years were marked by experiments, successes, losses, and reassessments. Simons's mathematical background gave him no immediate advantage on Wall Street.

Simons had not yet found the working method that would make the company famous under its later name, Renaissance Technologies. He had, however, identified the principle that would guide its development: market decisions would be based on patterns that could be measured, tested, and repeated, not on stories about what prices ought to do.

Simons later described how the sharp swings of trading affected him. One day, the positions might move in the right direction and make him feel like a genius. The next day, the market might reverse and produce the opposite feeling. It was, as he put it, no way to live one's life.

Trend following therefore became a starting point, but not the final objective. Simons gradually moved toward shorter-term statistical relationships that appeared more frequently and reduced dependence on a handful of major market moves.

To succeed, he needed more data, more powerful computers, and colleagues who could view the market from perspectives different from Wall Street's traditional approaches.

Scientists Instead of Market Gurus

Jim Simons's most important insight may not have been that mathematics could be applied to financial markets, but that he did not need to solve the problem alone. If the markets contained identifiable patterns, the best people for the task would be researchers accustomed to working with complex systems, large datasets, and uncertain relationships.

He did not seek help from traditional fund managers, equity analysts, or economists with fixed views about interest rates and the business cycle. Instead, he recruited mathematicians, statisticians, physicists, codebreakers, and computer scientists.

Many had no experience of financial markets when they were hired. To Simons, that was an advantage. They had no fixed ideas about how markets ought to work and could therefore investigate what the data actually showed.

Leonard Baum became one of the first important colleagues. He was one of the creators of the Baum–Welch algorithm, a method for estimating hidden processes behind observed data. Baum suspected that financial markets could be analyzed in a similar way: prices were visible, while the underlying states had to be estimated.

Another significant recruit was James Ax, a distinguished algebraist whom Simons knew from Stony Brook. Ax had built a successful academic career but was attracted by the opportunity to apply advanced mathematics in an entirely new field. He developed Baum's models further and attempted to adapt them

to more currencies, commodities, and futures markets.

As the models developed, the ambitions also grew. In 1982, Monometrics changed its name to Renaissance Technologies. The business would operate like a scientific laboratory: hypotheses would be formulated, tested against data, and discarded if the results failed to hold up.

In 1988, Renaissance Technologies registered as a CTA and formally became part of the Managed Futures industry. That same year, the quantitative Medallion Fund was launched, using Axcom's models as the basis for its trading. The name Medallion referred to the mathematical awards that Simons and Ax had received during their academic careers.

The initial results were promising but uneven. In 1989, the fund suffered a drawdown of about 30 percent. Ax believed the decline was within the model's expected risk and wanted to continue. Simons instead chose to halt trading and re-examine the system. The conflict ended with Ax leaving the firm.

To rebuild the model, Simons turned to Elwyn Berlekamp, a professor at Berkeley and an expert in information theory and coding. His work on distinguishing information from noise had clear parallels with financial markets.

Berlekamp was skeptical of models that attempted to explain the market's long-term development. He preferred a large number of shorter trades in which each individual position needed only a small statistical edge. Together with Simons, Henry Laufer, and other researchers, he worked to make trading faster, more consistent, and less dependent on individual judgment.

Henry Laufer had previously been a mathematician at Stony Brook and became one of Renaissance's longest-serving and most important researchers. He helped combine the firm's many separate models into a coherent structure.

The signals could be weighted together so that different rules did not need to compete over which trades should be executed. The system evaluated the portfolio as a whole and sought the combination of positions that best exploited the statistical edges identified by the models.

Under Berlekamp's leadership, Medallion's trading was re-designed. When the fund resumed operations in 1990, it produced a net gain of more than 50 percent for the year. Berlekamp then returned to academia, but the changes he and the research team had implemented became an important part of the firm's continued development.

The next major step came through Peter Brown and Robert Mercer. They had previously worked in computerized speech recognition at IBM's research division. Their task was to teach computers to interpret speech without first teaching them the grammatical rules of language. Instead, the systems were fed large quantities of text and speech, after which the algorithms estimated which words and word combinations were most likely.

The approach suited Simons's view of the market. A relationship did not need a complete economic explanation, but it had to recur often enough and survive rigorous testing.

Brown and Mercer joined Renaissance in 1993. Their experience with language models and statistical probabilities made it possible to analyze more signals and combine them in a common system.

No single signal needed to be particularly strong. The edge could be very small and sometimes exist only briefly. When hundreds or thousands of such signals were combined across many markets and trades, however, they could produce a more stable result that was also difficult to replicate.

The organization was built to support this work. The re-

searchers did not work in isolated groups that concealed their ideas from one another. Results, data, and methods were made available throughout the firm so that a discovery in one area could improve models in another. A new signal had to fit into the common system and be evaluated according to how it affected the portfolio as a whole.

Simons's role became increasingly unlike that of a traditional fund manager. He did not sit alone making decisions about the dollar, gold, or the bond market.

His main task was to find unusually talented scientists, persuade them to collaborate, and create an environment in which ideas could be tested without regard to prestige. A model was judged by how well it survived testing, even if it had been developed by the firm's most distinguished mathematician.

Renaissance's most important asset therefore became not a single model, but a way of working that made it possible to search constantly for new relationships and abandon those that had stopped working. Many of the patterns were too weak to appear on an ordinary price chart and emerged only when large quantities of historical data were compared.

When thousands of relationships were tested, some could appear profitable purely by chance. Renaissance therefore had to determine which patterns survived outside the period in which they had been discovered and discard those that merely described historical noise.

This differed from long-term trend following. A trend model can pass through several weak years because the market lacks clear movements, without the underlying principle having stopped working. A short-term statistical relationship, by contrast, may disappear completely when the market's structure changes, more participants begin exploiting it, or costs become

greater than the edge.

The work was never finished. The models had to be continually reassessed and new opportunities sought. Renaissance's success lay as much in its ability to keep searching as in the discoveries the firm had already made.

Exceptional Returns

Between 1988 and 2018, Medallion is reported to have generated an average annual gross return of about 66 percent. After fees, approximately 39 percent a year remained. The fund's fees eventually reached around five percent of assets and 44 percent of profits, far above the hedge fund industry's traditional model of a two percent management fee and a 20 percent performance fee.

An annual return of nearly 40 percent after fees is remarkable. When repeated over several decades, the compounding effect becomes staggering.

Over the 31 years from the beginning of 1988 to the end of 2018, one dollar invested in Medallion would theoretically have grown to about 21,000 dollars after fees, if all profits could have been reinvested. Medallion is also reported to have generated more than 100 billion dollars in cumulative trading profits.

The year 2018 does not mark the end of Medallion, but the final year in the most frequently cited compilation of the fund's long-term results. Since the last outside investors were bought out in 2005, Renaissance has not published an ongoing performance record for outsiders.

The history through 2018 became known primarily through Gregory Zuckerman's research for the highly readable book *The Man Who Solved the Market: How Jim Simons Launched the*

Quant Revolution. Information about the fund's subsequent performance has become public only sporadically.

The outside investors were bought out because Medallion had a clear capacity limit. The fund traded frequently and attempted to exploit small price movements that could disappear quickly. If orders became too large, they affected prices and reduced performance. A model that worked exceptionally well with one hundred million dollars could not be expected to produce the same percentage return with ten billion.

Renaissance protected Medallion's returns by limiting the fund's size. After the outside investors were bought out, the fund became available mainly to the firm's employees and owners. When capital reached the maximum that the models could handle, profits were distributed rather than reinvested. The firm thus declined additional capital in order to preserve the strategy's conditions.

Two Very Different Renaissance Funds

Medallion was not Renaissance Technologies' only fund. The firm also launched several programs for outside investors, including the Renaissance Institutional Equities Fund, RIEF. These funds were designed to manage substantially more capital and therefore relied on different models, longer time horizons, and more liquid positions.

The results were also entirely different. During the 2008 financial crisis, Medallion is reported to have gained about 80 percent after fees, while RIEF lost approximately 16 percent. The difference became clear again in 2020. Medallion rose by about 76 percent, while RIEF fell nearly 20 percent and Renaissance Institutional Diversified Alpha lost almost 32 percent.

The same organization could therefore produce both an exceptional gain and substantial losses in the same year. This shows that Renaissance's success cannot be explained solely by access to skilled mathematicians, advanced computers, or large datasets. What mattered was which models were used, how quickly they traded, and how much capital the strategies had to carry.

Medallion's returns should therefore not be attributed solely to the mathematical signals. The results were created through the interaction of research, data quality, diversification, leverage, risk control, efficient order management, and unusual discipline regarding capacity. If any of these elements had been missing, the same models would probably have produced a very different result.

Jim Simons did not discover a formula that showed where the market would go. Renaissance instead built an organization capable of detecting and exploiting many small statistical edges. Medallion's record became possible because the entire chain worked and the fund remained small enough.

Epilogue

At a Distance

Successful speculation in financial markets often requires standing apart from the crowd and following one's own path. The great irony is that trend following is fundamentally based on the opposite. When enough market participants drive a price in the same direction, the trend follower tries to follow along.

At the same time, choosing to invest in Managed Futures means breaking away from the traditional investment crowd. Despite its long history and established role in institutional portfolios, this form of management has never gained widespread acceptance. The trend follower therefore follows the crowd in the market, while the investor who chooses Managed Futures goes their own way.

The trend follower also follows the crowd's movement without having to share its conviction. There is no need to believe that gold will continue to rise, that interest rates will fall, or that a particular currency is undervalued.

The model observes that the price is moving in a particular direction and takes a position accordingly. When the movement ends or reverses, the position is closed. In this way, the model follows the crowd without being drawn into the narrative surrounding it.

This distance matters. Anyone who becomes convinced of a market's long-term potential risks holding the position even after the trend has reversed and thereby giving back a large share of the profit. The investor then follows the market both up and down in the same long position. Such investors are commonly called *permabulls* – perpetual optimists who remain positive regardless of what the price shows.

Even a seemingly broad portfolio can rest on the same economic conditions. When a crisis changes those conditions, previously weak correlations can strengthen and several assets can fall at the same time. The investor's response and the time required to recover then become at least as important as the percentage decline.

Managed Futures does not solve all these problems and does not automatically rise whenever the stock market falls. Instead, it adds a source of returns that can benefit from movements in either direction without depending on the same economic developments as traditional investments.

The manager does not need to know where the next major move will emerge. The first clear movements may appear in commodities as inflation rises, in interest-rate markets during a recession, and in currencies or equities during a financial crisis.

The next significant trend may just as easily begin in a market currently receiving little attention – cocoa, coffee, cotton, sugar, orange juice, cattle, or hogs. What matters is that the model can detect the movement and adjust its positions accordingly.

The principles often appear simple. Buy markets that are rising, sell markets that are falling, limit losses, and let profits grow. The difficulty lies in living with the consequences.

Most signals do not develop into major trends. Many positions therefore end with small losses, and several such losses may

occur in succession. During prolonged periods without trends, the strategy lacks the movements needed to generate new returns.

It can be tempting to shorten losing periods and increase the proportion of winning trades. Such adjustments, however, risk making the strategy less robust and overfitted to historical data.

The real test comes during drawdowns, losing years, and long periods without new highs, particularly when other investments are performing strongly. The manager knows the model and its historical weak periods. The investor, however, mainly sees capital standing still. The decision to leave can therefore feel rational even when the strategy is working as intended.

Following the crowd from a distance means respecting the market's movement without being consumed by it. The trend follower does not try to predict where the crowd will go. The trend follower waits until it moves, follows along while the direction persists, and steps aside when it reverses.

That is what it means to follow the crowd from a distance.